

Anticyclic Mutation

Eva Zimmermann

Eva.Zimmermann@uni-leipzig.de

Jochen Trommer

jtrommer@uni-leipzig.de

Institut für Linguistik

Universität Leipzig

Beethovenstraße 15

D-04107 Leipzig

Germany

Contents

1	Introduction	1
2	Basic Types of Non-Base Mutation	4
2.1	The Sample	4
2.2	Root-to-Affix Mutation	5
2.3	Affix-to-Affix Mutation	9
2.4	Affix-to-Word Mutation	13
2.5	Root-to-Word Mutation	15
3	Alternations between Mutation Types	22
3.1	Alternations between different Types of Anticyclic Mutation	23
3.2	Alternations between Cyclic and Anticyclic Mutation	24
3.3	Alternations Involving Tautomorphemic Mutation	26
3.4	Summary	28
4	A Concatenative Analysis of Anticyclic Mutation	29
4.1	A Concatenative Approach to Cyclic Mutation	29
4.2	GNA-Analysis of Anticyclic Mutation	34
5	Discussion: Concatenative vs. Construction Approaches to Mutation	39
5.1	Alternative Interpretations of the Cycle	40
5.1.1	Phase-based Cycles: Sande et al. (2020)	40
5.1.2	Cyclicity via postsyntactic C-command: Rolle (2018)	42
5.2	SBM as a Restriction on Subtractive Mutation (Alderete 2001, Rolle 2018) . . .	44
5.3	Locality of Mutation in a Concatenative Approach	46
6	Conclusions	48
7	References	48

Anticyclic Mutation

Abstract

A major argument for process-based approaches to nonconcatenative morphology is the observation that mutation morphology (morphological categories expressed by changes in tone, stress, segmental quality or quantity) seem to exhibit a systematic asymmetry: Affixal morphology cyclically triggers changes in the bases to which it attaches, but bases don't trigger comparable 'anticyclic' changes in affixes (Inkelas 1998, Alderete 2001, Rolle 2018). Intuitively, processes (morphology) can affect objects (stems), but objects cannot affect processes. Alderete (2001) calls this generalization 'Strict Base Mutation'. Inkelas (1998) argues that it follows from approaches where mutation is the effect of morphophonological constraints but is unexpected in item-and-arrangement models where mutation is the effect of defective floating material (Lieber 1992, Wolf 2007, Bye & Svenonius 2012), in the terms of Bermúdez-Otero (2012) 'Generalized Nonlinear Affixation'. This paper has three goals: *First*, we provide extensive typological evidence that Strict Base Mutation is empirically wrong presupposing a standard lexicalist cophonology approach: Anticyclic mutation exists for all major phonological modalities. *Second*, we demonstrate that the attested types of anticyclic mutation directly follow from a standard Generalized Nonlinear Affixation approach to cyclic mutation couched in the formal framework of Colored Containment Theory (van Oostendorp 2008, Trommer 2011, Zimmermann 2017). *Third*, we show that recent reformulations of the SBM in non-lexicalist models such as Cophonologies by Phase (Sande et al. 2020), and the model developed in Rolle (2018), while solving some of the problems raised by the Strict Base Mutation hypothesis in standard cophonology theory, still make problematic empirical predictions.

1 Introduction

There are two major types of theoretical approaches to nonconcatenative morphology. *Generalized Phonology* approaches basically treat nonconcatenative morphology with formal mechanisms also capturing phonological processes but triggered directly by morphological features. This holds true for derivational Item-and-Process accounts such as Amorphous Morphology (Anderson 1992) where processes such as the vowel fronting ('umlaut') in German diminutives (e.g., [valt] 'forest' and [vɛlt-çən] 'small forest') are achieved by rules like $V \rightarrow [-back] / __ [+diminutive]$. Such rules are formally of the same format as classical SPE rules (modulo the morphological conditioning). But the same is true for many OT-based approaches such as Cophonology Theory where mutation follows from construction-specific constraint rankings. In contrast, *Generalized Affixation* accounts extend the classical Item-and-Arrangement approach and maintain that mutation is the effect of affixing subsegmental or suprasegmental material. Thus German diminutive umlaut can be captured by suffixing a floating $[-back]$ feature to the final vowel in addition to the segmental affix which leads then to overwriting of the underlying $[+back]$ specification of the base by general phonological processes. This approach has been highly influential in the autosegmental literature on morphological length and tone, and has seen a recent revival under the heading 'Generalized Nonlinear Affixation' (GNA, Bermúdez-Otero 2012).

While arguments against an Item-and-Arrangement approach traditionally highlight on the claim that it is too restricted to account for the full complexity of attested nonconcatenative processes (Anderson 1992, Inkelas (1998), Alderete (2001), and, more recently, Rolle (2018) identify an important formal difference between the two approaches where a GNA account actually predicts the *richer* typology. Under the GNA approach, mutation follows simply from the possibility that one morpheme (e.g., the affix) contains an incomplete piece of phonology which is realized on another morpheme (e.g., the root) by general phonological processes. In this account, there is no inherent root-affix asymmetry. The affix as well as the root could contain defective phonological structure that happens to be realized on an adjacent morpheme (say an affix). We would hence expect a process such as 'anti-umlaut', where specific roots

(or complex bases of affixation) trigger segmental changes on affixes. In an amorphous system eschewing the autosegmental apparatus of GNA, there is no obvious way to model anti-umlaut: Morphological processes cyclically change the underlying forms of stems (either simplex roots or morphologically complex stems) but stems have no symmetric means to influence the output of morphological processes in an anticyclic way. Alderete (2001) calls this prediction the principle of ‘Strict Base Mutation’ (SBM) which we cite in (1) in the formulation of Rolle (2018).

- (1) *Strict Base Mutation Hypothesis (SBM, Rolle 2018:93)*
 Alternations triggered by morphophonological operations are found exclusively in the stem
 (simplex or complex) which serves as the base of a morphological process.

We will focus our discussion of the SBM on the original version of Cophonology Theory (variably also called ‘Sign-Based Phonology’ or ‘Construction Phonology’) as developed by Orgun (1996), Inkelas (1998), Inkelas & Zoll (2005), Inkelas (2014) because it embraces the most explicit claim that nonconcatenative morphology is generally process-based (see especially Inkelas 2008). However, most of our arguments extend mutatis mutandis to other Generalized Phonology approaches (see section 5 for some discussion).

Crucially, in Cophonology Theory, the SBM follows from the two assumptions in (2):

- (2) *Theoretical Assumptions deriving SBM in Cophonology Theory*
 (i) Roots are phonological objects, but morphology consists of morphophonological processes
 (ii) Phonological evaluation is cyclic, i.e., recursively triggered by morphological constructions

Assumption (2-i) directly captures the asymmetry imposed by the SBM for roots (indicated here and in the following by “√”) and affixes under simple affixation.¹ Only the pattern in (3-a), but not the one in (3-b) should be attested according to Strict Base Mutation:

- (3) *Root-affix interactions in mutation*
 a. *SBM-compliant Affix-to-Root mutation* b. *SBM-violating Root-to-Affix mutation*
- [afx [√] afx]

[afx [√] afx]

However, the SBM as formulated in (1) is not restricted to roots, but applies to all bases. The cyclicity assumption in (2-ii) generalizes the claim to complex bases as in (4). Since morphological structure-building in Cophonology Theory is incremental and applies in tandem with phonology, mutation cannot affect affixes added on later cycles (4-b) and applies strictly to inner morphological material which has been added on previous cycles (4-a):

- (4) *Affix-affix interactions in mutation*
 a. *SBM-compliant Affix-to-Affix mutation* b. *SBM-violating Affix-to-Affix mutation*
- [afx [afx [√] afx] afx]

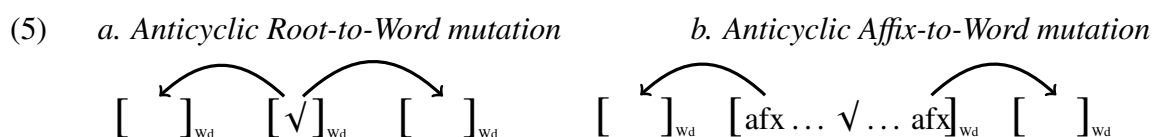
[afx [afx [√] afx] afx]

If we understand selection of a root morpheme as the initial cycle of a morphophonological derivation, what singles out the mutation patterns in (4-a) and (3-a) as conforming to the SBM is that mutation triggered by a morphological process in a given cycle applies strictly to material

¹The traditional notion of cyclicity, as employed in Lexical Phonology is derivational interleaving of morphological and phonological operations. A process-based formalism which uses this approach is the Amorphous Phonology framework of Anderson (1992). However, as shown by Orgun (1996), this notion of cyclicity can also be captured in a declarative constraint-based framework. This underlies the notion of cyclicity in Cophonology Theory, but also Antifaithfulness Theory (Alderete 1999), and most other current work in process-based morphology such as Paradigm Function Morphology (Stump 2001).

introduced by previous cycles. In this sense, we will call mutation compliant with the SBM *cyclic*, and the patterns where roots or affixes trigger mutation on outer material (i.e., on material added in the current or later cycles) *anticyclic*.

The nature of cycles of course depends crucially on the overall architecture of the grammar and the interface between morphology, syntax, and phonology. Cophonology Theory presupposes the classical lexicalist architecture also found in Lexical Phonology (Kiparsky 1982) and Stratal OT (Bermúdez-Otero 2018): cycles are restricted to a morphology component which strictly precedes phrasal syntax and phonology. On the other hand, the interaction of morphosyntax and phonology in phrasal phonology is non-cyclic (or, put differently, is restricted to a single cycle of phonology following all syntactic structure building). Accordingly, the formulation of SBM in (1) implies that roots and affixes should not be able to trigger mutation processes in phrasal phonology. We call these patterns excluded by the SBM *Root-to-Word mutation* and *Affix-to-Word mutation* shown schematically in (5-a) and (5-b):²



Whereas SBM seems to be a central means to distinguish Generalized Phonology and GNA accounts, this claim has to the best of our knowledge never been investigated systematically. We aim to fill this gap with this paper by (1) presenting extensive empirical crosslinguistic evidence against the SBM hypothesis establishing a systematic typology of attested anticyclic mutation and (2) developing analyses for representative cases of anticyclic mutation in Colored Containment Theory, a major recent framework embracing an item-and-arrangement approach (Zimmermann & Trommer 2014, Paschen 2018). These theoretical accounts demonstrate that a GNA account is not only in principle compatible with anticyclic morphology but parsimoniously derives all attested types of the phenomenon by the same constraints previously motivated for cyclic mutation.

Further below (in section 5), we will also address the predictions of cophonology approaches which embrace SBM, but assume a non-lexicalist approach to cyclicity without a strict separation of word-internal and phrasal grammar, and notions of the phonological cycle which also extends to phrases. Thus Rolle (2018) suggests that SBM applies in a model where cycles are defined over asymmetric c-command relations in a Distributed-Morphology architecture. Sande et al. (2020) propose that a version of the SBM holds in a Cophonology model where cycles are defined by Minimalist phases (‘Cophonologies by Phase’). These variants of the SBM will be discussed in section 5.1. Finally, we will argue that the GNA approach, while inherently incapable to capture SBM, still imposes substantial restrictions on mutation phenomena due to general restrictions on possible affixation patterns and phonological representations. Thus in section 5.3 we will show that the generalization in (6) holds true for our language sample:

- (6) *Phonological Locality Generalization on Anticyclic Mutation:*
Anticyclic mutation is phonologically adjacent to the position of its segmental anchor (modulo general phonological alternations).

The paper is structured as follows: In sections 2 and 3, we provide extensive empirical evidence that anticyclic mutation is attested in natural languages. Section 2 focusses on different basic patterns of anticyclic mutation, and section 3 extends the argument to more complex cases where anticyclic patterns alternate among each other or with cyclic mutation. In section 4.1, we present our theoretical background assumptions for a containment-based GNA account of

²Note that the structures in (5) don’t have a cyclic counterpart simply because under a strictly lexical architecture there are no morphological operations in syntax. Word-constituting constructions cannot have an effect on other words because they are part of morphology, not of syntax.

cyclic mutation. Finally, in section 4.2, we show that this analysis directly transfers to anticyclic mutation. A critical discussion of construction approaches to mutation relating to the SBM is given in section 5 where we argue that the GNA approach is closer to the typological facts than recent non-lexicalist modifications of the cophology approach (Rolle 2018, Sande et al. 2020). We conclude in section 6.

2 Basic Types of Non-Base Mutation

Following Wolf (2007), we subsume under ‘mutation’ all types of non-concatenative morphology where morphological contrasts are expressed by modifying phonological features; i.e., segmental quality (7-a+b), length (7-c+d), tone (7-e), and stress (7-f). These are illustrated in (7) with standard cases of cyclic non-concatenative morphology. The examples in (8) show that exactly the same types of mutation also cooccur with segmental affixation.³

(7) *Types of cyclic mutation: non-concatenative morphology without concomitant segmental affixation*

- | | | | | | |
|---|---------|--------------|---|---------|--------------|
| a. <i>V Quality</i> (German) | bru:de | ‘brother’ | → | bry:de | ‘brothers’ |
| b. <i>C Quality</i> (Texistepec Popoluca) | dastah | ‘to dig’ | → | nastah | ‘I dig’ |
| c. <i>V Length</i> (Hausa) | gudù | ‘walk’ | → | gudù: | ‘walking’ |
| d. <i>C Length</i> (Shizuoka Japanese) | katai | ‘hard’ | → | kat:ai | ‘hard!’ |
| e. <i>Tone</i> (Ngbandi) | gwè | ‘swam’ (sg.) | → | gwé | ‘swam’ (pl.) |
| f. <i>Stress</i> (Upriver Halkomelem) | ts’etém | ‘crawl’ | → | ts’étəm | ‘crawling’ |

(a. Helbig & Buscha 2001:241, b. Reilly 2002:23), c. Schuh 1989:38, d. Davis & Ueda 2002:2, e. Nida 1949:63-64, f. Galloway 1993:56)

(8) *Types of cyclic mutation: mutation with segmental affixation*

- | | | | | | |
|---------------------------------|------------|--------------|---|-----------|----------------|
| a. <i>V Quality</i> (German) | hu:t | ‘hat’ | → | hy:t-ə | ‘hats’ |
| b. <i>C Quality</i> (Fula) | re:du | ‘belly’ | → | de:di | ‘bellies’ |
| c. <i>V Length</i> (Tarahumara) | to | ‘take’ | → | to:ru | ‘take’ (pass.) |
| d. <i>C Length</i> (Päri) | cam | ‘eat’ (tr.) | → | cam:u | ‘eat’ (intr.) |
| e. <i>Tone</i> (Kanuri) | tádà | ‘boy’ | → | tàdà-á | ‘boys’ |
| f. <i>Stress</i> (Modern Greek) | ˈanθrop-os | ‘man’-NOM.SG | → | anˈθrop-u | ‘man’-GEN.SG |

(a. Helbig & Buscha 2001:241, b. Laughlin 2015:422, c. Caballero 2008:154, d. Andersen 1988:91, e. Cyffer 1998, f. Revithiadou 1999:19)

Accordingly, we will show in this section that anticyclic mutation effects are attested for all modalities listed in (7) and (8). To do this, we will first outline basic properties of our language sample (section 2.1), and then go step-by-step through different anticyclic configurations exhibiting anticyclic mutation: Root-to-Affix Mutation (section 2.2), Affix-to-Affix Mutation (section 2.3), Affix-to-Word Mutation (section 2.4), and Root-to-Word Mutation (section 2.5).

2.1 The Sample

The data in this paper are from a sample of anticyclic mutation data that we have collected, containing 138 distinct mutation patterns from 70 languages across linguistic macroareas and language families. See appendix A and B on the structure of the sample and basic statistics, and appendix C for schematic classification of all languages in the sample for anticyclic mutation and documentation of the data not explicitly discussed in the main text. In this section, we focus on representative examples of anticyclic mutation to illustrate the major structural types of the phenomenon across different phonological modalities.

³Abbreviations and glosses follow the Leipzig Glossing Rules wherever possible. Additional abbreviations are listed in appendix D.

Note that our data are based on two types of sources. On the one hand, we draw on theoretical work, especially from the classical autosegmental literature. These sources typically provide detailed arguments for an analysis in terms of defective phonological structure (e.g., Clements 1990 on Kikuyu, Noske 1996 on vowel features in Turkana, and Remijsen & Ayoker 2020 on lengthening in Shilluk), which effectively anticipates our interpretation as anticyclic mutation. Adding to this body of work, our sample also contains many cases of anticyclic mutation we have identified in descriptive sources on specific languages. This part of our sample is more subject to the inherent limitations of typological research. Language sources are of very different detail and quality, and it is not possible to show in detail for every data set that the categorization of a given pattern (in our case as anticyclic mutation) is preferable to possible alternative interpretations. To mitigate these limitations we have added an extensive online appendix which provides further information on all languages in our sample, our specific categorization, and in many cases, additional data. The appendix entries also specify if the interpretation on the data in the sample is building on previous theoretical work by other scholars. For a few patterns in our sample, there is also a broader theoretical literature including alternative interpretations of the data (most prominently postaccenting in Russian). We could have omitted these from the sample, but have considered it more informative for the reader to include them and to explicitly indicate published alternative analyses in the text and in the single language entries of the appendix.

The most systematic analytic alternative to the data in our sample is of course allomorphy. Virtually all cases of mutation (whether cyclic or anticyclic) could in principle also be captured by positing different allomorphs for the targets of mutation. To minimize the possibility of an interpretation as allomorphy, we have only included patterns in our sample, where anticyclic according to our sources mutation applies to *all* eligible target morphemes of a given type. Moreover, in the big majority of our data, mutation applies to multiple different targets. Thus Greek postaccenting puts stress on all following inflectional suffixes, and Kipsigis Causative morphology lengthens all short prefix vowels preceding it (see section 2.2). This point is especially clear in the cases of anticyclic mutation targeting independent words (see sections 2.4 and 2.5), where the set of mutation targets is an open class of items including nouns and verbs.

2.2 Root-to-Affix Mutation

We start our survey with anticyclic Root-to-Affix mutation since this is basically the mirror image to the cyclic cases of mutation discussed in the literature. The pattern is schematically summarized in (9).

- (9) *Anticyclic Root-to-Affix mutation, abstract*



Consonant Quality: Several examples of anticyclic Root-to-Affix mutation for consonantal features are found in Korean where specific vowel- and sonorant-final roots trigger aspiration or glottalization of following obstruent-initial suffixes often leading to minimal pairs of roots differing according to their mutation behavior. (10) illustrates this with two suffixes, the declension marker /-ta/ and the coordinating suffix /-ko/, but, as Gleim & Lee (p.c.) confirm, the same effects are also found with all other obstruent-initial suffixes such as negative /-ci/, subordinative/gerundive /-ke/ and /-tolok/ ‘in order to’. Here and in the following we indicate target morphemes (or target words) of mutation by boxes and mutation triggers by underlining (hence by the pure underscore “_” in cases where the trigger morpheme doesn’t have segmental

content). sg and cg abbreviate the features [spread glottis] and [constricted glottis] respectively.⁴

(10) *Consonantal Root-to-Affix mutation in Korean (Gleim & Lee 2019)*

a. *Aspiration*

/na ^{sg} -ta/	→	na- t^ha	give.birth-DEC	/na ^{sg} -ko/	→	na- k^ho	give.birth-COOR
/co ^{sg} -ta/	→	co- t^ha	good-DEC	/co ^{sg} -ko/	→	co- k^ho	good-COOR
/al ^{sg} -ta/	→	al- t^ha	suffer-DEC	/al ^{sg} -ko/	→	al- k^ho	suffer-COOR

b. *Glottalization*

/na ^{cg} -ta/	→	na- tʔa	get.better-DEC	/na ^{cg} -ko/	→	na- kʔo	get.better-COOR
/i ^{cg} -ta/	→	i- tʔa	tie-DEC	i ^{cg} -ko/	→	i- kʔo	tie-COOR
/nam ^{cg} -ta/	→	nam- tʔa	remain-DEC	nam ^{cg} -ko	→	nam- kʔo	remain-COOR

c. *No Mutation*

/na-ta/	→	nata	occur-DEC	/na-ko/	→	nako	occur-COOR
/p ^h i-ta/	→	p ^h ita	blossom-DEC	/p ^h i-ko/	→	p ^h iko	blossom-COOR
/al-ko/	→	alko	know-DEC				

See appendix C for more consonantal Root-to-Affix mutation from Fula, Tauya and Acoma.

Vowel Quality: As far as we know, the earliest evidence in the literature for floating vocalic features on roots has been identified by Kenstowicz (1979) for Chukchee where roots containing only consonants may have a lowering or raising effect on affixes (e.g., the lowering root /-rw-/ ‘to split’ in [ya-rw-ə-len] PRF-split-Ø-PRF ‘he has split’ vs. the non-lowering root /-rw-/ ‘to dig/scratch’ in [ye-rɣ-ə-lin] PRF-dug/scratch-Ø-PRF ‘he has dug/scratched’, Wolf 2007:51-52, Krause 1976:13). Similar cases are found in the Kordofanian language Moro (see appendix C). However, vocalic Root-to-Affix mutation is by no means restricted to defective vowel-less bases. In Turkana (Noske 1996), most suffix vowels harmonize with the vowel of the lexical root in [ATR] (e.g., the Allative suffix in [a-dɔk-un] INF-climb-ALL ‘to climb down’ vs. [a-gol-un] INF-discover-ALL ‘to close in’, p. 67). As expected, the low [-ATR] vowel /a/ in roots triggers [-ATR] in most contexts (e.g., [a-rap-un] INF-close-ALL ‘to discover’, p. 85), whereas /a/ in suffixes typically shifts to [+ATR] [o] after [+ATR] stems since the language doesn’t have a low [+ATR] vowel (e.g., the singular suffix in [ɛ-tim-at] MSG-hair-SG ‘hair of head’ vs. [e-pipj-ot] MSG-firestick-SG ‘firestick’, p. 80). The same holds for nouns with the plural suffix /-(i)a/ after non-low vowels (11-a,b) and after most roots with final /a/ (11-c,d). However, for a number of noun roots with final /a/, the suffix appears in its [+ATR] form [-(j)o] (11-e,f).

(11) [+ATR] *Root-to-Suffix mutation in Turkana (Noske 1996:85,93,94)*

a. ɲi-riɟj- o	b. ɲa-pɛdi- a	c. ɲɪ-malɟ- a	d. ɲɪ-qaz- ia	e. ɲɪ-lap- jo	f. ɲa-xabat- jo
M.PL-N-PL	F.PL-N-PL	M.PL-N-PL	M.PL-N-PL	M.PL-N-PL	F.PL-N-PL
‘leopards’	‘piles’	‘arrows’	‘works’	‘moons’	‘cupboards’

Following Noske, we take this as evidence that the irregular /a/-vowel verbs are characterized by a final floating [+ATR] feature following the [-ATR] specification of the root vowel. Independent evidence for this interpretation comes from the fact that these roots while enforcing [+ATR] suffixes still trigger regular [-ATR] harmony on the high-vowel class prefix [ɲɪ,ɲi-] (cf. (11-e) vs. (11-a) with non-low [+ATR] root). This makes it unlikely that the pattern results from an underlying [+ATR] low vowel in these roots which would then neutralize to [a]. Anticyclic vocalic mutation is found for virtually all basic vowel features: ATR as in Turkana,

⁴The phonetic nature of Korean laryngeal contrasts is controversial, but orthogonal to their role in mutation. We simply adopt here the phonetic characterization assumed by Gleim and Lee.

Yoruba and Ik, backness (Ainu), [+high] and [+round] (Bari), and [+low] (Hungarian, Neve'i, and Koryak), documented in appendix C.

Tone: Root-to-Affix mutation for tone is found in Kinande where specific verb roots impose a H-tone on preceding prefixes such as the Infinitive prefix /ri/- and the tense markers /ndi/- and /a/-. (12) shows this effect for the verb root /tum/ ‘send’ (12-a’,b’,c’) in contrast to the minimally different root /hum/ ‘hit’ which doesn’t trigger tonal mutation on prefixes (12-a,b,c):

(12) *Root-to-Prefix mutation for tone in Kinande (Black 1995:9)*

- a. e-ri-hum-a ‘to hit’ IV-INF-hit-FV a’. e-rí-tum-a ‘to send’ IV-INF-send-FV
- b. tu-né-mu-ndi-hum-a ‘we will hit’ S1PL-TNS-TNS-hit-FV b’. tu-né-mu-ndí-tum-a ‘we will send’ S1PL-TNS-TNS-send-FV
- c. tú-li-a-hum-a-a ‘we hit’ S1PL-TNS-hit-PF-FV c’. tú-li-á-tum-a-a ‘we send’ S1PL-TNS-send-PF-FV

Whereas in Kinande, tonal mutation adds a fixed tone on affixes, there are also cases where Root-to-Affix mutation leads to register changes for single tones. Thus in the Tibeto-Burman 4-tone language Tenyidie, many suffixes such as the Present Continuous affix /-ba/ show register harmony with the tone of the preceding verb base. If the root has a higher-register tone (i.e., Mid or Extra-high), the suffix is also higher, i.e., has an Extra-high tone (13-a,b). On the other hand, if the root has a lower register tone (High or Low), the suffix tone is High (13-c,d). In addition there is a small irregular class of High tone (i.e., Low-register) roots such as (13-e) which impose the high register on the suffix.

(13) *Subtonal root-to-suffix mutation in Tenyidie (Meyase 2015:67)*

- a. pētá ‘drive’ pētá-bá ‘drive’-PRS.CONT
- b. rādī ‘change’ rādī-bá ‘change’-PRS.CONT
- c. rālī ‘rest’ rālī-bá ‘rest’-PRS.CONT
- d. pelè ‘believe’ pelè-bá ‘believe’-PRS.CONT
- e. kēlé ‘heat’ kēlé-bá ‘heat’-PRS.CONT
- | | | Register | |
|--------|------|----------|------|
| Melody | | low | high |
| | High | √ | √ |
| | Low | √ | √ |

The same effects as for the Present Continuous /-ba/ are reported by Meyase for the Imperative suffix /-hie/, the Imperative Negative suffix /-bie/, the Progressive suffix /-ʒə/, and the Perfective suffix /-ta/ (Meyase 2015:66+67).

As argued by Meyase, this can be captured by assuming that /kēlé/ (13-e) has a final floating high register feature which associates to the tonal root node of /-ba/.

Length: In the Southern Nilotic language Kipsigis, specific verbs trigger the lengthening of preceding agreement prefixes as shown by the contrast between /tʃam/ ‘like’ and /kat/ ‘to greet’ in (14).

(14) *Root-to-prefix lengthening in Kipsigis (Non-Past Imperfective forms, Kouneli 2022:6)*

	tʃám ‘to like’	kat ‘to greet’
1sg	á-tʃám-é	â:-kàt-í
2sg	í-tʃám-é	î:-kàt-í
3	Ø-tʃám-è	í:-kàt-ì
1pl	kí-tʃám-é	kî:-kàt-í
2pl	ó-tʃám-é	ô:-kàt-í
	AGR-V-Tns	

Kouneli argues that this is the effect of a floating mora associated with the verb which either associates to a preceding vowel (if available), or is otherwise (in the 3sg) filled by epenthetic [i]. Independent evidence for this analysis comes from the fact that underlying /i/ in Kipsigis behaves phonologically in a different way. For example, it coalesces with /a/ resulting in [ɛ:] (e.g., /kà-í-tʃám/ CURR.PST-2SG-like → [kɛ:-tʃám] ‘you liked’, Kouneli 2022:7). In contrast, the 1sg prefix /á-/ in (14) is simply lengthened without a change in vowel quality.

See appendix C for another case of vocalic Root-to-Affix lengthening in Cherokee and for examples of consonantal Root-to-Affix lengthening (gemination) in Finnish and Shoshoni.

Stress: Postaccentuation, i.e., Root-to-Affix mutation for stress has been diagnosed by many authors for Greek and Russian.⁵ Thus in Modern Greek (Revithiadou 1999), there seem to be roots with fixed underlying accents in different positions (15-a,b) and accentless roots with variable accent (15-c). In contrast, nouns like /uranos/ according to Revithiadou are postaccentuating and cause stress to be realized on the following suffix (15-d).

⁵Alderete (2001) argues that post-stem accent in Russian is not the effect of root-specific post-accenting, but the default pattern of stress in Russian. We include Russian here as a (potential) case of postaccenting since Alderete’s arguments seem to be far from decisive (see, e.g., Dubina 2012 for detailed criticism, and Molczanow et al. 2019 of a summary of the literature showing that the theoretical and experimental arguments for different claims on Russian remain indecisive). Note also that Alderete’s argument doesn’t extend to Greek postaccentuation where there is broad evidence against a post-stem default (see Apoussidou 2003).

(15) *Modern Greek stress patterns (Revithiadou 1999:93+94)*

	<i>Nominative Singular</i>	<i>Genitive Singular</i>	<i>Accusative Singular</i>	<i>Nominative Plural</i>	<i>Genitive Plural</i>	<i>Accusative Plural</i>	
a. <i>Fixed σ1 stress:</i>	klívan-os	klívan-u	klívan-o	klívan-i	klívan-on	klívan-us	‘kiln’
b. <i>Fixed σ2 stress:</i>	fantár-os	fantár-u	fantár-o	fantár-i	fantár-on	fantár-us	‘soldier’
c. <i>Alternation penult/σ1:</i>	ánθrop-os	anθróp-u	ánθrop-o	ánθrop-i	anθróp-on	anθróp-us	‘man’
d. <i>Postaccenting:</i>	uran- <u>ós</u>	uran- <u>ú</u>	uran- <u>ó</u>	uran- <u>í</u>	uran- <u>ón</u>	uran- <u>ús</u>	‘sky’

See appendix C for more cases of postaccentuation and preaccentuation in Mayo, Tagalog, Aklan, and Russian.

Summary of Anticyclic Root-to-Affix Mutation: All cases of Root-to-Affix mutation in our sample are summarized in (16). Asterisks (‘*’) point to languages documented in appendix C. As the table shows, there are multiple examples for all mutation modalities introduced in (7).

(16) *Sample of Root-to-Affix mutation patterns*

<i>Tone</i>			<i>Vowel Quality</i>			<i>Consonant Quality</i>
Tenyidie	Kunama*	Kpelle (3.1)	Turkana	Moro*	Bari*	Korean
Bangime (3.3)	Mao*	Bari*	Chukchee	Ainu*		Tauya*
Barasana*	Korean*	Awa*	Hungarian*	Ik*		Fula*
Anywa*	Bari*	Cherokee*	Neve’i*	German*		Acoma*
Chichewa*	Kɔnnɪ*	Kinande	Yoruba*	Koryak*		Chimila*

<i>Length</i>		<i>Stress</i>	
Kipsigis (V)	Cherokee* (V)	Greek	Tagalog*
Finnish* (C)		Aklan*	Russian*
Shoshoni* (C)		Mayo*	

2.3 Affix-to-Affix Mutation

Whereas some cases of Root-to-Affix mutation have been discussed before as counterevidence against the SBM hypothesis (see Wolf 2007 on Chukchee, Apoussidou 2003 on Greek stress, and Rolle 2018 on tonal cases), there is a second major type of anticyclic mutation which has to our knowledge been so far ignored in the literature, namely a pattern where a more inwards affix triggers mutation on a more outwards affix, instantiating the structure in (17), repeated from (4).

(17) *Anticyclic affix-affix mutation, abstract*



Vowel Quality: Affix-to-Affix mutation for [±ATR] is found in the Kuliak language Ik, where several suffixes shift all more peripheral suffix vowels to [+ATR] while not affecting preceding vowels. An example is the Realis suffix /-a/ triggering a change in the Past tense suffix /-nɔkɔ/ (18). As in Turkana, the effect is unidirectional to the right. In line with the otherwise general [ATR] harmony of the language, all preceding vowels remain [-ATR] in agreement with the low vowel of the Realis suffix.

(18) [+ATR] Affix-to-Affix mutation: *Ik Realis* /-a/ (Schrock 2014:90)

- | | | | | | |
|------------|---|-------------|--|----------------------------|---------------------|
| a. bɛr-a- | nok^o | ‘he built’ | | bɛr-ʊkɔ́-nɔk ^o | ‘he built with it’ |
| b. kɔ́ɗ-a- | nok^o | ‘he cried’ | | kɔ́ɗ-ʊkɔ́-nɔk ^o | ‘he cried from it’ |
| c. dób-a- | nok^o | ‘he caught’ | | dóɓ-ʊkɔ́-nɔk ^o | ‘he caught with it’ |

The [+ATR] mutation not only extends to both vowels of the Past suffix, but also to multiple more peripheral suffixes. This is shown for another outwards mutating suffix, Distributive /-a:k/ in (19) which turns the Intransitive Infinitive suffix /-ɔn/ (cf. [ʔɛɓ-ɔn] ‘to fly away’, p.320) and the Andative suffix /-ʊkɔt/ (cf. [bʊr-ɔn-ʊkɔt] ‘to fly away’, p.332) into [+ATR].

(19) [+ATR] Affix-to-Affix mutation: *Ik Distributive* /-a:k/ (Schrock 2014:90)

- | | | |
|--------------|--|--|
| a. ci-a:k- | ón-ʊkɔt | ‘to become full (of multiple people)’ |
| b. dɔ́k-a:k- | -ón-ʊkɔt | ‘to become wet (of multiple entities)’ |
| c. ɪlɪb-a:k- | ón-ʊkɔt | ‘to become green (of multiple entities)’ |

A more complex Affix-to-Affix mutation pattern is found in the Papuan language Nimboran (Anceaux 1965, Inkelas 1993). Several suffixes such as the derivational particle suffix /-N/ trigger fronting of back affix vowels with concomitant raising and unrounding of these vowels, as shown in (20). In (20-a) the Locative suffix /-bá/ ‘above’ and the final 1st-person agreement suffix /-ú/ surface in their underlying forms. In (20-b), /-N/ triggers fronting of /-ba/ to [be] and unrounding for /-u/. Example (20-c) shows that vowels other than /u/ are raised to [i] in this position (and may be followed by word-final consonants). In (20-d), only the last vowel undergoes mutation because it immediately follows /-N/. Finally, (20-e,f) show that neither front vowels in general nor the subject plural suffix /-i/ trigger fronting or raising/unrounding.

(20) Vocalic Affix-to-Affix mutation in Nimboran (Inkelas 1993:369,574,576,562,564)

- | | |
|-----------------------------------|--|
| a. ŋgua-bá-k-u | → ŋguábáku (p. 562) |
| bite-LOC-PST-1 | ‘I bit above’ |
| b. sá-i-N-bá-r-u | → sáim- béri (p. 576) |
| drive.away.PL-SPL-PART-LOC-FUT-1 | ‘we will drive away from above’ |
| c. sá-N-maN-d-ám | → sáim- endím (p. 575) |
| drive.away-PART-INCL.SDU-FUT-INCL | ‘you and I will drive away (from here)’ |
| d. sá-N-d-u | → sán- d̥i (p. 564) |
| drive.away.SG-PART-FUT-1 | ‘I will drive away from here’ |
| e. ŋgedóu-k-se-p-am | → ŋgedóukesepám (p. 563) |
| draw-SDU-LOC-R.PST-3.M | ‘they two (m.) drew recently from here to there’ |
| f. ŋgedói-i-rár-d-u | → ŋgedóiráru (p. 569) |
| draw.PL-SPL-LOC-FUT-1 | ‘We (pl.) will draw them from here to above’ |

Affix-to-Affix mutation for multiple vocalic features is also found in Koryak (see appendix C).

Tone: Anticyclic Affix-to-Affix mutation is also attested for tone. An example can be found in the suffix system of Margi in one of its possessive constructions (see the Chichewa and Bakweri data in appendix C for mirror image patterns with prefixes). The pronominal suffixes in point are used in different constructions, especially in predicative forms as in (21-a). In this context, the pronouns reveal an underlying tonal contrast: 1sg /-jú/ and 3sg /-ɔ́á/ have a consistent H-tone whereas the other suffixes show tone polarity (a H after a preceding L, and

a L after a preceding H), plausibly because they underlyingly lack a tonal specification. The /a/-possession construction is formed by attaching the suffix /-á/ to a noun and adding the same set of pronominal suffixes (21-b). The central difference is that the tone of these suffixes is now uniformly L-toned. While this might be expected for the polar set of agreement affixes (they show L after the suffix-H), it neutralizes the underlying specification of the H-tone affixes. Thus /-á/ triggers tone lowering on more peripheral affixes.

(21) *Affix-to-Affix-tone: Margi Predicative forms (Hoffmann 1963:74+78)*

a. Predicative forms		b. a-Possessive	
	H-final base	L-final base	
1sg	Màrjì-jù ‘I am Margi’	Hèjì-jù ‘I am Higi’	kór-á-jù ‘my head’
2sg	Màrjì-gù ‘you (sg) are Margi’	Hèjì-gù ‘you (sg) are Higi’	kór-á-gù ‘your (sg) head’
3sg	Màrjì-ɕá ‘he is Margi’	Hèjì-ɕá ‘he is Higi’	kór-á-ɕà ‘his head’
1du	Màrjì-mà ‘we (du) are Margi’	Hèjì-má ‘we (du) are Higi’	kór-á-mà ‘our (du) head’

An example of Affix-to-Affix mutation *without* a concomitant segmental affix is described for the Ghanaian Kwa language Gã in Paster (2000, 2003). As can be seen in (22), tense-aspect marking is structurally inside subject agreement. For example, the 1sg-prefix /mí-/ precedes the progressive prefix /n-/ in (22-a) and the 3sg-prefix /e-/ precedes the future prefix /baá-/ in (22-b).

(22) *Verb inflection in Gã (Paster 2000, 8+10)*

- a. mí-n-cha 1sg-PROG-dig ‘I’m digging’ b. e-baá-cha 3sg-FUT-dig ‘he will dig’

Gã has various constructions where morpho-syntactic features are only expressed by changing the tonal melody of the base. Interestingly, this includes some tense-aspect markers as can be seen in (23) where partial paradigms for three verbs inflected for different person and tense categories are given. The forms for Habitual, Perfective, and Simple Past only differ in the tone realized on the subject prefixes /mí-/ and /o-/. These have different tone specifications in the Habitual but are both H-toned in the Perfective, and L-toned in the Simple Past. These facts according to Paster follow straightforwardly from assuming that the Perfective is marked by realizing an additional H and the Simple Past by realizing an additional L. The different tone specifications for the subject prefixes in the habitual then simply show the underlying tone specification for these prefixes. Crucially now, if we recall that tense-aspect marking is morphologically more inwards than subject agreement (22), this is an anticyclic mutation: the tense inflection triggers a (tonal) mutation on a morphologically more outwards marking for agreement (we notate vowel nasalization here by a tilde below vowels to keep it from interfering with tone diacritics).

(23) *Gã: Affix-to-Affix tone overwriting (Paster 2003, 28-30)*

	<i>Habitual</i> ('sing') <i>Underlying H/L</i>	<i>Perfective</i> ('dig') <i>Grammatical H</i>	<i>Simple Past</i> ('cultivate') <i>Grammatical L</i>
1SG	mí- [↓] lá-a	mí-__-cha	mí-__-dú
2SG	ò-lá-a	ó-__-cha	ò-__-dú
3SG	è-lá-a	é-__-cha	ò-__-dú
1PL	wò-lá-a	wó-__-cha	wò-__-dú
2PL	ɲè-lá-a	ɲé-__-cha	ɲè-__-dú
3PL	àmè-lá-a	àmé-__-cha	àmè-__-dú
	SAGR-√-HAB	SAGR-PFV-√	SAGR-SPST-√

See appendix C for a similar though more restricted case in Gaahmg.

Consonant Quality: Anticyclic Affix-to-Affix mutation for consonantal quality is found in the Ethiopian-Semitic language Chaha. Chaha Malefactive and Benefactive object agreement markers are summarized in (24), subsegmented into a consonantal part marking case and a remainder marking person and number of the object. These object suffixes now crucially surface in one of two versions, the choice being dependent on the preceding subject marking. The so-called ‘heavy’ form surfaces after verbs with plural subject affixes, the 2.Sg.f, or the impersonal (Rose 2007, 38) and the ‘light’ form elsewhere. These forms are systematically related phonologically. Thus the Malefactive markers differ only on their feature values for [±voice] and [±cont]. The difference between the heavy and the light form of the object markers can easily be derived from assuming that the trigger suffix impose the features [-continuant-voiced] on an obstruent following them, partially ignoring intervening sonorants for which these features are non-distinctive (see Rose 2000 for evidence that [β] in Chaha is a sonorant). For example, the subject 3pl suffix /-o/ triggers hardening in the following Malefactive suffix /-β/ in [ji-rəxiβ-o-p-a] S3-find-S3MPL-MAL-O3F ‘they find (something) to her detriment’ (cf. [ji-rəxiβ-β-a] S3-find-MAL-O3F ‘he finds (something) to her detriment’ Rose 2007:40).

(24) *Consonantal suffix-to-suffix mutation: object marking in Chaha* (Rose 2007, 39)

	<i>Malefactive</i>		<i>Benefactive</i>	
	<i>Light</i>	<i>Heavy</i>	<i>Light</i>	<i>Heavy</i>
1.SG	-β-i	- <u>p</u> -i	-n-i	-n-i
1.PL	-β-ndə	- <u>p</u> -ndə	-n-ndə	-n-ndə
2.SG.F	-β-x ⁱ	-β- <u>k</u> ⁱ	-n-x ⁱ	-n- <u>k</u> ⁱ
2.PL.F	-β-xma	-β- <u>k</u> ma	-n-xma	-n- <u>k</u> ma
3.SG.F	-β-a	- <u>p</u> -a	-r-a	-r-a
3.PL.F	-β-əma	- <u>p</u> -əma	-r-əma	-r-əma

Consonantal Affix-to-Affix mutation also occurs for multiple features in Acoma (appendix C).

Length: In addition to the cases cited in section 2.2, where lexically specified verbs trigger lengthening, Kouneli (2022) also discusses a morphological derivation which changes intransitive verbs into causative verbs, and in turn triggers the same lengthening effect. In effect, a causative mora anticyclically triggers lengthening on more peripheral agreement affixes. Since Kouneli doesn’t provide minimal pairs, we give here examples from the closely related language Nandi, which exhibits exactly the same alternations:

(25) *Nandi Causative lengthening* (Creider & Creider 1989:93)

- a. kir-á-ŋɛ̀t áne: ‘I woke up’
 PST-1SG-wake I
 b. ki:-[a:]—ŋɛ̀t kipet ‘I woke Kibet up’
 PST-1SG-wake I

Stress: In parallel to postaccenting roots (cf. 2.2), Modern Greek according to Revithiadou (1999) also has a postaccentuating suffix. As can be seen in (26) the derivational suffix /-ik/ causes stress to be realized on a following suffix.

(26) *Modern Greek postaccenting suffixation (Revithiadou 1999:204)*

- a. ɣál-os ‘Frenchman’ ɣal-ik-ós ‘French’
 b. elvet-ós ‘Swiss’ elvet-ik-ós ‘Swiss’
 √/-NOM.SG √/-DER-NOM.SG

Summary of Anticyclic Affix-to-Affix Mutation: Table (27) summarizes our sample of anticyclic Affix-to-Affix mutation. Although rarer than anticyclic Root-to-Affix mutation, there are again examples for all phonological mutation modalities introduced in (7).

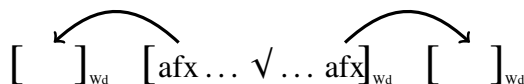
(27) *Sample of Affix-to-Affix mutation patterns*

Tone		Vowel Quality	Consonant Quality	Length	Stress
Margi	Gã	Ik	*Ket	Chaha	Tamil (C)
Bakweri*	Chichewa*	Nimboran	Acoma*	Kipsigis* (V)	Greek
Gaahmg*	Tetsót’iné*	Koryak*	Kashaya*		
Uspanteko*					

2.4 Affix-to-Word Mutation

The constitutive axiom of lexicalist approaches to morphophonology is that Phrase-level phonology strictly follows all cyclic word-internal phonology. Words are only concatenated after the application of all word-level morphology and phonology, and subsequently subject to a single joint cycle of phonology. A theorem of this architecture is that word-internal phonology cannot apply across words. Since standard Cophonology Theory adopts this strict separation and equates mutation with (construction-specific) word-internal phonology, it follows that mutation triggered by morphology inside a word *Word*₁ should never apply to any other word *Word*₂. We will call this pattern, schematically shown in (28) *Affix-to-Word mutation*. Again, we will provide crosslinguistic evidence that this type of mutation is actually attested across different phonological modalities.

(28) *Anticyclic Affix-to-Word mutation, abstract*



Paster & Kim (2011) describe different cases of this phenomenon in the Bantu language Tiriki. Several inflectional prefixes impose a High tone, not on their bases, but on the last syllable of a preceding word (29) (with concomitant insertion of a downstep triggered by the adjacent High tones). (29) illustrates this with two noun class prefixes: /à/- which marks verb-subject-agreement on verbs (29-a), and /và/- marking concord of modifiers with nouns (29-b). In both cases, the H-tone triggered by the prefix shows up on the preceding noun:

(29) *Affix-to-Word mutation in Tiriki (Paster & Kim 2011:80+85)*

- a. mú-línà ‘friend’ ^Hà-rhumúl-â: ‘(s)he is hitting’ → mú-lí^lná à-rhumúl-â: ‘a friend is hitting’
 CLS-friend CLS-hit-FV CLS-friend CLS-hit-FV
- b. vá-línà ‘friends’ ^Hvà-rhá:nô ‘five’ → vá-lí^lná vâ-rhá:nô ‘five friends’
 CLS-friend CLS-five CLS-friend CLS-five

A near mirror image of the Tiriki pattern is found with definiteness affixes in Supyire (30) that impose a Low tone on a following word. While the segmental part of definiteness marking consistently shows up as a suffix on the head noun of a DP, the L tone it triggers may appear on any right-adjacent word, a separate noun (30-a), a verb (30-b), a conjunction (30-c), or a preposition (30-d):

(30) *Affix-to-Word mutation in Supyire (Carlson 1994:59+60)*

- a. $\leadsto$ possessed noun: nà-ŋīⓁ ḡwᵇ-ᵇ-ní → nà-ŋī ḡwᵇ-ᵇ-ní
 man-DEF knife-3SG-DEF ‘the man’s knife’
- b. $\leadsto$ verb cī-réⓁ pá:n → cī-ré pà:n
 tree-DEF chop ‘chop a tree’
- c. $\leadsto$ conjunction cēè-ŋīⓁ ná ū pjà-ŋī → cēè-ŋī nà ū pjà-ŋī
 woman-DEF and her child-DEF ‘the woman and her child’
- d. $\leadsto$ postposition ná cēè-ŋīⓁ ī → ná cēè-ŋī ī
 with woman-DEF with ‘with the woman’

See appendix C for similar cases in Miya, Ciyao and Mao. That Affix-to-Word mutation is not restricted to tone is exemplified by recent work of Remijsen & Ayoker (2020) on Shilluk. Remijsen & Ayoker show that several nominal inflectional categories in the language are expressed by lengthening of the initial vowel in a right-adjacent word. The examples in (31) demonstrate this pattern for the plural. (31-a,b,c) show verb forms and a nominal modifier in their default form with an initial short vowel (the prefix /á-/ in (31-c) derives cardinal numbers from ordinal ones). Following plural nouns, these vowels become long (31-a’,b’,c’):

(31) *Shilluk Plural Affix-to-Word lengthening (Remijsen & Ayoker 2020:141-142,145, Remijsen & Ayoker 2019:7)*

- a. kùl á-lî:dà b. kùl ú-lî:dò c. á-dàk
 warthog IPFV-look warthog NEVP-look CRD-third
 ‘I looked at the warthog’ ‘Sb. apparently looked at the warthog’ ‘three’
- a’. tùl-__ ä-lî:dà b’. tòk-__ ù-lî:dò c’. tòŋ-__ á-dàk
 edge-PL IPFV-look edge-PL NEVP-look spear-PL CRD-third
 ‘I looked at the foreheads’ ‘Sb. apparently looked at the edges’ ‘three spears’

Remijsen & Ayoker argue that lengthening is morphologically a mora which functions as a plural suffix on the noun, but is phonologically realized on the first syllable of the following word. See appendix C for more Shilluk data and additional examples of Affix-to-Word gemination from Finnish and Afar. The appendix also describes Table (32) summarizes all the Affix-to-Word mutations in our sample.

(32) *Sample of Affix-to-Word mutation patterns*

Tone				Length	
Supyire	Tiriki	Kisi (3.2)	Miya*	Shilluk	Afar*
Kuria (3.2)	Ciyao*	Mao*		Finnish*	

2.5 Root-to-Word Mutation

In the lexicalist approach of Standard Cophonology theory, there is one further logical possibility that would violate the SBM, mutation triggered by a root in Word_i on a different Word_j, as shown in (33):

(33) *Anticyclic Root-to-Word mutation, abstract*



Intuitively, this is the least expected mutation effect under the SBM in a Cophonology approach. Roots should not trigger mutation (since they are phonological objects, not constructions), and the lexicalist architecture of Cophonology Theory predicts that phonological processes triggered inside of words should never apply to material of other words. Again there are well-documented counterexamples to this prediction.

Probably the most detailed and prominent case of Root-to-Word mutation in the literature is tone in the Bantu language Kikuyu, introduced by the classical work of Clements and Ford (see Gjersøe 2015 for recent instrumental confirmation of Clements and Ford’s fieldwork). In Kikuyu, specific nouns such as /βiriβiri/ ‘chillies’ (34-c) or /mwàýáhìná/ ‘weakling’ (34-d) trigger downstep on the first syllable of a following word, whereas other words of similar tonal shape do not. Thus /mòβàkè/ in (34-a) has only Low tones like /βiriβiri/, but doesn’t trigger downstep. Similarly, both /βàngírí/ (34-c) and /mwàýáhìná/ (34-d) end in a H-toned syllable but only the latter triggers downstep. The glossing here reflects the proposal of Clements (1984) already implicit in Clements & Ford (1981) that downstep is triggered by a floating L:

(34) *Kikuyu downstep in subject + verb (Clements & Ford 1981:321)*

- | | | | | |
|----|-------------------------|---|---------------|-----------------------------|
| a. | mòβàkè
tobacco.plant | né
cop | mòèýá
good | ‘the tobacco plant is good’ |
| b. | βàngírí
bangles | né
cop | ɲjèýá
good | ‘bangles are good’ |
| c. | βiriβiri
chillies | ↓né
cop | ɲjèýá
good | ‘chillies are good’ |
| d. | mwàýáhìná
weakling | ↓né
cop | mòèýá
good | ‘the weakling is good’ |

Similar lexeme-specific Root-to-Word mutation is also extensively documented for Otomanguean languages where it is described as “tonal perturbation” (see also Cahill 2011:18 for similar data in a Papuan language, Gadsup). McKendry (2013) gives a survey of the broad literature on the phenomenon in Mixtec. We illustrate it here with data from San Miguel el Grande Mixtec (Pike 1944, 1948, Mak 1953, 1958, McKendry 2013). The examples in (35) contrast two homophonous verbs where only the one in (35-b) causes an additional H-tone on the initial TBU of the following word. The data in (36-b) and (37)-b) show that tonal perturbation is also triggered

by specific determiners and nouns. The minimally different examples in (36-a) and (37)-a) illustrate again that this is a lexeme-specific effect, not a construction-specific or phonologically predictable process. Thus both /βēʔē/ ‘house’ and /nāʔā/ ‘hand’ are bisyllabic Mid-tone nouns preceding a possessor, but only the latter imposes a H-tone on the following noun.

(35) Verb^H Noun: ML → HL (Pike 1948:81)

a.	kēē ‘will go away’	kōò ‘snake’	kēē kōò ‘the snake will go away’	Verb _{MM} Noun _{ML}
b.	kēē ‘will eat’	kōò ‘snake’	<u>kēē</u> kóò ‘the snake will eat’	Verb _{MM(H)} Noun _{ML}

(36) Det^H Noun: LH → HH (Pike 1948:80)

a.	tàká ‘all’	sùcí ‘child’	tàká sùcí ‘all the children’	Det _{LH} Noun _{LH}
b.	máá ‘that’	sùcí ‘child’	<u>máá</u> súcí ‘that child’	Det _{HH(H)} Noun _{LH}

- (37) Noun^H Noun: ML → HL (McKendry 2013:134-136)
- | | | | | | |
|----|---------|----------|-------------------|--|------|
| a. | βēʔē | jājàn | βēʔē jājàn | Noun _{MM} Noun _{ML} | p.80 |
| | ‘house’ | ‘coyote’ | ‘house of coyote’ | | |
| b. | nāʔā | jājàn | nāʔā jājàn | Noun _{MM(H)} Noun _{ML} | p.80 |
| | ‘hand’ | ‘coyote’ | ‘hand of coyote’ | | |

What seems to be characteristic to both Kikuyu and Otomanguean is that mutation is largely insensitive to the morphosyntactic relation between trigger and target. Thus a downstep-triggering noun in Kikuyu will induce downstepping not only on copulas as in (34), but also on full verbs (38-a), conjunctions (38-b), other nouns (38-c) and prepositions (38-d).

(38) *Kikuyu downstep* (Clements & Ford 1981:315+321+322)

- a. *Noun + verb*
- | | | |
|----------------|----------------|---------------|
| <u>mwànékì</u> | <u>↓-nír-é</u> | ‘Mwaneki saw’ |
| Mwaneki | saw | |
- b. *Noun + conj (+noun)*
- | | | | |
|----------------|------------|-----------|-----------------------|
| <u>mwànékì</u> | <u>↓nà</u> | njòyóná-é | ‘Mwaneki and Njuguna’ |
| Mwaneki | and | Njuguna | |
- c. *object noun + object noun*
- | | | | |
|---------|-----------------------|---------------|-------------------------------|
| áhèiré | <u>mwàyáhìpá</u> | <u>↓njátá</u> | ‘he gave the weakling a star’ |
| he-gave | weakling _Ⓢ | star | |
- d. *object noun + preposition (+noun)*
- | | | | | | |
|------------------------|-----------------------|---------------|-------|-------|--|
| ndjóní [↑] ré | <u>mwàyáhìpá</u> | <u>↓ǒéínè</u> | wá | nómbà | ‘I didn’t see the weakling inside the house’ |
| I-didn’t-see | weakling _Ⓢ | inside | ASSOC | house | |

Note that the data in (38) would be problematic for cophology theory even if it were to abandon the assumption that postlexical phonology is non-cyclic. A central axiom of the framework is that domains of cophonologies are basically constructions corresponding to units of morphosyntax. But it is far from obvious why the occurrence of a specific noun would define a construction. In none of the cases in (38), the trigger is the head of its mutation target. Moreover in (38-c,d) the trigger and the target of downstepping don’t form an exclusive constituent. Thus even if we assume that the prepositional phrase in (38-d) is attached to the object NP, the minimal domain containing both [mwàyáhìpá] and [ǒéínè] would comprise the noun and the complete PP (hence: [mwàyáhìpá ↓ǒéínè wá nómbà], roughly [[weakling]_N [inside of [house]_{NP}]_{PP}]_{NP}. This would incorrectly predict that downstepping should apply to all words of this domain, not just to [ǒéínè].

This makes it highly unlikely that downstepping in Kikuyu could be modeled by a rule or a cophonology applied to a specific syntactic domain in parallel to word-internal mutation. On the other hand, in an autosegmental GNA approach, downstep-triggering nouns can be modeled in just the same way as affixes triggering downsteps on their bases: by floating tones. See appendix C on more Kikuyu data, and further cases of tonal Root-to-Word mutation in Lango, Kɔnni, and Awa, and section 3.1 on Kpelle.

Root-to-Word mutation also occurs for length. A well-known example is *Radoppiamento Fonosintattico* in Italian dialects, word-initial gemination which is partially triggered by phonological factors, but in many cases also by an arbitrary set of function words.⁶ Thus in Tuscan, the preposition /a/ triggers gemination on following nouns (39-d) whereas neither the preposi-

⁶Phonological Radoppiamento occurs predictably after word-final stressed vowels, whereas idiosyncratic Radoppiamento is triggered by items with final unstressed vowels and cannot be predicted by phonological factors (compare, e.g., /a/ and /la/ in the examples).

tion /in/ (39-b) nor the minimally different definite article /la/ (39-c) do. Note that initial /k/ undergoes intervocalic lenition in (39-c) as other singleton plosives do.

(39) *Root-to-Word length mutation: Italian Radoppiamento Fonosintattico (Amato 2019)*

- a. kasa → 'ka:sa 'house' c. la kasa → la'xa:sa 'the house'
 b. in kasa → iŋ 'ka:sa 'in the house' d. a kasa → a 'kʰa:sa 'at home'

See appendix C for additional cases of Root-to-Word lengthening in Finnish and Shilluk.

Extensive patterns of Root-to-Word consonant mutation for segmental features have been reported for Celtic languages. Thus in Welsh, the possessive pronoun /i/ triggers lenition (here: voicing) on a following word (40-b) such as the noun for 'dog', which is [ki] in isolation, whereas the numeral /tri/ triggers spirantization (40-c). Crucially, if both are combined every mutating item triggers mutation on the right-adjacent word as in (40-d).⁷

(40) *Root-to-Word consonant mutation: Welsh (Pyatt 2004:1)*

- a. ki 'dog' b. i gi 'his dog'
 c. tri xi 'three dogs' d. i dri xi 'his three dogs'

See appendix C for a case of Root-to-Word postaccentuation in Kashaya.

Just as affixes triggering mutation might lack segmental content or not, there seem to be syntactically independent triggers of mutation without a segmental component. Thus the Vocative in Shilluk is marked by a High tone at the right edge of noun phrases. That this is not an affix, but a syntactically independent element is evident from the fact that it surfaces as the last tone of whatever word is rightmost in the phrase, whether it is the head noun (41-a), an adjective (41-b), or a verb (41-c) (note that the HL falling contour in (41-b) is simplified to Mid before the morphological H, resulting in a MH contour).

(41) *Shilluk Vocative H (Remijsen & Ayoker 2019:16)*

	<i>Noun phrase without vocative</i>	<i>Noun phrase (vocative)</i>	<i>Translation</i>
<i>NP final Constituent</i>			
a. Noun	jùr	jǔr —	'Woman!'
b. Adjective	jùr à tē:k	jùr à tē:k —	'Strong woman!'
c. Verb	jùr à nèn:nò	jùr à nèn:nǒ —	'Woman who is watching!'

A similar pattern involving tone has been recently established with meticulous empirical scrutiny for an otherwise non-tonal language, Samoan, by Yu (2021). In Samoan, Absolutive case is encoded by a High tone on the syllable immediately preceding the marked DP, regardless of the morphosyntactic affiliation of the host word (e.g., the subject in (42-a) but the finite verb in (42-b)).

(42) *Samoan Absolutive H (Yu 2021:6)*

- a. na tala e le tamá [— le faitotoʔa].
 PST open ERG DET boy ABS DET door
 'The boy opened the door'

⁷Celtic consonant mutation is a highly complex and controversial area. Our interpretation of Celtic consonant mutation here follows the work of Wolf (2005, 2007) and especially Iosad (2012, 2014), who argues that different types of consonant mutation require substantially different analyses. Some cases are fully phonological, some instantiate cyclic mutation via featural affixes, and some are triggered by floating features on lexical items in phrasal phonology, as we assume here for the Welsh case at hand. See section 5.3 for more discussion.

- b. na talá [— le faitotoʔa] e le tama .
 PST open ABS DET door ERG DET boy
 ‘The boy opened the door’

See Yu (2021:19ff) for extensive data which show that the Absolutive-H is not tied to any specific trigger word, but appears before any determiner, (proper) noun, or pronoun which starts an absolutive DP.

A straightforward concatenativist interpretation of these cases is that the Shilluk Vocative and the Samoan Absolutive tones are morphosyntactically independent words which are phonologically defective, and hence associated to the preceding TBU by general phrase-level phonology. They might be called ‘clitics’, but only in the sense that their underlying form is defective by lacking segments. See Remijsen (2021) for a similar case involving length in Shilluk Associative plurals and section 3 for more tonal examples. Segmental examples might involve cases of syntactically conditioned consonant mutation in Celtic which lack an overt trigger (see, e.g., Borsley & Tallerman 1996).

Root-to-Word mutation is hence attested for all four major modalities, tone, length, segmental quality, and stress, as summarized in (43).

(43) *Sample of Root-to-Word mutation patterns*

Tone				Consonantal Quality	Length	Stress
Kikuyu	Mixtec	Mao*	Izon	Welsh	Shilluk (V)	Kashaya*
Samoan	Supyire (3.3)	Kpelle (3.1)	Guébie*		Italian (C)	
Seenku*	Gadsup*	Awa*	Igbo*		Finnish* (C)	
Aghem*	Lango*	Urarina*	Kɔnnɪ*		Shoshoni* (C)	
Makaa*						

What is the significance of this result for the SBM hypothesis? A possible reaction by proponents of the SBM might be to consider it a hypothesis which is strictly limited to lexical (word-level) morphophonology, but this would imply that root-to-Word mutation must be achieved by a substantially different formal mechanism from the one triggering morphological mutation. This provides effectively a parsimony argument for a concatenativist approach to mutation which would extend straightforwardly to phrasal phonology.

Alternatively, a proponent of SBM might assume an extension of the SBM to syntactic constructions, where cyclicity is generalized to phrasal phonology. Thus we might assume that functional projections play a similar role in syntax as morphological constructions including affixation in morphology. This would fit cases where functional elements such as prepositions and numerals trigger mutation on their complements (cf. the examples from Italian and Welsh above). However, there are also clear examples of root-to-Word-mutation which do not fall under this generalization. Thus the Italian *wh*-word /ke/ ‘what’ triggers *Radoppiamento Fonosintattico* (gemination) on a following verb as in [ke fɾaj] ‘what are you doing?’ (Passino 2013:315), /ke/ is a function word, but in no obvious sense a functional modifier of the verb. It is rather a complement of the verb which appears in a preverbal position only in virtue of its *wh*-feature.⁸ In fact, Kikuyu and Shilluk seem to provide near mirror-images to the Welsh and Italian cases where lexical (nominal) arguments trigger mutation on their head. In Kikuyu, as we have seen, a noun triggers downstep on a preposition and moreover on one modifying a different noun. In Shilluk, in addition to the lengthening pattern discussed in section 2.4 there is a second pattern of morphological lengthening, where specific nouns induce lengthening on a modifying numeral (see appendix C). Also root-to-Word mutation without segmental morphemes seems to exhibit this kind of symmetry: where the Shilluk *Vocative-H* is realized on the DP it modifies, the Samoan *Absolutive-H* surfaces on material preceding the *Absolutive DP*.

A further obvious possibility to introduce cyclicity into phrasal phonology would simply assume that phonological cycles apply generally bottom-up, where syntactic material which is embedded deeper is evaluated first. This amounts to a notion of cyclicity originally proposed in Chomsky & Halle (1968).

Counterevidence against this assumption is provided by Rolle (2018) based on data from Izon, where specific overwriting patterns are triggered not by affixes or constructions, but by arbitrary classes of lexical words, in particular, nouns and nominal modifiers (e.g., adjectives). In specific domains, only the leftmost element keeps its lexical tones, whereas it imposes an overwriting pattern on lexical material to its right. (44) shows this for the combination of attributive nominal modifiers and nouns, where the floating melody of the (leftmost) modifier replaces the underlyingly associated tones of the head nouns. Low, High, and Low-High are the only tonal overwriting melodies employed in the language.

(44) *Modifiers overwriting nouns (Rolle 2018:214)*

⁸Note that gemination is not a general feature of *wh*-words in Italian. Besides /ke/, also /dove/ (‘where’) and /come/ (‘how’) trigger it. However, /kwando/ (‘when’), /kosa/ (‘what’) and /ki/ (‘who’) fail to trigger gemination (Irene Amato, p.c.)

			N _{(L)H}		N _H		N _L	
			bùrù	‘yam’	námá	‘meat’	wàrí	‘house’
Mod _{(L)H}	èbì	‘good’	èbì	bùrú	èbì	námá	èbì	wàrí
Mod _H	èndì	‘that’	èndì	búru	èndì	námá	èndì	wárí
Mod _L	kálá	‘small’	kálá	bùrù	kálá	nàmà	kálá	wàrí

The nouns overwritten in (44) also have floating tones which in turn can overwrite when the nouns are leftmost in the appropriate phrasal context, for example if they function as objects preceding verbs:

(45) *Object noun overwriting verb (Rolle 2018:226)*

- a. N_{(L)(H)} bùrù ‘yam’ + fɛ ‘buy’ → bùrù fɛ ‘buy a yam’
 b. N_(H) námá ‘meat’ + fɛ ‘buy’ → námá fɛ ‘buy a meat’
 c. N_(L) òró ‘mat’ + fɛ ‘buy’ → òró fɛ ‘buy a mat’

Complex noun phrases show that in bigger domains, it is systematically the leftmost element which imposes its floating tones on the following words:

(46) *Overwriting by left-most modifier (Rolle 2018:217)*

- a. ìnè tàrà díbá búrú
 ìnè_{(L)(H)} tárá_(H) dibà_{(L)(H)} bùrù_{(L)(H)}
 my three big yam
 ‘my three big yams’
 b. béí mà: bùrù
 béí_(L) mà:_{(L)(H)} bùrù_{(L)(H)}
 this two yam
 ‘these two yams’

Crucially, in a constellation where a modified noun precedes a verb, the tone imposed by the modifier extends to the verb:

(47) *Non-scopal overwriting of noun+verb by adjective (Rolle 2018:226)*

			N _{(L)(H)} ‘yam’	V ‘plant’	
a.	Mod _{(L)(H)}	‘big’	<u>òpù</u> bùrù	gbóró	‘plant a big yam’
b.	Mod _(L)	‘small’	<u>kálá</u> bùrù	gbòrò	‘plant a small yam’

Under the standard assumption that nominal modifiers form a constituent with their noun ([Adj Noun] Verb]), the prediction of Strict Base Mutation is that neither the adjective nor the noun should be able to trigger overwriting on the verb, counter to fact.

The same point can again be made with Samoan which also shows that it is not generally the syntactically less embedded words which trigger mutation on other words. Recall that it is the absolutive marker of the object which triggers mutation on a preceding subject. However, in standard analyses of VSO languages, whether they employ movement of the verb or of a VP, the object is embedded deeper than verb and subject, i.e., the structure is roughly [Verb ... [Subject ... [Abs-Object]]] (see Collins 2017 for recent discussion on Samoan syntax).

In section 3, we will also see evidence for phonologically predictable alternations in the attachment of non-segmental clitics, providing further evidence against a version of the SBM generalized to syntactic constructions. The topic will be taken up more systematically in section 5 where we discuss the recent proposal by Rolle (2018) who proposes to subsume word-internal and external mutation by a general SBM requirement based on asymmetric c-command relations.

3 Alternations between Mutation Types

Under a concatenativist approach, the mutation types we have discussed so far are just descriptive categories. An affix tone attaches to another affix, not because there is any explicit,

construction-specific imperative to do so but simply as a result of concatenating floating affixal structure under adjacency to other affix material and independent phonological constraints (e.g., the requirement to associate to the left as in Gã). For this reason, we also expect cases where mutation crosscuts these categorizations and affects different targets (e.g., other words vs. other affixes) depending on the phonological context. In the following, we show that this prediction is borne out and virtually all conceivable types of such alternations are found across languages. Section 3.1 discusses alternations between different types of anticyclic mutation, section 3.2 cases where anticyclic mutation alternates with cyclic mutation. Finally, in section 3.3 we turn to a type of anticyclic mutation which is otherwise difficult to diagnose – mutation on the trigger morpheme itself – in alternation with cyclic mutation. All these alternations provide important evidence that mutation is not just allomorphy of single morphemes since it involves different target morphemes. They also demonstrate that anticyclic mutation cannot be conceptually separated from cyclic mutation. This constitutes a further variant of the empirical problem for the SBM hypothesis which assigns a special theoretical status to cyclic mutation.⁹

3.1 Alternations between different Types of Anticyclic Mutation

Root-to-Affix ~ Root-to-Word Mutation The simplest type of anticyclic mutation alternations is found with roots comprising floating features which attach to affixes or adjacent words on a first-come-first served basis. A straightforward example for this pattern is the Mande language Kpelle which has two types of low-toned nouns (48-a,b), one of which (48-b) triggers raising of low-toned suffixes to a Fall. The same effect is regularly observed after H-toned nouns (48-c) while all other nouns ending in L (or Fall, i.e., HL) (48-a,d) don't affect suffix tone. This distribution is naturally captured by assuming that nouns like /jàlà/ carry a final floating H as indicated in (48-b). Crucially, the same effect obtains if the singular (suffixless) noun is followed by a low-toned independent word such as the verb /kàà/ 'see'.

(48) *Noun classes in Kpelle (Konoshenko 2008, 24)*

		<i>Singular</i>	<i>Plural</i>	'I saw a ...'	
a.	L.L	gbònò	gbònò-ɣàà	ɛ́ gbònò kàà	'ring'
b.	L.L ^H	jàlà	jàlà- yáà	ɛ́ jàlà káà	'lion'
c.	H.H	wúlú	wúlú- yáà	ɛ́ wúlú káà	'tree'
d.	L.HL	yòwô	yòwô-ɣàà	ɛ́ yòwô kàà	'axe'

This is an expected state of affairs in a floating-feature approach since it simply requires that the phrasal phonology treats lexical floating H's in the same way as the lexical phonology. However, it remains surprising in an amorphous account where mutation is the effect of constructions.

Alternations between Different Word Targets: Purely featural syntactic formatives, as the Shilluk Vocative discussed in section 2.5 might also alternate in the directionality of their attachment, in parallel to featural affixes. Under the lexicalist approach we presuppose here, where all mutation in phrasal phonology is anticyclic (see section 1), this is also a case of alternation between different anticyclic targets. But even if this assumption were to be abandoned, these patterns would be particularly interesting: In syntactic constructions, it is often unclear or theory-dependent what relations would count as cyclic or anticyclic (see section 5 on different proposals). Here cases of alternation are revealing since at least one of the cases will be typically anticyclic, independently of the definition of phrase-level cyclicity.

⁹Note that our use of *alternation* here is a slight extension of its standard use in theoretical linguistics. In its textbook use, *alternation* designates the fact that an abstract category (like a morpheme) has different realizations in its specific form which are predictable from its context. Here we talk about different realizations with respect to the position of a morpheme.

Tonal examples of this pattern can be found in cases of tonal associative marking. Associative (possessive) constructions in African languages are often marked by partially or purely tonal means either on the possessor or the possessum, as documented in the comprehensive typological survey of Cahill (2000). Crucially, two of the languages discussed by Cahill show a positional alternation. In Igbo (Goldsmith 1976:180ff., Clark 1990:ch.7), the possessive linking element is a High tone which shows up on the second (possessor) noun if this is LH (49-a,b), but not if it is all-L (49-c,d) and on the first noun if it ends in a low-toned syllable (49-b,d). If this syllable is already High (49-a,c), there is no overt change on the first noun. In effect, the associative tone might be realized overtly on the first (49-d) or second noun (49-a), on both nouns (49-b), or on none of them (49-c) (see appendix C for similar data from Makaa).

(49) *Igbo Associative construction (Clark 1990:253-254)*

a.	ísí	+	òké	→	ísí —	[↓] óké	b.	óðù	+	òké	→	ó [↓] ðù —	[↓] óké
	tail		rat			‘tail of a rat’		head		rat			‘head of a rat’
c.	ísí	+	èṅwò	→	ísí èṅwò	d.	óðù	+	èṅwò	→	ó	[↓] ðé	— èṅwò
	head		monkey		‘head of a monkey’		tail		monkey				‘tail of a monkey’

A case of alternating different word-target mutation outside of the DP domain is identified in work by Sande (2018) and Trommer (to appear) for Guébie where tonal change alternates between verbs and subjects. A similar case is again found in Igbo (see appendix C).

3.2 Alternations between Cyclic and Anticyclic Mutation

Affix-to-Base ~ Affix-to-Affix Mutation: The formation of Intransitive verbs from Transitive ones in Tamil (Dravidian) is achieved by consonant gemination either on the verb root or a following verbal suffix (Schiffman 1999, Sundaesan & McFadden 2014). The forms in (50) contrast the bare stem and the Past tense form for Transitive–Intransitive pairs. Note that the final [-u] in the bare stems in (50-a,b) is an epenthetic segment regularly found in consonant-final words. The alternations in (50-a,b) involve gemination of the root-final consonant in the Intransitive form and the allomorph [in] as Past marker. Note that there are no voiced geminates in Tamil, gemination hence involves devoicing for some sounds, as in (50-a). In (50-b), since there is nor geminated [r] in Tamil, [t:] surfaces instead. In the forms in (50-c,d), a different allomorph for the Past – /-ndʒ/ – can be found and gemination now affects the tense suffix, not a stem consonant, resulting in [tʃ:]. As pointed out by Sundaesan & McFadden, in such forms, gemination always affects the first suffix directly adjacent to the stem (aspect or tense), not only the past tense suffix.

The Intransitive morpheme in Tamil hence triggers either cyclic mutation on the morphologically more inward stem or anticyclic mutation on a morphologically more outwards affix. Note that in contrast to the epenthetic [u], the underlying final stem vowels /æ/ and /i/ in (50-c,d) block gemination of the root consonant. If gemination is attributed to mora affixation, this plausibly follows from intervention of the vocalic mora.

(50) *Gemination in Tamil (Sundaesan & McFadden 2014, 2+3)*

<i>Transitive</i>			<i>Intransitive</i>		
<i>Bare stem</i>	<i>Past</i>		<i>Bare stem</i>	<i>Past</i>	
a. a:g-u	a:(g-i)n-	‘become’	a:k:—-u	a:k:—-in-	‘make’
b. u:r-u	u:r-in-	‘ooze’	u:t:—-u	ut:—-in-	‘pour’
c. oɖæ	oɖæ-ndʒ-	‘break’	oɖæ—	oɖæ —- [tʃ:]	‘break’
d. vedʒi	vedʒi-ndʒ-	‘burst’	vedʒi—	vedʒi—- [tʃ:]	‘burst’

See appendix C on a tonal example of Affix-to-Base mutation alternating with Affix-to-Affix Mutation from the Mayan language Uspanteko.

Affix-to-Base ~ Affix-to-Word Mutation: In Kisi (Childs 2011), proper nouns are marked by a suffixal High. If a proper noun with an underlying final Low such as /tàmɓà/ is in preverbal subject position, this H docks on the following verb, as shown by the contrast in (51).

(51) *Kisi: Raising on verb after proper noun (Childs 2011:94)*

- | | | | | |
|----|--------|-----------|------|----------------------------|
| a. | ò | kpè:ŋiá | liá | ‘She shook off the bees’ |
| | she | shake.off | bees | |
| b. | tàmɓà— | kpé:ŋiá | liá | ‘Tamba shook off the bees’ |
| | Tamba | shake.off | bees | |

Whereas the pattern in (51) instantiates anticyclic Affix-to-Word mutation, proper noun formation in other contexts results in cyclic Affix-to-Base mutation. In isolation, utterance-finally, and in other syntactic environments, the suffix-H is realized on the proper noun itself, as can be seen in (52-a,b), where both /tàmɓà/ and /sàà/ show a final H in non-subject position (the underlying tone of /sàà/ emerges in (52-c)).

(52) *Kisi: Raising of proper noun in other positions (Childs 2011:97)*

- | | | | | | |
|----|-------|-------|--------|------------|---------------------------------|
| a. | hàlí | cúkál | sàá | tàmɓá— | ‘Hali pricked Tamba for Saa’ |
| | Hali | prick | Saa | Tamba | |
| b. | hàlí | cúkál | tàmɓá— | sàá | ‘Hali pricked Saa for Tamba’ |
| | Hali | prick | Tamba | Saa | |
| c. | sàà | cúkál | hàlí | á jùèi | ‘Saa pricked Hali with a thorn’ |
| | Saa | prick | Hali | with thorn | |
| d. | jùmá | cò | jòmndé | kilió | ‘Njuma is cutting wood’ |
| | Njuma | AUX | wood | cut | |

The examples in (52) also show that lexically H-final proper nouns such as /hàlí/ and /jùmá/ don’t participate in the alternation. /hàlí/ does not appear with a final L in non-subject position (52-c), and /jùmá/ doesn’t trigger raising on the following auxiliary (52-d). Childs explains this fact by the natural assumption that the suffix-H is absorbed by the final H of the base before it can be associated to a following word. Absorption (to the right) also accounts for the fact that /sàà/ doesn’t host the floating H in (52-c).

In the examples so far, a non-segmental affix alternates between full realization on one morpheme (e.g., the root) and full realization on another morpheme. However, with morphological tone melodies comprising several tones, we expect that part of the alternation could also extend to more than one morpheme simultaneously. In fact, such a pattern seems to be attested in a notoriously challenging case of tonal morphology in the Bantu language Kuria. As can be seen in (53), Kuria expresses the remote future by a High tone on the third mora of the stem (which spreads to the penultimate mora by an independent phrase-level process).

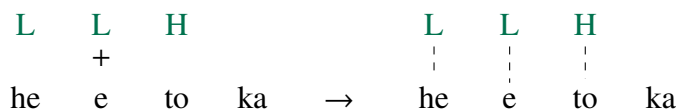
(53) *Kuria Remote Future ‘we will ...’ (Marlo et al. 2015:254)*

- | | | | | | |
|--------------|---------------------|------------|--------------|------------------------|------------|
| a. 3μ-Stems: | n-to-re-[—saamb-á] | ‘burn’ | c. 5μ-Stems: | n-to-re-[—koondókór-a] | ‘uncover’ |
| b. 4μ-Stems: | n-to-re-[—heetók-a] | ‘remember’ | d. 6μ-Stems: | n-to-re-[—hootótér-a] | ‘reassure’ |
| | FOC-1PL-√-FV | | | FOC-1PL-√-FV | |

A natural interpretation for this distribution is given by Trommer (2022): the remote future affix is a Low Low High melody which is mapped one-by-one from left-to-right to the stem moras,

as in (54).

(54) *Autosegmental analysis of Kuria tone morphology*



That the tonal formative is an affix, not some type of clitic is evidenced by the fact that it applies to the stem constituent (indicated by square brackets in (53)) excluding subject agreement and tense prefixes. If the stem is too short to accommodate three tones (55), one or two of the affix tones are lost if the tensed verb isn't followed by other morphosyntactic material.

(55) *Kuria Remote Future – short stems ‘we will ...’ (Marlo et al. 2015:254)*

- | | |
|--|---|
| <p>a. 2μ-Stems: n-to-re-[romǎ] ‘bite’</p> <p style="text-align: center; font-size: small;">FOC-1PL- $\sqrt$-FV</p> | <p>b. 1μ-Stems: n-to-re-[rj-a] ‘eat’</p> <p style="text-align: center; font-size: small;">FOC-1PL- $\sqrt$-FV</p> |
|--|---|

However, surprisingly, the LLH melody can be fully realized if the short verb is followed by additional material in the same clause (56).

(56) *Kuria Remote Future ‘we will ...’ (Marlo et al. 2015:254)*

- | | |
|---|---|
| <p>a. 2μ-Stem: n-to-re-[rom-a] éyétóóke ‘bite a banana’</p> <p style="text-align: center; font-size: small;">FOC-1PL- $\sqrt$-FV</p> | <p>b. 1μ-Stem: n-to-re-[rj-a] eyétóóke ‘eat a banana’</p> <p style="text-align: center; font-size: small;">FOC-1PL- $\sqrt$-FV banana</p> |
|---|---|

This set of data has been taken by Sande et al. (2020) as evidence that morphosyntactic domains must straddle the boundaries between morphology and syntax. However, as shown in a detailed reanalysis by Trommer (2022), in a concatenativist approach there is a simple more conservative approach based on an alternation between affix-base mutation and partial affix-base/affix word mutation. This is based on the assumption that in the lexical phonology, there is one-by-one left-to-right mapping (57-a) such that surplus tones remain until the phrasal phonology where they are then associated to following TBUs (57-b). Under this approach, Kuria is just minimally different from the Kisi case discussed above.

(57) *Reanalysis of Kuria phrase straddling*

a. *Word Level:*



→



b. *Phrase Level:*



→



to

3.3 Alternations Involving Tautomorphemic Mutation

There is one further type of non-base mutation. Floating material may overwrite the specification of tautomorphemic segmental material. Thus imagine an underlying root /^Hbà/ (i.e., a segmental morpheme associated to a Low tone with a preceding floating High tone) in a language where H tones overwrite right-adjacent L-syllables. Thus the hypothetical root would consistently surface as [bá] – with a High tone. The problematic aspect about this non-alternating type of tautomorphemic docking is that there is no obvious way to distinguish it from the result of unaltered underlying representations as in our case an underlying suffix already associated to a High tone (/bá/). However, given the possibility of alternating mutation, i.e., floating material

surfacing on different morphemes contingent on phonological context, we expect to find cases where tautomorphemic mutation can be diagnosed and in fact such cases are also attested.

Root-to-Affix ~ Tautomorphemic Root Mutation: The language isolate Bangime has productive plural inflection for nouns with a suffix showing tone polarity: It bears H if the immediately preceding base tone is L and bears L if the preceding tone is H, as in (58).

(58) *Bangime Plurals (regular polarity, Hantgan 2009:77)*

	<i>Singular</i>	<i>Plural</i>		<i>Singular</i>	<i>Plural</i>		
a.	b ^w ɛ	b ^w ɛ-ndɛ́	‘mosquito’	c.	ké	ké-ndè	‘thing’
b.	gíjɔ̃ ⁿ	gíjɔ̃ ⁿ -ndɛ́	‘shadow’	d.	dú:ɡú	dú:ɡú-ndè	‘forest’

In a special class of nouns, the plural also involves a change of the final root tone. Thus /kònó/ in (59-b) becomes [kòndè-ndɛ́], not *[kònó-ndè] (as in [kùwó] → [kùwó-ndè]).

(59) *Bangime Plurals (alternating polarity, Hantgan 2009:77)*

	<i>Singular</i>	<i>Plural</i>		<i>Singular</i>	<i>Plural</i>		
a.	<u>pí</u> nà	píná- <u>ndè</u>	‘fear’	b.	<u>kò</u> nó	kòndè- <u>ndé</u>	‘doorway’
c.	<u>dó</u> ò	dóó- <u>ndè</u>	‘paper’	d.	<u>bw</u> ó	bwò- <u>ndé</u>	‘field’

As shown by Hantgan, the most natural categorization of these nouns is as carrying a final floating tone. In the plural (60-a), this tone attaches to the tonally unspecified plural affix, but in the singular the floating tone can only be realized by incestuous association to the root itself (60-b).

(60) a. <i>Plural</i>		b. <i>Singular</i>
$\begin{array}{c} \text{L} \quad \textcircled{\text{H}} \\ \swarrow \quad \searrow \\ \text{kò} \quad \text{nò} \quad \text{ndɛ} \end{array} \rightarrow \begin{array}{c} \text{L} \quad \textcircled{\text{H}} \\ \swarrow \quad \searrow \\ \text{kò} \quad \text{nò} \quad \text{ndɛ} \end{array}$		$\begin{array}{c} \text{L} \quad \textcircled{\text{H}} \\ \swarrow \quad \searrow \\ \text{kò} \quad \text{nò} \end{array} \rightarrow \begin{array}{c} \text{L} \quad \text{H} \\ \swarrow \quad \searrow \\ \text{kò} \quad \text{nó} \end{array}$

See appendix C on more tonal examples where Root-to-Affix mutation alternates with tautomorphemic root mutation in Anywa, South Kyungsang Korean, Bari, and Chichewa. Also cases of post- and preaccentuation typically involve alternations with tautomorphemic realizations (see appendix C on Aklan, Mayo, Russian and Tagalog).

Affix-to-Root ~ Tautomorphemic Affix Mutation: Under the description of Uchihara & Cano (2019), the 3pl agreement prefix in Tlapanec has a Mid tone followed by floating Low tone, which typically shows up on the following verb stem. This can be seen from comparing the 3pl form with an almost homophonous 3sg form where the prefix also has a Mid tone but no floating Low tone (61-a,b). However, with H.M roots there is free variation between realizing the L on the prefix (61-c).

(61) *Tlapanec L alternation (Uchihara & Cano 2019:33,42)*

	3sg	3pl	
a. <i>Ø.H-Root</i>	nī-rāhnjú:ʔ	nī-rāhnjú:ʔ	‘feel spicy’
b. <i>H.Ø-Root</i>	nī-víjú	nī-víjú	‘whistle’
c. <i>H.M-Root</i>	nī-xpáthū	nī-xpàthū / nì-xpáthū	‘hit’

This is plausibly a cumulative effect of the marked status of LH sequences in the language (unattested in underived verb stems) and the preference to maintain stem H’s (which is achieved

in a simpler way in (61-a,b). See appendix C for a similar segmental case in German.

Root-to-Word ~ Tautomorphemic Root Mutation: Alternations with tautomorphemic root docking are also found for mutation across word boundaries. This is shown by mid-toned nouns in Supyire which have a floating final L overwriting the initial syllable of a following word in parallel to the floating L's of definiteness affixes discussed in section 2.4.

(62) *Supyire M_Ⓛ nouns: Root-to-Word mutation (Carlson 1994:57+58)*

- | | | | | | | | |
|----|-----------------------|-----------------------|-------------------|------|----------|---|---|
| a. | <i>Possessor noun</i> | sāɲɲcǝ̃n _Ⓛ | mēː | → | sāɲɲcǝ̃n | mèː | |
| | ↪ possessed noun | bird | song | | | | 'bird's song' |
| b. | <i>Object noun</i> | cēwē _Ⓛ | wíː | → | cēwē | wìː | |
| | ↪ verb | woman | look.at | | | | 'look at the woman' |
| c. | <i>Noun</i> | cēwē _Ⓛ | ná | nò | → | cēwē | nà nò |
| | ↪ conjunction | woman | and | man | | | 'woman and man' |
| d. | <i>Noun</i> | ná | pwūn _Ⓛ | ī | → | ná pwūn | ì |
| | ↪ postposition | with | dog | with | | | 'with a dog' |

In isolation, the Low tone is unpronounced, but before a suffix with preassociated tone, it shows up on the noun root itself, either as part of a Mid-Low contour if the final syllable is stressed (63-a,c) or as a register Low supplanting the Mid on the final syllable if this is unstressed (63-b).

(63) *Supyire M_Ⓛ nouns: tautomorphemic ~ root mutation (Carlson 1994:54)*

- | <i>Bare noun</i> | <i>+ def. suffix</i> | | <i>Bare noun</i> | <i>+ def. suffix</i> | |
|------------------|---|-----|------------------|----------------------|--|
| a. sɪˈkā | sɪˈkā | -ɲɪ | '(the) goat' | b. ˈkērēgē | kērēgè |
| c. ˈpwūn | pwūn | -ɲɪ | '(the) dog' | | |

A simple explanation for this pattern is that Supyire only allows floating tones at the right edge throughout the grammar. The final Low in a bare noun thus remains floating in lexical and postlexical phonology. However, the phonological systems diverge as to how they repair *non-final* floating tones. These are associated to the right in phrasal phonology but to the left in lexical phonology, resulting in tautomorphemic root docking. See appendix C for a similar non-tonal alternation involving nasalization in Seenku.

3.4 Summary

The anticyclic mutation data in our sample that involve alternation are summarized in (64).

(64) *Alternation data*

	<i>Tone</i>	<i>Length</i>	<i>Segmental</i>	<i>Stress</i>
<i>Root-Affix ~ Root-Word:</i>	Kpelle	Finnish* (C)		
	Awa*	Shoshoni* (C)		
<i>Affix-Base ~ Affix-Affix:</i>	Uspanteko*	Tamil (C, 2.3)	Koryak*	
<i>Affix-Base ~ Affix-Word:</i>	Kuria			
	Kisi			
	Ciyao*			
<i>Different Word targets:</i>	Guébie			
	Makaa*			
	Igbo(*)			
<i>Root-to-Affix ~ Tautomorphemic Root:</i>	Bangime			Aklan*
	Anywa*			Mayo*
	Chichewa*			Russian*
	Korean*			Tagalog*
	Bari*			
<i>Affix-to-Root ~ Tautomorphemic Affix:</i>	Tlapanec		German*	
			Koryak*	
<i>Affix-to-Affix ~ Tautomorphemic Affix:</i>			Nimboran(*)	
<i>Word-Word ~ Tautomorphemic Root:</i>	Supyire		Seenku*	

4 A Concatenative Analysis of Anticyclic Mutation

In this section, we show how anticyclic mutation can be straightforwardly captured in a concatenative approach to non-concatenative morphology. As background, we lay out in section 4.1 how the GNA approach that we adopt handles the better-known *cyclic* variant of mutation which will then be applied to anticyclic mutation in section 4.2.

4.1 A Concatenative Approach to Cyclic Mutation

Just as segmental affixation, many mutation patterns can be understood as addition of phonological material in parallel to segmental affixation. Thus the Eastern Sudanic language Gaahmg (Stirtz 2011) expresses subject agreement by suffixing (i.e., adding) tones at the right edge of the base tone, namely mid tone for non-third person, high tone for 3sg, and low tone for 3pl. This tonal addition is only obscured by general tonological alternations of the language, for example the lowering of M to L after another L (65-d). Tones affected by alternations are given in square boxes and brackets indicate phonetically vacuous repetition of tones, thus H(M)M is pronounced as HM.

(65) *Tone suffixation in Gaahmg (Incomplete) (Stirtz 2011:195)*

(Infinitive)		1sg + M	3sg + H	3pl + L	
a. fír	H	fír H-M	fír (H)-H	fír H-L	‘to smell’
b. bÉĪ	HM	bÉĪ H(M)-M	bÉĪ HM-H	bÉĪ H(L)-L	‘to name’
c. cōr	M	cōr (M)-M	cōr M-M	cōr M-L	‘to help’
d. dūr	L	dūr (L)-L	dūr L-M	dūr M-L	‘to bury’

Gaahmg person marking directly instantiates some of the classical arguments for the autosegmental representation of tone morphemes. The tone changes cannot be represented in a unified way by assuming tonal features such as [$\pm$ rising] or [$\pm$ falling], the resulting contours are simply a function of adding the tone for the person/number category to the one of the base. Similar

points can be made for other types of mutation. In Dinka, for example, 3sg is expressed by lengthening short vowels to long (e.g., [wèc] → [wè:c] ‘(s)he kicks’), and long vowels to super-long (e.g., [lè:r] → [lè::r] ‘(s)he rolls’, Andersen 1992:16,28). This can be captured in a unified way by a mora affix as in (66).

$$(66) \quad \text{a. } we_{\mu}c + \mu \rightarrow we_{\mu\mu}c \quad \text{b. } le_{\mu}r + \mu \rightarrow le_{\mu\mu}r$$

Additive mutation for segmental features can be found in Seenku where plural formation involves adding a suffixal [-back] at the right edge of the rightmost base vowel, as shown by McPherson (2017). This mutation does not trigger any change in front vowels that are already specified as [-back] (67-a,b) and results in diphthongal glide+full vowel sequences in the case of non-low back vowels (67-c,d).

(67) *Vocalic [-back] suffixation in Seenku (McPherson 2017:223+231)*

		<i>Sg</i>	<i>Pl</i>		<i>Sg</i>	<i>Pl</i>	
<i>Front V stems:</i>	a.	bi ²¹	bi ³	‘goat(s)’	b.	ke ¹ re ¹	ke ² re ² ‘day(s)’
<i>Back V stems:</i>	c.	fu ³	fwi ³	‘hare(s)’	d.	so ²¹	sw ^{e3} ‘horse(s)’
<i>Low V stems:</i>	e.	ka ²¹	ke ³	‘yam(s)’	f.	kjə ² ba ²¹	kjə ³ be ³ ‘orphan(s)’

In contrast, replacive mutation arises if featural affixation leads to phonotactically illicit feature combinations. Thus for low vowels, the Seenku [-back] affixation leads not to a glide+vowel sequence but to a [-back] monophthong as in (67-e+f). Similarly, the Irrealis prefix-H in San Miguel El Grande Mixtec (68) replaces the tone in the first syllable of the verb (McKendry 2013:45) since the language doesn’t allow for contour tones.

(68) *Irrealis prefix-H in San Miguel El Grande Mixtec (McKendry 2013:45)*

	<i>Irrealis</i>		<i>Imperfective</i>			<i>Irrealis</i>		<i>Imperfective</i>			
a.	MM	kānī ⁿ	HM	hānī ⁿ	‘hit’	c.	LM	kìkū	HM	kíkū	‘sew’
b.	ML	kāʔà ⁿ	HL	káʔà ⁿ	‘talk’	d.	LH	ʃíkó	HH	ʃíkó	‘sell’

Since the goal of this paper is to show that anticyclic mutation can be captured under conservative theoretical assumptions (pace recent work in Cophonology Theory, Sande et al. 2020), we adopt a substantially restricted implementation of Optimality Theory – Colored Containment Theory – in line with much recent work on nonconcatenative morphology (e.g., Zimmermann 2017, Paschen 2018). See Wolf (2005, 2007) for a similar concatenativist approach to mutation couched in Correspondence Theory.

Colored Containment Theory is a conservative extension of the original implementation of OT in Prince & Smolensky (1993) with a more limited set of possible structural changes than Correspondence Theory – basically just insertion and marking for non-pronunciation – and principled modularity restrictions on the phonology-morphology interface.¹⁰

In the following, we provide representative analyses for the examples of replacive tone mutation in Mixtec and additive tone mutation in Gaahmg that illustrate both the Colored-Containment formalism and its application to mutation. The crucial constraints, all in substance standard constraints from the OT-literature on tone (cf. Yip 2002), are given in (69). Formally, they make use of a central distinguishing feature of Containment Theory: Markedness constraints may either refer to material marked as phonetically realized in a given candidate (indicated by underlining in constraint names as in (69-b)) or to all material in an output representation, whether marked as phonetically realized or not (‘generalized’ markedness constraints as (69-a)).

¹⁰We adopt here the autosegmental version of Colored Containment Theory (Trommer 2011), which is slightly more restrictive than versions based on Goldrick’s 2001 Turbidity theory, as assumed in much early work in (Colored) Containment (see, e.g., Revithiadou 1999, Finley 2008, van Oostendorp 2007). All analyses developed here could also be implemented in a Turbidity-based version of Containment.

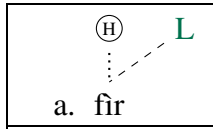
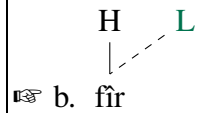
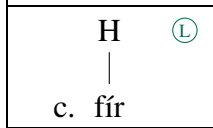
(69) *Constraints*

- a. $\tau \triangleright \sigma$: Assign * to every tone which is not associated to a syllable.
- b. *CONT(OUR): Assign * to every syllable associated phonetically with two different tones.
- c. MAX: | Assign * to every morphological association line which is not phonetic.

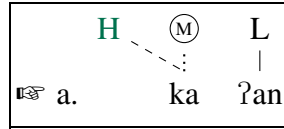
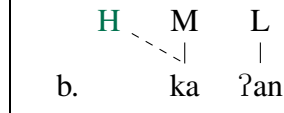
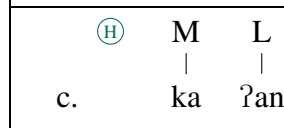
The tableaux in (70) show how different rankings of these constraints derive additive and replacive tone mutation in Gaahmg and Mixtec. In both cases, $\tau \triangleright \sigma$ guarantees association of a floating affix tone. Since in Gaahmg, MAX | is ranked above *CONTOUR, the affix tone is simply added to the underlying tone of the base, whereas higher ranking of *CONTOUR in Mixtec leads to phonetic deassociation of the underlying M. Recall that under Containment, association lines cannot be literally deleted but just marked as phonetically invisible. Those are indicated here by dotted lines ‘...’ whereas inserted association lines are dashed ‘- - -’ and elements that are not integrated into the structure by phonetically visible lines are circled in all following depictions to ease readability. Since $\tau \triangleright \sigma$ in contrast to *CONTOUR is a generalized markedness constraint, it is not violated by ‘deassociating’ the lexical M because the association line is still present (though not pronounced), turning the floating tone’s lack of association into a strategic advantage. The Containment approach thus directly derives the potential of floating features to ‘dispossess’ associated features, a fact which must be captured by specific constraints in Correspondence Theory (e.g., the constraints MAX-FLT and *FLOAT in Wolf (2007) and much more literature cited there).

(70) *Analyzing tone mutation*

i. *Additive mutation (Gaahmg)*

Input: = c.	$\tau \triangleright \sigma$	MAX	* <u>CONT</u>
a. 		*!	
b. 			*
c. 	*!		

ii. *Replacive mutation (Mixtec)*

Input: = c.	$\tau \triangleright \sigma$	* <u>CONT</u>	MAX
a. 			*
b. 		*!	
c. 	*!		

The tableaux in (70) also illustrate the second crucial ingredient of Colored Containment Theory, namely the representation of morpheme affiliation by color, a substantial restriction on the visibility of morphological structure in phonological computation. Thus the meaning or status as being a root or affix of the involved morphemes is invisible for phonology. The only information transferred from morphosyntax to phonology by colors is which phonological elements are part of the same morpheme or not (or – in the case of epenthetic material – are not part of any morpheme at all and hence colorless). Morphological color is thus broadly equivalent with the traditional notion of morpheme ‘boundaries’ but more adequate in an autosegmental context since it naturally allows to capture identical or different morpheme affiliation of elements on different tiers, which is difficult to represent with boundary symbols. Conceptually, colors provide a morpheme-level implementation of the Indirect Reference Hypothesis (Selkirk 1986, Nespor & Vogel 1986, Inkelas & Zoll 2005, Bermúdez-Otero 2012), which disallows direct reference to morphosyntactic information in phonological rules and constraints.

We will illustrate the use of color here with two other crucial aspects of our analysis of

mutation morphology. First, it has been often observed that floating features typically don't associate to segmental affix material that accompanies them. A simple example is tone mutation triggered by the indefinite plural suffix in Dzùùngoo (McPherson 2017, Solomiac 2014), shown in (70).

(71) *Dzùùngoo overwriting with segmental affix (McPherson 2017:222, Solomiac 2014)*

a. mòdzīn + ^Hì → [mòdzín]-ì/*mòdzìn-í 'people' b. vī + ^Hì → [vī]-ì/*vì-í 'dogs'

If, as argued by McPherson, raising is triggered by a floating High tone, it could in principle dock either to the suffix or the root, illustrated in (71). In both cases, it would overwrite a tone underlyingly associated to a TBU. This tie is naturally resolved by the constraint ALTERNATION developed by van Oostendorp (2007) to account for Derived Environment effects (see Trommer 2022 for discussion of this constraint in accounting for mutation), defined in (72).¹¹

(72) ALTERNATION: Assign * to every epenthetic association line which connects two nodes of the same color.

The tableau in (73) shows how ALTERNATION determines directionality of association by penalizing association of the floating H to the tautomorphemic suffix (73-c) which shares the same color. Since contour tones are illicit in Dzùùngoo (73-b), stem overwriting emerges (73-a).

(73) *Heteromorphemic overwriting in Dzùùngoo*

Input: d.	τ ▷ σ	ALTERNATION	*CONTOUR	MAX
a.				*
b.			*!	
c.		*!		*
d.	*!			

Crucially, the preference for heteromorphemic feature association is not special to mutation. ALTERNATION also explains the often made observation that autosegmental material stranded by deletion relinks to the 'segment which triggered it' (Clements & Ford 1979, p.207/fn.18). As shown by van Oostendorp (2007), ALTERNATION captures a substantial part of what has been traditionally called *Nonderived Environment Blocking*. An example is the fact that tone spreading is often restricted to apply across morpheme or word boundaries (Myers 1997).

Morphological colors also allow for a simple account of different types of across-the-board mutation. Thus Trommer (2024) shows that the standard cases of tonal overwriting for entire stems and syntactic phrases by a single tonal melody can be directly captured by tonal circumfixes, a phenomenon also robustly attested outside of overwriting (see, e.g., Pulleyblank 1986 on Tiv, Clark 1990 on Igbo, Hyman & Ngunga 1994 on Ciyao, and Trommer 2024 on Asante Twi). In tandem with a high-ranked CONTIGUITY constraint penalizing phonetically realized tones of

¹¹In contrast to the ALTERNATION account, Wolf (2005, 2007) captures the ban on tautomorphemic docking by a specific constraint. NoTAUTOMORPHEMICDOCKING much in the spirit of Colored Containment Theory; NoTAUTOMORPHEMICDOCKING refers to both morphemic affiliation and underlying representations. In contrast to ALTERNATION, it doesn't generalize to Derived-Environment effects outside of mutation.

a different color between tautomorphemic tones, tonal circumfixes are effectively enforced on embedded constituents. While circumfixation seems to be the right approach for standard cases of tonal overwriting, it doesn't extend to anticyclic overwriting since circumfixation is inherently cyclic. Moreover, there are also cases of cyclic overwriting in segmental phonology which have a directional character that cannot be captured naturally by circumfixes.

Thus in Terena, 1sg possession or agreement is expressed by nasalizing the first segment of the base word. Crucially, the morphological nasal feature spreads up to a blocking oral stop (74-a,b), or, in absence of the latter, up to the end of the word (74-c,d) although the language otherwise doesn't have nasal harmony (Bendor-Samuel 1960, Akinlabi 1996).

(74) *Terena 1sg nasalization*

- a. ituke 'his (pron.)' — ĩⁿduke 'my (pron.)' c. ajo 'his brother' — ājō 'my brother'
b. owoku 'his house' — ōwōⁿgu 'my house' d. emo?u 'his word' — ēmō?ū 'my word'

'Morphemic harmony' (Finley 2009) of this kind has been a major argument for positing morpheme-specific constraints as in the Terena analyses of Finley (2009) and Akinlabi (1996). In contrast, the only additional constraint type we need to account for it is one against proper spreading of underlyingly associated features, formulated for [nasal] in (75). As ALTERNATION, it refers to abstract color, not to particular morphemes or constructions.

- (75) *SPREAD [nasal]: Assign * to every [nasal] feature which is linked
to at least one segment by a colored (underlying association line)
and to at least one segment by a colorless (epenthetic) association line.

With *SPREAD [nasal] ranked above the harmony-triggering constraint (here: SHARE [nasal]: 'Assign * to every pair of adjacent segments not sharing a nasal feature', McCarthy 2011, Kimper 2011), nasal agreement is effectively restricted to mutation contexts as in (76-i) but excluded for underlyingly associated [nasal]-features (76-ii). $n \triangleright \bullet$ is defined absolutely parallel to $\tau \triangleright \sigma$ and demands that every [nasal] is associated to a segmental root node (= $\bullet$). DEP |, on the other hand, is the counterpart to MAX | penalizing inserted association lines between underlying elements.

(76) *Terena*

i. *Spreading under mutation*

ii. *No spreading for underlying nasal*

Input: = c.	$n \triangleright \bullet$ *SPR [n]	SHR [n]	DEP	Input: = c.	$n \triangleright \bullet$ *SPR [n]	SHR [n]	DEP
 a. ā j ō			***	 a. ũ n ã	*!		**
 b. ā j o		*!	*	 b. ũ n a	*!	*	*
 c. a j o	*!			 c. u n a		**	

After demonstrating how the GNA approach accounts for both additive and replacive cyclic mutation, the following section will show how it naturally predicts different types of anticyclic mutation and hence patterns that violate the SBM.

4.2 GNA-Analysis of Anticyclic Mutation

Crucially, anticyclic mutation in a GNA account follows from the same mechanisms as its cyclic counterpart. Tableau (77-a) illustrates this for the Kinande verb root /tum/ which imposes a H-tone on the preceding prefix (Root-to-Affix mutation) resulting in [e-**rí**-tum-a] ‘to send’ (cf. (12)) and (77-b) for the Margi form /kór-á-**ᵀ**jú/ → [kór-á-**jù**] ‘my head’ (Affix-to-Affix mutation) where the possession affix /-a/ triggers overwriting of the underlying H on 1sg /-ju/ by L. The rankings achieving this are only minimal variants of the ones outlined for cyclic mutation in section 4.1.

(77) *Anticyclic heteromorphemic overwriting*

i. *Kinande*

ii. *Margi*

<i>Input: d.</i>	$\tau \triangleright \sigma$	ALT	*CONT	MAX
				*
			*!	
		*!		*
	*!			

<i>Input: d.</i>	$\tau \triangleright \sigma$	ALT	*CONT	MAX
				*
			*!	
		*!		*
	*!			

Root-to-Word and Affix-to-Word mutation raise the question of how lexical and phrasal phonology interact. Recall from section 1 that we (as the Cophonology literature) adopt the conservative Lexical Phonology position that they form different subgrammars as in Stratal Optimality Theory. Hence the constraint rankings of lexical and phrasal phonology may differ, and the outputs of lexical phonology are fed into phrasal phonology (Kiparsky 2000, Bermúdez-Otero 2018, see also McCarthy & Prince 1993, Myers 1997 for the same assumption). Further, following Trommer (2011, 2024), Paschen (2021) we adopt the Colored-Containment equivalent of Bracket Erasure – Monochromization: At the transition from Word Level to Phrase Level all morphemes of a single word are assigned a common but unique new color. In effect, color at the Phrase Level does not distinguish roots and affixes, but (possibly complex) different words.

In (78-a), we illustrate this with Root-to-Word mutation in Mixtec. This takes place in the phrase-level phonology presupposing previous runs through the lexical phonology for the single involved words which has effectively preserved the floating features from their lexical entries. At the phrase level, ALT is ranked above $\tau \triangleright \sigma$ to ensure that floating features are not associated to the same word in forms where the word is utterance-final, as shown here for $nā?ā^H + jājàn$ ‘the coyote’s house’ and $nā?ā^H$ ‘house’ in isolation (78-b).

(78) *Mixtec floating tone*

i. *Heteromorphemic association*

ii. *Non-realization in 1-word utterance*

Input: d.	ALT	* <u>CONT</u>	$\tau \triangleright \sigma$	MAX
a.				*
b.		*!		
c.		*!		*
d.			*!	

Input: b.	ALT	* <u>CONT</u>	$\tau \triangleright \sigma$	MAX
a.		*!		
b.			*	

Whereas the direction of association in all examples so far has been fully determined by ALTER-NATION, it isn't sufficient for cases where a featural affix without segmental content is 'trapped' between two comparable hosts, as in Gã /mí- $\ominus$ -dú/ → [mì-dú] 'I cultivated' (section 2.3). We assume that ties like this are resolved by constraints on linear directionality of docking such as the one in (79) deriving left-association of the tense- $\ominus$ in (80).

- (79) *Assoc-R: Assign * for every pair of elements A and X that are associated in the output but not in the input if A precedes B and B is associated to X in the input.

- (80) *Gã*: Outwards mutation due to directionality

Input: d.	ALT	* <u>CONT</u>	*Assoc-R	$\tau \triangleright \sigma$	MAX
a.					*
b.		*!			
c.			*!		*
d.				*!	

In fact, floating material in Gã associates to the left more generally. Thus Paster (2003) shows that the floating negative H suffix always associates to the left (the root), not the right (the TAM suffix), e.g., negative perfective /é-kánè- $\ominus$ -kò/ → [é-kán¹é-kò] 'he has not counted' (Paster 2003:34)

Also anticyclic across-the-board mutation can be modelled in parallel to its cyclic counterpart. We illustrate this point with Nimboran. The constraints and the ranking here are almost completely analogous to the Terena evaluations in section 4.1. Floating features result in multiple association, not available to underlyingly associated features due to the definition of *SPREAD constraints.

Tableau (81-i) is a simplified evaluation (omitting consonants and root vowels) for across-the-board mutation examples like ((20)-c) /sá-N-maN-d-am/ → [sá-m-endím] where the floating [-b(ack)] affix fronts the penultimate and ultimate vowel. Association of [-b(ack)] to

a segmental root node is enforced by the constraint $-b \triangleright \bullet$ (excluding (81-i-d)), which in turn activates the necessity for multiple linking by virtue of $\text{SH(A)R(E)} [-b(\text{ack})]$ (excluding (81-i-c)). The only independent complication here is the additional raising of the ultimate vowel in Nimb-
 oran, which we capture as a phonological side effect of fronting. This fronting is triggered by the constraint ANCHOR_R (-back,+high) which requires the rightmost segment of a [+back] span to also carry the feature specification [+high], violated by (81-i-b).¹² See Smolensky & Legendre (2006) for similar conditions on high vowels in [ATR]-harmony. Tableau (81-ii) is the schematic evaluation of examples such as ((20)-e) [ɪŋgedókeseɓám] where underlying front vowels (i.e., preassociated [-back]) do *not* spread.

(81) *Nimboran apophony*

i. Multiple association of floating [-b(ack)]
 [-b(ack)]

ii. No spreading of underlyingly associated [-b(ack)]

Input: = d.	$-b \triangleright \bullet$	*SPR [-b]	ANCH _R (-b,+h)	SHR [-b]	DEP
a.					**
b.			*!		**
c.				*!	*
d.	*!				

Input: = c.	$-b \triangleright \bullet$	*SPR [-b]	ANCH _R (-b,+h)	SHR [-b]	DEP
a.		*!			*
b.		*!	*!		*
c.					*

Again this is completely analogous to the corresponding Terena evaluation in section 4.1. *SPREAD [-back] blocks further association of a feature which already has an underlying as-
 sociation line (cf. the explicit definition of the analogous constraint *SPREAD [+nasal] in (75)).

Under our approach, also alternating patterns directly follow from the same constraints as consistent cyclic and anticyclic mutation. In fact, we assume that in many cases the absence of alternations is just a collateral effect of morphological and syntactic combinatorics. Thus (non-alternating) Root-to-Affix mutation in Korean is due to the fact that verbal roots in the language are generally bound – they are always followed by at least one suffix, i.e., a potential host for its floating features. The same picture also holds, although for slightly different morphological reasons, for the Margi possessive suffix and the Gã negation tone and of course for the bulk of cyclic Affix-to-Root mutation since bound affixes by definition require a host. On the other hand, alternating cases like Kisi exhibit morphosyntactic contexts, where the floating elements lacks a host (the floating H of noun roots associates to the root itself in absence of an affixal host).

The affix tone associates to the right if possible but leftwards in absence of an adequate host, it is crucial that $\tau \triangleright \sigma$ outranks *Assoc-L (= *| L) so that association is rightwards if there is an appropriate segmental host (82-i), but otherwise to the left as a last resort (82-ii).

(82) (i) *Kisi anticyclic mutation*

(ii) *Kisi cyclic mutation*

¹²A span is understood here as feature with epenthetic association lines plus all the segments to which it is associated.

Input: d.	* <u>CONT</u> $\tau \triangleright \sigma$	* L	MAX
a.			*
b.	*!		
c.		*!	*
d.		*!	

Input: c.	* <u>CONT</u> $\tau \triangleright \sigma$	* L	MAX
a.		*	*
b.	*!		
c.		*!	

In Kisi as in several other cases prosodic boundaries seem to play a role in alternations (see, e.g., Tebay 2019 on the claim that in Nimboran vocalic mutation doesn't extend to root morphemes since these constitute independent feet). Thus the fact that the Proper Noun-H doesn't spread to a right-adjacent object is plausibly due to an intervening phonological phrase boundary. We assume that blocking of association in these cases is due to high-ranked CRISP-EDGE constraints (Ito & Mester 1999, Kaplan 2018) but won't go into any details here due to reasons of space.

Alternations between word-internal and -external mutation also follow under the assumption made here that they form consecutive OT-evaluations. We illustrate this with Supyire. Recall that in Supyire, roots with floating L-tones show additive heteromorphemic association if they are followed by a right-adjacent morpheme (an affix) word-internally (e.g., /pwūn_L-ŋi/ → [[pwūn_L]-ŋi] 'the dog', ((63)-c)), but overwriting rightwards association if they are followed by an independent word (/ná pwūn_L ī/ → [ná pwūn_L ī] 'with a dog (with)', ((62)-d)). Otherwise, if no other word follows the floating L is not overtly realized (/pwūn_L/ → [pwūn] 'dog', ((63)-c)). We capture this by different constraint rankings in the respective phonological grammars. Thus we assume that at the phrase level, ALT is undominated and hence above $\tau \triangleright \sigma$. This predicts rightwards association on a following word (83-i) and otherwise non-realization of the floating _L (83-ii).

(83) *Supyire phrase-level phonology*

(i) $M_{\textcircled{L}} M \rightarrow M L$

Input: c.	ALT	$\tau \triangleright \sigma$	MAX
a.			*
b.	*!		*
c.		*!	

(ii) $M_{\textcircled{L}} \rightarrow M$

Input: c.	ALT	$\tau \triangleright \sigma$	MAX
a.	*!		*
b.	*!		*
c.		*	

At the word level we face an apparent paradox: Word-final $\textcircled{L}$ obviously remains unassociated at this stratum, hence $\tau \triangleright \sigma$ must thus be ranked relatively low. This effect, however, cannot be due to ALT since homomorphic self-association is obligatory in suffixed forms. Instead, this follows naturally from the constraint $*_{\textcircled{\tau}} \tau$ ('Assign * to every floating tone followed by a non-floating tone'), introduced by Trommer (2022) to capture autosegmental left-to-right association (Goldsmith 1976, Yip 2002). In conspiracy with high-ranked MAX | and *Assoc-R, $*_{\textcircled{\tau}} \tau$ leads to additive self-association (84-i). Crucially, $*_{\textcircled{\tau}} \tau$ remains unviolated in the absence of a suffix, and hence $\textcircled{L}$ stays afloat word-finally (84-ii).

(84) *Supyire word-level phonology*

(i) $M_{\textcircled{L}} X \rightarrow M L X$

Input: c.	$*_{\textcircled{\tau}} \tau$	MAX	*Assoc-R	ALT	$\tau \triangleright \sigma$
a.		*	*!		
b.		*		*	
c.	*!				*

(ii) $M_{\textcircled{L}} \rightarrow M_{\textcircled{L}}$

Input: b.	$*_{\textcircled{\tau}} \tau$	MAX	*Assoc-R	ALT	$\tau \triangleright \sigma$
a.				*!	
b.					*

5 Discussion: Concatenative vs. Construction Approaches to Mutation

In the preceding sections, we have shown that SBM is extensively violated crosslinguistically and across all relevant phonological modalities. There are not only cases of outright anticyclic mutation (section 2), but also alternations between the anticyclic and cyclic variant (section 3), indicating that they are not formally different phenomena. Maybe most strikingly, some cases of mutation just apply according to simple phonological directionality, regardless of hierarchical structure. Thus in Gã, tonal mutation applies always to the left whether it targets more internal or more peripheral material (see appendix C on Anywa tone which exhibits basically the mirror generalization to the left and Acoma which shows the same phenomenon for seg-

mental features). Analytically, we have shown that a simple autosegmental approach based on non-segmental affixation derives all the observed patterns in our language sample without any additional assumptions and fully in parallel to cyclic mutation.

Our data also strengthen an argument against optimality-theoretic construction-grammar approaches raised by Zimmermann (2017) against the claim of Inkelas (1998) where mutation is the effect of phonological markedness constraints: Many cases of anticyclic mutation are phonologically non-optimizing. For example, tone mutation as in Supyire leads to marked word-internal contour tones and Acoma and Korean mutations create marked (aspirated, palatalized, glottalized) consonants. Similarly, Ik and Turkana vowel mutation creates round mid vowels, which are arguably more complex than the otherwise expected low unround vowels and these patterns also break up the otherwise general pattern of ATR-harmony in the two languages. In Construction Phonology, all these patterns are unexpected since mutation is supposed to be derived by exactly the same phonological constraints which also capture regular phonological alternations. On the other hand, in a concatenativist setting, markedness increase and decrease should both be possible as long as they correspond to a possible underlying representation.

To complete our argument against SBM and construction-grammar approaches to mutation, we will now discuss recent proposals trying to salvage the SBM either by assuming a non-lexicalist notion of the cycle (section 5.1) and/or by the assumption that SBM is not an absolute restriction on all possible mutation processes but only limits a well-defined subset of mutation phenomena (5.2). We will consider several closely related versions of these claims and argue that they are also untenable given the empirical evidence presented in this paper.

5.1 Alternative Interpretations of the Cycle

In our discussion above, we have presupposed three of the classical premises of Lexical Phonology. (i) Word-level morphology strictly precedes syntax and Phrase-Level phonology (ii) Word-internal morphophonology is internally cyclic. Word-internal cycles are defined by affixes successively added to roots. (iii) Phrase-level phonology is non-cyclic. These hypotheses are adopted in the classical work on Cophonology Theory leading to the problems for the SBM identified above. However, recent work in Cophonology has argued that non-lexicalist notions of phonological cycles may obviate these problems.¹³

5.1.1 Phase-based Cycles: Sande et al. (2020)

The most recent variant of construction phonology, Cophonologies by Phase (Sande et al. 2020), provides a radical departure from earlier versions which embraced SBM. Sande et al. explicitly take the Kuria data discussed above (section 3.2) as evidence for a model where inner material *can* have effects on outer material – as long as they are introduced in the same syntactic phase. At the same time, they take these very data to show that domains of cophonologies may be nonisomorphic to the ones predicted by classical lexical architectures (words and phrases). This follows in the account from the combination of phases with the second – logically independent – assumption that words are not formed in the lexicon but in the syntax. There are thus syntactic phases contained inside of words and phases that are limited simultaneously by word-internal and phrasal boundaries, especially a phase triggered by the category-defining head ‘v’ (‘little v’, see Embick 2021 and references cited there), or Voice, which might include a verb root with lower affixes and complements but exclude affixes corresponding to higher functional structure in the clause

¹³We do not discuss here the possibility to formulate Strict Base Mutation in a version of Stratal OT where cycles coincide with strata (i.e., where there are no stratum-internal cycles, Bermúdez-Otero 2018). To our knowledge this has never been proposed as a general approach to mutation morphophonology in the literature, and it is unclear to us how it would look like in detail. However, it seems that this approach would incorrectly predict that Affix-to-Word mutation (section 2.4) and Root-to-Word mutation (section 2.5) should be impossible since word-internal phonology in Stratal OT cannot have an effect across word boundaries.

This is a special case of Sande et al.’s general claim on morphophonology in (85), which for anticyclic mutation amounts to the corollary in (86). This is in effect the phase-based residue of the SBM-hypothesis endorsed by Sande et al..

- (85) *The Phase Containment Principle (Sande et al. 2020:19)*
Morphological operations conditioned internal to a phase cannot affect the phonology of phases that are not yet spelled out.
- (86) *Phase-based Locality Generalization on Anticyclic Mutation:*
Targets of anticyclic mutation must be part of the phase where the mutation trigger is spelled out.

The data in our survey provide evidence that anticyclic mutation is not generally bound by phases. We illustrate this with the two types of phases invoked by Sande et al., which can be most easily identified empirically: major lexical categories and DP’s.

Lexical Categories and Roots: In the grammatical architecture adopted by Sande et al., roots of major lexical categories (i.e., nouns, verbs, etc.) are assumed to be categoryless and receive a specific morphosyntactic category only by syntactic merge with a category-defining functional head (little *n*, little *v*, etc.). These category-defining heads according to Sande et al. trigger spellout (see their analysis of Hebrew nouns and verbs), hence induce phases including their roots before higher-level material is added. This directly implies that roots should not be able to trigger mutation effects outside of minimal stems. But, as we have seen in our survey (see section 2.2), roots have effects on a wide variety of affix types, which should be outside the category-defining heads. Examples include:

- Greek noun roots trigger stress on case suffixes
- Kipsigis verb roots trigger lengthening on agreement prefixes
- In Tenyidie, verb roots trigger subtonal changes on TAM suffixes
- Kɔnnɪ noun roots trigger tone mutation on nominal number suffixes (appendix C)
- Yoruba verb roots trigger [+ATR] on nominalizing prefixes (appendix C)

Noun Phrases (DP’s): With much of the minimalist syntactic literature, Sande et al. posit that full nominal projections, DP’s, are phases and should hence disallow DP-internal material to trigger mutation on outside material. Virtually all cases of Root-to-Word mutation in section 2.5 which are triggered by nouns or determiners are counterexamples to this prediction, e.g.:

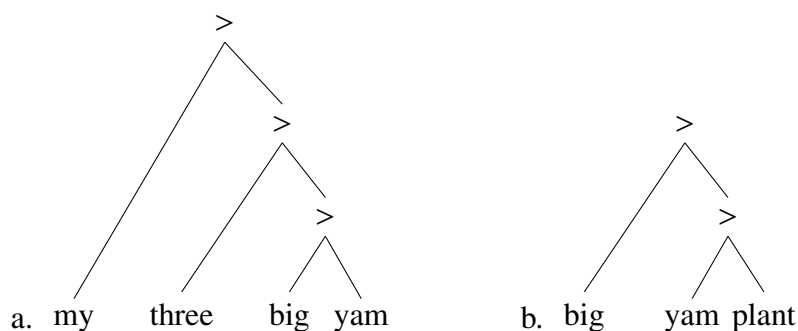
- In Supyire, determiners which are final in their DP trigger tone changes on following nouns, verbs or conjunctions, hence material clearly outside their DP’s.
- The Italian case where a *wh*-pronoun, presumably also of the category *D* triggers gemination on a following verb, externally to its own, minimal, DP projection.
- In Kikuyu, the head noun of a DP triggers downstep on a following conjunction, preposition, or the head noun of a following independent DP, as discussed in detail above.
- In Kpelle, object nouns trigger alternations on following verbs.

Thus in many cases, Phases are apparently a too small domain for capturing anticyclic mutation. Complementarily, Trommer (2022) shows for one of Sande et al.’s major case studies of anticyclic mutation, verbal tone in Kuria, that the Cophonologies by Phase approach predicts a too wide domain (a CP, hence an entire clause) for the application of tonal mutation. Taken together, the phase-based version of the SBM while avoiding specific pitfalls of the lexicalist version seems to create new empirical problems for the hypothesis.

5.1.2 Cyclicity via postsyntactic C-command: Rolle (2018)

Rolle (2018) adopts a Distributed Morphology approach where phonological cyclicity is defined not on syntactic representations, but on a postsyntactic spellout tree encoding only the asymmetric c-command relations between vocabulary items ($\approx$ the postsyntactically inserted morphemes). Thus the syntactic representation of Izon [ɪnè tàrà díbá búrí] ‘my three big yam’ discussed in section 2.5 (see example (46)) is converted to the simplified structure in (87-a), where ‘>’ indicates asymmetric c-command. Crucially, in the algorithm he proposes, vocabulary items corresponding to heads are higher than their complements, specifiers higher than the phrases to which they are adjoined. Also specifiers of specifiers are higher in the spellout tree than their complements (see Rolle 2018 on a detailed derivation by asymmetric c-command in the sense of Kayne 1994). Under this analysis, the Izon case discussed above [òpù bùrù gbóró] ‘plant a big yam’ (47) has no longer the problematic cyclic structure [[big yam] plant] for the purposes of phonology, where the adjective /òpù/ ‘big’ imposes tone anti-cyclically on the verb /gbóró/. Under the plausible assumption that the adjective is a specifier of the noun and the NP comprising both is in a specifier position above the verb, this structure results in the representation in (87-b), where the effect of ‘big’ on ‘plant’ is in accordance with Strict Base Mutation:

(87) *Asymmetric c-command tree for Izon (Rolle 2018:218)*



Rolle’s proposal elegantly solves the problem for SBM in the Izon case. Still we remain skeptical about postsyntactic spellout trees as a general approach to cyclicity since it implies that phrasal phonology is inherently cyclic. However, as decades of research in Lexical Phonology and Stratal OT have shown, there is only little *prima facie* evidence for cyclicity in Phrase Level phonology (see, e.g., Bermúdez-Otero 2018 for an overview).¹⁴ Virtually all cases of putative phrase level cyclicity reported in the literature are tonal or at least prosodic, most of them cases of tonal overwriting triggered by morphosyntactic categories or specific lexical items (see, e.g., McPherson & Heath 2016 on Dogon and Harry & Hyman 2014 on Kalabari). Rolle’s cyclic approach naturally predicts grammatical tone of this type. However, Trommer (2024) demonstrates that tonal overwriting can be captured fully in parallel in Colored Containment. Thus cyclic tonal overwriting per se doesn’t provide clear evidence for cyclicity in phrasal phonology. On the other hand, there are no reported cases of fully general *segmental* sandhi processes in phrasal phonology which exhibit signs of cyclicity. Here there is a clear typological gap which is unexpected under phrasal cyclicity.

Even in tonal phonology, the most plausible cases for cyclic phrasal sandhi processes reported in the literature don’t seem to fall in line with the c-command-based version of cyclicity proposed by Rolle. We will illustrate this with the best-established case for Phrase-Level cyclic-

¹⁴The major argument in SPE (Chomsky & Halle 1968) for phrase level cyclicity is the fact that phrasal stress reflects syntactic constituency. Prosodic Phonology (see Selkirk 2011 for an overview) derives these effects without cyclicity from mapping syntactic structure to units of the prosodic hierarchy. Note also that the issue of phrasal cyclicity is in principle independent from recursivity. Thus the Match Theory version of Prosodic Phonology advocated by Selkirk (2011) embraces fully recursive prosodic representations which allow for phonetic sensitivity to recursive embedding (see, e.g., Wagner 2005), but the phonological evaluation of constraints applies in parallel.

ity, Mandarin tone-3 sandhi, where the tone of a tone-3 word is changed to tone 2 if immediately followed by another tone-3 word (e.g., *hao3 jiu3* → *hao2 jiu3* ‘good wine’, all data from Chen 2021). This process apparently applies cyclically. However, structures with internally modified preverbal modifiers exhibit left-branching cycles, as expected by traditional constituent-based cyclicity. Thus in the example in (88), where the verb phrase should have essentially the same structure as in the Izon case in (87-b), sandhi applies first in the modifier phrase [*hen3 hao3*] (88-a), and only afterwards on the entire VP changing the tone of *hao3* before *xie3* (88-b) (*bi3* keeps its tone because at this point the following word has tone 2). In contrast, under the c-command-based algorithm, we would expect that the relevant structure for cyclic evaluation is the tree in (88-c), since *hen* is presumably the specifier of the specifier of *xie* in parallel to ((87)-b). This incorrectly predicts that [*hen3* > [*hao3* > *xie3*]] becomes *[*hen3* > [*hao2* > *xie3*]] which is not further modified by sandhi because the only tone-3 word to its right ([*xie3*]) is separated from it by an intervening tone-2 syllable:

(88) *Cyclic application of Tone-3 Sandhi (Kaisse 1985:173)*

	[[[zhei zhi] bi] _{NP} [[hen hao] xie] _{VP}]						
	this	CLAS	pen	very	good	writes	
Underlying tone:	4	1	3	3	3	3	
a. Cycle 1:	4	1	3	[2	3]	3	
b. Cycle 2:	4	1	3	[2	2	3]	(= output tone)

‘This pen writes very well’

The same problem with left branching for a c-command-based approach to cyclicity also shows up in compounds. In left-branching compounds in Izon, the left-most noun imposes its tone pattern on the remainder of the compound.

(89) *Non-scopal overwriting of NP by embedded noun (Rolle 2018:219)*

[[òsǐ _(L) ná má _(H)] wá rì _(L)] → òsǐ nà mǎ wá rí ‘a snail-meat house’			
snail	meat	house	(a house/place where there is snail-meat)

Rolle derives this by the assumption that the non-head modifier of a compound is syntactically its specifier. Hence *snail* in (90) is the specifier of *meat* and the specifier of the specifier of *house*, resulting in the c-command tree ([*snail* > [*meat* > *house*]]), ensuring overwriting by the tonal melody of snail. However, again, the cyclic structure of Mandarin tone-3 sandhi is markedly different, as shown by the example in (90), which follows standard cyclicity. Tone-3 sandhi is apparently first applied to the complex non-head constituent ([[*water3* *color3*]] and subsequently to the entire compound:¹⁵

¹⁵Note that the standard type of cyclicity apparent in (90) is also the one which is argued for by Duanmu (1999) for Shanghai in one of the most stringent arguments for systematic cyclicity in compounding found in the linguistic literature. The same version of cyclicity is also applied in SPE (Chomsky & Halle 1968) to account for compound (and phrasal) stress. Finally, this approach to cyclicity is also assumed in arguments that applications of the Rhythm Rule in English compounds and phrases are cyclic (Prince 1983, Kaisse 1985, Gussenhoven 1991).

(90) *Cyclic application of Tone-3 Sandhi in compounds (Kaisse 1985:170)*

	[[shui cai] bi] ‘water color pen’		
	water color pen		
Underlying tone:	3	3	3
a. Cycle 1:	[2	3]	3
b. Cycle 2:	[2	2	3] (= output tone)

Under cyclicity based on post-syntactic c-command trees, we would incorrectly expect the tone sequence 3-2-3 ([water3 > [color3 > pen3]] → *[water3 > [color2 > pen3]]).

In effect, the c-command-based trees extend the coverage of the SBM to the type of tonal overwriting found in Izon, but this seems to come at the cost of positing a new type of cyclicity not found elsewhere in grammar.

Empirically, Rolle’s claim for Strict Base Mutation is much more restricted in scope than the one by Inkelas (1998) since he limits it to cases of across-the-board mutation, which we will discuss below in section 5.2. For simple mutation there are cases in our sample where mutation clearly violates syntactic c-command based SBM. In Samoan, as discussed in section 2.5, the case marking of object nouns triggers a H-tone on preceding subjects, which are arguably higher in the syntactic structure than objects (see Collins 2017). The same holds for Tense marking in Guébie (appendix C). Another example is Aghem, where specific nouns trigger downstep on following modifiers such as /wó/ ‘hand’ in (91-b’). This contrasts with /fú/ ‘rat’ which lacks the downstepping effect, but is otherwise tonally identical (91-b):

(91) *Root-to-Word downstepping in Aghem (Hyman 2001:3 Hyman 1979)*

a.	kí-fú		‘rat’	a’.	kí-wó		‘hand’
	CLS-rat			CLS-hand			
b.	fú	kín	‘this rat’	b’.	<u>wó</u>	↓kín	‘this hand’
	rat	DEM		hand	DEM		

The same holds for Root-to-Word length mutation in Shilluk (see appendix C).

5.2 SBM as a Restriction on Subtractive Mutation (Alderete 2001, Rolle 2018)

Both Alderete (2001) and Rolle (2018) claim that phonologically additive mutation might be anticyclic (≈ triggered by floating features), but that subtractive and replacive mutation is strictly cyclic (≈ triggered by construction-specific phonologies).

The data in our survey provide clear counterexamples against this typological claim. Virtually all cases of anticyclic mutation involving segmental features are descriptively replacive (e.g., Turkana) and the Gã data in section 2.3 instantiate the same point for tone: Structurally internal prefix tones completely replace tone on more outwards prefixes. Since both the overwriting tone and the tone of the target prefix may be H or L, this cannot be the result of tonal underspecification on the overwritten syllable. Our data from alternations between different mutation targets also show that there is no strict boundary between replacive and non-replacive mutation, as shown for example by Supyire (section 3.3) where L mutation triggered by roots is replacive if affecting following words but is additive if the tone associates tautomorphemically to a stressed syllable. A cyclic counterpart are the Seenku data in section 1 where the same vocalic features lead either to diphthongization or to overwriting depending on the quality of the base vowel.

A methodological problem with any typological claim on featural subtraction is that putative examples of non-replacive deletion as in Japanese (Alderete 2001, Rolle 2018) or Vedic (Halle & Vergnaud 1987, Inkelas 1998, Rolle 2018) are plausible in the theoretical contexts where

they have been formulated, but not surface-evident from the data. ‘Accent deletion’ in Vedic results not in accent-less outputs but in forms with *initial* accent. While this arguably is the default position for accentuation in the language, there is no principled reason to exclude an analysis where it is added by ‘dominant’ affixes. Tonal ‘subtractive’ affixes in Japanese lead to the same pitch realization as in underlyingly tone-less bases. However, the Word-Level tone of single words is made opaque by phrasal tonology. The clearest case of truly subtractive prosody would be affixes which render content words unstressable or affixes which transform lexical bases that have a fixed tonal target into items that are tonally unspecified in the sense that their pitch is linearly interpolated from their phonetic context. Whereas there have been convincing arguments for surface underspecification of tone in the phonetic interpretation of underlyingly tone-less morphemes (see, e.g., McPherson 2011 and Myers 1998 on Chichewa), there is to the best of our knowledge no documented case where the same effect is triggered by morphological processes.

Rolle (2018) makes an innovative proposal to circumvent the methodological problem to identify subtractive mutation by using across-the-board mutation as a litmus test for genuine subtraction. He argues that tonal overwriting where just a single TBU is affected (like Gã in our sample) may be amenable (as in our analysis above) to a different mechanism (phonology and floating features) not subject to the SBM, whereas overwriting of multiple syllables or words must be due to subtraction implemented by cyclic cophonologies.

In a thorough typological survey of similar tonal overwriting patterns, Rolle finds only one potential case of anticyclic across-the-board overwriting, the Mande language Jalkunan. With respect to tone, our results confirm Rolle’s finding. All cases of anticyclic tonal overwriting in our sample affect single, not multiple tones. On the other hand, our sample contains clear counterevidence against Rolle’s generalization for segmental features. Thus Moro (section 2.2), Nimboran, Ik (section 2.3) and Koryak (appendix C) all exhibit anticyclic across-the-board mutation in vocalic mutation. In the cases of Nimboran and Koryak this also applies in languages which otherwise do not exhibit vowel harmony.

Whereas much more systematic work is needed on the crosslinguistic distribution of across-the-board Mutation, we think that Rolle’s generalization on *tonal* across-the-board mutation points to a significant typological bias. However, the fact that it doesn’t extend to other phonological features seems to indicate that the typological gap for anticyclic tone mutation is not due to general formal properties of the morphosyntax-phonology interface, but to differences in substance, i.e., to functionally grounded phonological preferences for tonal and segmental features. This would not be an unexpected result: As the typological literature shows (see e.g., Hyman 2008), tone and segmental features have typologically different preferences with respect to the directionality of spreading. Similarly there is an ongoing debate on long-distance effects in spreading which seems relatively frequent in tone, but rare (and possibly absent) for segmental harmony (see Jardine 2016, McCollum & Essegbey 2018 for controversial discussion). In fact, Trommer (2024) shows that Rolle’s generalization can be implemented as an emergent ‘soft’ constraint in a concatenative account. If, as argued by Trommer, most cases of cyclic across-the-board overwriting for tone fall out from tonal circumfixes which do not tolerate heteromorphemic tones between their prefixal and suffixal portion, this would account for the lack of an anticyclic counterpart. The latter would “require a kind of circumfixal (discontinuous) root morpheme, a ‘circum-root’ or, conversely, a tonal infix. Such structures, if existing at all, would be extremely infrequent. It is well known that even segmental infixes are much rarer than prefixes and suffixes (Greenberg 1963), and the status of discontinuous segmental roots is at best marginal. Given that floating material in roots, while clearly attested, is also considerably less frequent than in affixes (the statistical basis of the SBM), a tonal circum-root (a tonally specified root with a discontinuous floating tone) would be the coincidence of several phenomena which are already independently rare typologically. Thus it is likely that greedy inside-out overwriting

is simply accidentally unattested” (Trommer 2024:52+53).

5.3 Locality of Mutation in a Concatenative Approach

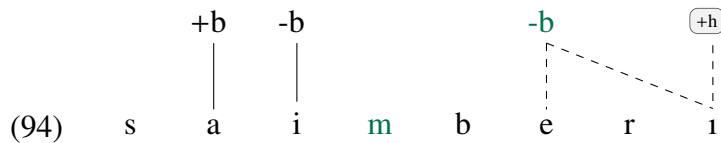
In the preceding subsections, we have argued that Strict Base Mutation under a cyclic construction grammar approach imposes typologically problematic locality predictions on possible mutation patterns. We conclude that mutation is not generally restricted by morphosyntactic structure. In contrast, the concatenative approach predicts a different kind of locality restriction because mutation is an effect of defective phonological material evaluated in the phonological grammar. We summarize this prediction in (92):¹⁶

- (92) *Phonological Locality Generalization on Anticyclic Mutation:*
Anticyclic mutation is phonologically adjacent to the position of its segmental anchor (modulo general phonological alternations).

By ‘segmental anchor’, we intend the segmental part of the morpheme inducing anticyclic mutation if there is one and otherwise the segmental position of the morpheme as it can be inferred from segmental affixes of the same morphological type. Thus the anchor location for the purely tonal tense prefix in Gã (section 2.3) would be the position between root and agreement morphemes where other segmental tense prefixes surface in the language. ‘Anticyclic mutation’ denotes the autosegmental melody (i.e., a string) of features on an autosegmental tier which is the output of the mutation process (in the case of foot structure: a foot node). Phonological adjacency is defined as in (93).

- (93) *Phonological Adjacency:*
A segmental morpheme position *M* and a mutation pattern *P* on autosegmental tier *T* are phonologically adjacent iff there is no feature *F* on *T* which would intervene between *M* and *P*

Let us briefly illustrate the application of (92) and (93) with a non-trivial example from Nimboran apophony (section 2.3) In examples such as ((20)-b) /sá-i-N-bá-r-u/ → [sáim-béri] drive.away.PL-PL.SUBJ-PART-LOC-FUT-1 ‘we will drive away from above’, the anticyclic mutation pattern is the feature [-back] on the vocalic [±back] tier (94). Although the vowel segments which undergo mutation are not string-adjacent to the triggering morpheme /-N/, the surfacing [-back] feature is not separated by any other [±back] feature from the affix vowels it associates to and thus adjacent to it in the sense of (93). The insertion of [+high] on the last vowel instantiates the phonological alternation clause in (92) since it is only indirectly triggered by the [-back] mutation pattern (see the analysis in section 4.2).



Similarly, in Kinande, no tone intervenes between roots like /tum/ and the High they trigger on preceding prefixes. And in Mayo, the accent of the accent-triggering roots is the only and thus closest accent to the root morpheme despite its dislocation to prefixes. The only case in our

¹⁶Note that we have formulated (92) cautiously as a generalization over *anticyclic* mutation. It also seems to be true for the overwhelming majority of cyclic mutation patterns, but, as pointed out by an anonymous reviewer, there are some potentially problematic cases of cyclic mutation involving ‘mobile’ palatalization and labialization in Ethiopian Semitic languages, especially in Chaha (Akinlabi 1996, Rose 1997, Wolf 2007, see also Zoll 1996 for a number of other cases which however are much less dramatic). Addressing these cases in detail is beyond the scope of this paper. Note however that long-distance labialization and palatalization in Ethiosemitic is not a feature specific to mutation, but also occurs in vowel-consonant harmony (see, e.g., Rose 2004 on palatalization in Harari triggered by overt high front vowels. It thus seems to be in line with the intuition encoded by (92) that mutation is the effect of concatenative affixation and independent phonological processes.

sample which might seem *prima facie* problematic for phonological locality is vowel mutation in Koryak where [+low] mutation skips intervening high vowels. However, transparent vowels are also found in many productive vowel harmony systems (Rose & Walker 2011), we thus take this pattern as actually confirming the validity of the proposed locality generalization. Apart from Koryak, and a handful of cases in our survey which lack evidence for a segmental anchor (and hence cannot be evaluated for the hypothesis), the Phonological Locality Generalization holds without exception for all anticyclic mutation patterns in our sample.

This generalization follows directly under a GNA account because ‘mutation’ is simply the spellout of defective phonological material for a morpheme *M* in the very position where *M* is concatenated. If this material is salvaged phonologically by epenthesis, it will show up in situ. Otherwise, if it is associated to underlying lexical material by the force of phonological constraints, it is subject to the standard locality conditions of (autosegmental) phonology, which typically involve adjacency.

We know of only two potential exceptions to the Phonological Adjacency generalization in anticyclic mutation. *First*, in some varieties of Mixtec (see section 2.5), H-tone mutation affects the second syllable of the following target word if its first syllable is a Mid tone. However, Tranel (1995) shows that this can be captured by assuming that the intervening Mid tones are default tones, hence underlyingly absent. This correctly predicts that for targets with initial L and H-tones it is the first syllable which shows the mutation tone. Moreover, even Mid-initial target words show phonological locality effects: if the second syllable of the target word starts with a glottal stop it blocks association of the mutation tone to the second syllable which instead shows up on the word-initial syllable.

Second, cases of anticyclic mutation under non-adjacency have been reported for Celtic consonant mutation (see Stump 1988, Green 2006). (95) shows a representative case from Breton based on the description of Stump (1988). The possessive pronoun /va/ triggers spirantization on the following noun (95-b), whereas the quantor /holl/ triggers voicing (95-c)). However if the noun is modified by both morphemes, spirantization triggered by /holl/ seems to apply across the intervening /va/ (95-d):

(95) *Breton consonant mutation under non-adjacency (Stump 1988)*

- | | | | |
|----|-------------|---|--|
| a. | tud | | ‘people’ |
| b. | <u>va</u> | zud | ‘my people’ |
| | my | people | |
| c. | <u>holl</u> | dud | ‘all people’ |
| | all | people | |
| d. | <u>va</u> | <u>holl</u> | zud (* dud) |
| | my | all | people |

Strikingly, like in (95), the strongest cases of apparent non-adjacency between overt words reported in the literature on Celtic instantiate relations between nouns (as target) and modifiers of nouns (as triggers). We hence follow the detailed arguments in Wolf (2005, 2007), Iosad (2012, 2014) and Laoide-Kemp (2024) that these patterns do not instantiate anticyclic mutation across words at all, but rather morphological marking of head constituents: nouns (or non-maximal projections of nouns) show agreement with modifiers by prefixes which cyclically trigger mutation on their host constituents. Thus the morphological formative triggering lenition in (95-b,d) would not be [va], but a prefix consisting solely of floating consonantal features, attached to [tud], and encoding agreement with the 1sg possessor. Since agreement is not a phonological, but a syntactic relation, this naturally explains that it can skip NP-internal material and that the non-adjacency effects are limited to these constructions.

Note finally that Cophonology approaches don't predict any form of phonological locality between segmental affixes and mutation. In pure Cophonology approaches, a segmental prefix or suffix is only connected to mutation (i.e., in cophonological terms: construction-specific phonology) by being part of a specific construction introducing both. But phonological computation in Cophonology Theory is completely independent from morphology, and in fact barred from accessing morphological boundaries or the affiliation of phonological material to specific morphemes (see Orgun & Inkelas (2002) for detailed elaboration). Thus we do not expect any systematic correlations between the position of affixes and the position of mutation effects they trigger. Overall, cophonologies predict tight morphosyntactic locality restrictions, and the lack of phonological locality restrictions. Our data suggest that the empirical situation is rather reversed.

6 Conclusions

In this paper, we have provided evidence that anticyclic mutation is a widespread and variegated phenomenon in the languages of the world: It is attested for all types of phonological modality and applies between roots and affixes as well as between affixes and more peripheral affixes, and roots or affixes across word boundaries. Moreover, anticyclic mutation alternates among different subtypes and with cyclic mutation which makes it unlikely that anticyclic mutation is a strictly morphological phenomenon substantially distinct from cyclic mutation. The sum of these data points constitutes a strong empirical argument against non-morphemic cyclic theories that are restricted by the SBM. On the theoretical side, we have shown that an analysis of mutation phenomena which combines autosegmental or prosodic affixation with purely phonological constraints avoids the limitations of strictly amorphous approaches and provides a unified account for the complex variety of its cyclic and anticyclic incarnations of mutation and their alternations. In a final step, we have demonstrated that the concatenativist approach also makes accurate typological predictions on the locality domains of anticyclic mutation, based on purely phonological restrictions which fit the data much better than existing variants of construction phonology, especially the recent Cophonologies-by-Phase approach. Where traditional cophonologies have too small domains to capture anticyclic mutation, phase-based cophonologies tend to have too big domains. On the other hand, we have argued that auto- and supra-segmental representations provide a direct way to solve apparent problems of a classical lexical – word-based – architecture since autosegmental material may naturally persist from the word level to the phrase level. Taken together, this accounts for the fact that regular phonological alternations are typically bounded to words or to phrasal contexts, whereas word-internally mutation might 'seep' into phrasal domains. The concatenativist account thus – we think, correctly – predicts a crucial distinction between phonological alternations and mutation which by definition cannot be captured in a cophonology approach equating mutation and productive phonology.

7 References

- Akinlabi, Akinbiyi (1996) Featural Affixation, *Journal of Linguistics* 32:239–289.
 Alderete, John (1999) Morphologically Governed Accent in Optimality Theory. PhD thesis, University of Massachusetts, Amherst.
 Alderete, John (2001) *Morphologically governed accent in Optimality Theory*. Routledge, New York.
 Amato, Irene (2019) Gorgia and Raddoppiamento Fonosintattico: when strength matters. Presentation at OCP 16.

- Anceaux, J. C. (1965) *The Nimboran Language: Phonology and Morphology*. Martinus Nijhof, 's Gravenhage.
- Andersen, Torben (1988) Consonant Alternation in the Verbal Morphology of Pări, *Afrika und Übersee* 71:63–113.
- Andersen, Torben (1992) Morphological Stratification in Dinka: On the Alternations of Voice Quality, Vowel Length and Tone in the Morphology of Transitive Verbal Roots in a Monosyllabic Language, *Studies in African Linguistics* 23:1–63.
- Anderson, Stephen R. (1992) *A-Morphous Morphology*. Cambridge University Press, Cambridge.
- Apoussidou, Diana (2003) The Deformity of Anti-Faithfulness. In: *Proceedings of the Stockholm Workshop on "Variation within Optimality Theory"*. 15–24.
- Bendor-Samuel, J. (1960) Some problems of segmentation in the phonological analysis of Terena, *Word* 16:348–355.
- Bermúdez-Otero, Ricardo (2012) The Architecture of Grammar and the Division of Labour in Exponence. In: J. Trommer, ed., *The Morphology and Phonology of Exponence*. Oxford University Press, Oxford, 8–83.
- Bermúdez-Otero, Ricardo (2018) Stratal phonology. In: S. Hannahs & A. R. K. Bosch, eds., *The Routledge Handbook of Phonological Theory*. Routledge, Abingdon, 100–134.
- Black, Cheryl A. (1995) Boundary Tones on Word-Internal Domains in Kinande, *Phonology* 12(1):1–38.
- Borsley, Robert D. & Maggie Tallerman (1996) Phrases and soft mutation in Welsh, *Journal of Celtic Linguistics* 5:1–49.
- Bye, Patrik & Peter Svenonius (2012) Exponence, Phonology, and Non-concatenative Morphology. In: J. Trommer, ed., *The Morphology and Phonology of Exponence*. Oxford University Press, Oxford, 427–495.
- Caballero, Gabriela (2008) Choguita Rarámuri (Tarahumara) phonology and morphology. PhD thesis, UC Berkeley.
- Cahill, Michael (2000) Tonal Associative Morphemes in Optimality Theory, *OSU Working Papers in Linguistics* 53:31–70.
- Cahill, Michael (2011) Tonal Diversity in Languages of Papua New Guinea. Technical Report Electronic Working Paper 2011-008, SIL.
- Carlson, Robert (1994) *A Grammar of Supyire*. Walter de Gruyter.
- Chen, Fulang (2021) On the Left-/Right-Branching Asymmetry in Mandarin Tone 3 Sandhi, *Proceedings of AMP 2020*.
- Childs, G. Tucker (2011) *A Grammar of Kisi*. Mouton de Gruyter, Berlin.
- Chomsky, Noam & Morris Halle (1968) *The Sound Pattern of English*. Harper & Row, New York.
- Clark, Mary M. (1990) *The tonal system of Igbo*. Foris, Dordrecht.
- Clements, G. N. & K. C. Ford (1981) On the phonological status of downstep in Kikuyu. In: D. L. Goyvaerts, ed., *Phonology in the 1980's*. Story-Scientia, Ghent, 309–357.
- Clements, George N. (1990) The role of the sonority cycle in core syllabification. In: J. Kingston & M. Beckman, eds., *Papers in Laboratory Phonology I: Between the Grammar and Physics of Speech*. University Press, Cambridge, Cambridge, 283–333.
- Clements, George Nick & Kevin C. Ford (1979) Kikuyu Tone Shift and its Synchronic Consequences, *Linguistic Inquiry* 10:179–210.
- Clements, G.N. (1984) Principles of Tone Assignment in Kikuyu. In: G. Clements & J. Goldsmith, eds., *Autosegmental Studies in Bantu Tone*. Foris Publications, Dordrecht, 281–340.
- Collins, James N. (2017) Samoan predicate initial word order and object positions, *NLLT* 35:1–59.
- Creider, Chet A. & Jane Tapsubei Creider (1989) *A Grammar of Nandi*. Helmut Buske Verlag,

- Hamburg.
- Cyffer, Norbert (1998) *A Sketch of Kanuri*. Köppe, Köln.
- Davis, Stuart & Isao Ueda (2002) The Typology of Mora Augmentation. Available under: <http://www.wata-net.com/proceedings/IsaoUeda/DaviesandUeda2.pdf>.
- Duanmu, San (1999) Alignment and the Cycle are Different. In: B. Hermans & M. van Oostendorp, eds., *The Derivational Residue in Phonological Optimality Theory*. Benjamins, Amsterdam, 129–152.
- Dubina, Andrei (2012) Towards a Tonal Analysis of Free Stress. PhD thesis, Radboud University Nijmegen.
- Embick, David (2021) The Motivation for Roots in Distributed Morphology, *Annual Review of Linguistics* 7(69-88).
- Finley, Sara (2008) Formal and cognitive restrictions on vowel harmony. PhD thesis, Johns Hopkins University.
- Finley, Sara (2009) Morphemic Harmony as Featural Correspondence, *Lingua* 119(3):478–501.
- Galloway, Brent (1993) *A Grammar of Upriver Halkomelem*. University of California Press, Berkeley.
- Gjersøe, Siri Moen (2015) Downstep and Phonological Phrases in Kikuyu. Master's thesis, Humboldt-Universität zu Berlin.
- Gleim, Daniel & Hyunjung Lee (2019) The Short Life Span of Laryngeal Sonorants in Korean. Talk at OCP 16 Verona, January 17 2019.
- Goldrick, Matthew (2001) Turbid output representations and the unity of opacity. In: *Proceedings of NELS 30*. GLSA, Amherst, MA, 231–245.
- Goldsmith, John A. (1976) Autosegmental Phonology. PhD thesis, MIT, Cambridge, MA.
- Green, Antony Dubach (2006) The Independence of Phonology and Morphology: The Celtic Mutations, *Lingua* 116(11):1946–1985.
- Gussenhoven, Carlos (1991) The English rhythm rule as an accent deletion rule, *Phonology* 8(1):1–35.
- Halle, Morris & Jean-Roger Vergnaud (1987) *An Essay On Stress*. MIT Press, Cambridge MA.
- Hantgan, Abbie (2009) Does Tone Polarity Exist? Evidence From Plural Formation among Bangime Nouns, *Indiana University Working Papers in Linguistics* 8:69–83.
- Harry, Otelemate & Larry M. Hyman (2014) Phrasal Construction Tonology The case of Kalabari, *Studies in Language* 38(4):649–689.
- Helbig, Gerhard & Joachim Buscha (2001) *Deutsche Grammatik – Ein Handbuch für den Ausländerunterricht*. Langenscheidt, Leipzig.
- Hoffmann, C. (1963) *A Grammar of the Margi Language*. Oxford University Press for International African Institute, London.
- Hyman, Larry M. (2001) Privative tone in Bantu. In: K. Shigeki, ed., *Proceedings of the Symposium “Crosslinguistic Studies of Tonal Phenomena: Tonogenesis, Japanese Accentology, and Other Topics”*. Tokyo University of Foreign Studies, Tokyo, 237–258.
- Hyman, Larry M. (2008) Directional Asymmetries in the morphology and phonology of words with special reference to Bantu, *Linguistics* 46(2):309–350.
- Hyman, Larry M. & Armindo Ngunga (1994) On the non-universality of tonal association conventions: evidence from Ciyao, *Phonology*. 11:25–68.
- Hyman, Larry. M, ed. (1979) *Aghem grammatical structure*. University of Southern California. SCOPIL 7, Los Angeles.
- Inkelas, Sharon (1993) Nimboran position class morphology, *NLLT* 11:559–624.
- Inkelas, Sharon (1998) The theoretical status of morphologically conditioned phonology: a case study from dominance, *Yearbook of Morphology* 1997 121–155.
- Inkelas, Sharon (2008) The Morphology-Phonology Connection. In: *Proceedings of the Berkeley Linguistics Society*. 145–162.

- Inkelas, Sharon (2014) *The Interplay of Morphology and Phonology*. Oxford University Press, Oxford.
- Inkelas, Sharon & Cheryl Zoll (2005) *Reduplication: Doubling in Morphology*. Cambridge University Press, Cambridge.
- Iosad, Pavel (2012) Representation and variation in substance-free phonology A case study in Celtic. PhD thesis, University of Tromsø.
- Iosad, Pavel (2014) The phonology and morphosyntax of Breton mutation, *Lingue e linguaggio* 13(1):23–42.
- Ito, Junko Itô & Armin Mester (1999) Realignment. In: R. K. Kager, H. van der Hulst & W. Zonneveld, eds., *The Prosody-Morphology Interface*. Cambridge University Press, Cambridge, 188–217.
- Jardine, Adam (2016) Computationally, tone is different, *Phonology* 33:247–283.
- Kaisse, Ellen M. (1985) *Connected speech: the interaction of syntax and phonology*. Academic Press., New York.
- Kaplan, Aaron (2018) Asymmetric Crisp Edge. In: R. Bennett, A. Angeles, B. Adrian, D. Buckley, N. Kalivoda, S. Kawahara, G. McGuire & J. Padgett, eds., *Hana-bana: A Festschrift for Junko Ito and Armin Mester*. .
- Kayne, Richard S. (1994) *The Antisymmetry of Syntax*. MIT Press, Cambridge MA.
- Kenstowicz, Michael (1979) Chukchee vowel harmony and epenthesis. In: P. R. Clyne, W. F. Hanks & C. L. Hofbauer, eds., *The Elements: A Parasession on Linguistic Units and Levels*. CLS, Chicago, 402–412.
- Kimper, Wendell (2011) Competing Triggers: Transparency and Opacity in Vowel Harmony. PhD thesis, University of Massachusetts, Amherst.
- Kiparsky, Paul (1982) Lexical phonology and morphology. In: *Linguistic Society of Korea: Linguistics in the morning calm*. Hanshin Publishing Company, Seoul, 3–92.
- Kiparsky, Paul (2000) Opacity and cyclicity, *The Linguistic Review* 17:351–67.
- Konoshenko, Maria (2008) Tonal systems in three dialects of the Kpelle language, *Mandenkan* 44:21–42.
- Kouneli, Maria (2022) Inflectional classes in Kipsigis, *Glossa: a journal of general linguistics* 7(1):1–33.
- Krause, Scott Russel (1976) Topics in Chukchee Phonology and Morphology. PhD thesis, University of Illinois at Urbana-Champaign.
- Laoide-Kemp, Anna (2024) Irish initial consonant mutation: disentangling phonology from morphosyntax. In: M. Jaurena & Z. Metzler, eds., *Proceedings of the Thirty-Fifth Western Conference on Linguistics*. Vol. 29, California State University, Fresno, 58–68.
- Laughlin, Fiona Mc (2015) Inflection in Pulaar. In: M. Baerman, ed., *The Oxford Handbook of Inflection*. Oxford University Press, Oxford, 419–445.
- Lieber, Rochelle (1992) *Deconstructing Morphology*. University of Chicago Press, Chicago.
- Mak, Cornelia (1953) A Comparison of Two Mixtec Tonemic Systems, *International Journal of American Linguistics* 19:85–100.
- Mak, Cornelia (1958) The tonal system of a third Mixtec dialect, *International Journal of American Linguistics* 24:61–70.
- Marlo, Michael R., Leonard Chacha Mwita & Mary Paster (2015) Problems in Kuria H tone assignment, *NLLT* 33:251–265.
- McCarthy, John & Alan Prince (1993) Prosodic Morphology. Constraint Interaction and Satisfaction. ROA 485-1201.
- McCarthy, John J. (2011) Autosegmental spreading in Optimality Theory. In: J. Goldsmith, E. Hume & L. Wetzels, eds., *Tones and Features (Clements memorial volume)*. Mouton de Gruyter, Berlin, 195–222.
- McCollum, Adam G. & James Essegbey (2018) Unbounded Harmony Is Not Always Myopic:

- Evidence from Tutrugbu. In: W. G. Bennett, ed., *Proceedings of the 35th West Coast Conference on Formal Linguistics*. Cascadia Proceedings Project, Somerville MA, 251–258.
- McKendry, Inga (2013) Tonal Association, Prominence and Prosodic Structure in South-eastern Nochixtlán Mixtec. PhD thesis, University of Edinburgh.
- McPherson, Laura (2011) Tonal Underspecification and Interpolation in Tommo So. Master's thesis, UCLA.
- McPherson, Laura (2017) Multiple feature affixation in Seenku plural formation, *Morphology* 27:217–252.
- McPherson, Laura & Jeffrey Heath (2016) Phrasal grammatical tone in the Dogon languages The role of constraint interaction, *NLLT* 34(2):593–639.
- Meyase, Savio (2015) The fifth tone of Tenyidie. In: L. Konnerth, S. Morey, P. Sarmah & A. Teo, eds., *North East Indian Linguistics*. Vol. 7, Australian National University: Asia-Pacific Linguistics Open Access, Canberra, 63–68.
- Molczanow, Janina, Ekaterina Iskra Olga Dragoy, , Richard Wiese & Ulrike Domahs (2019) Default stress assignment in Russian: evidence from acquired surface dyslexia, *Phonology* 36:61–90.
- Myers, Scott (1997) OCP effects in Optimality Theory, *NLLT* 15(4):847–892.
- Myers, Scott (1998) Surface underspecification of tone in Chichewa, *Phonology* 15:367–391.
- Nespor, Marina & Irene Vogel (1986) *Prosodic Phonology*. Foris Publications, Dordrecht.
- Nida, Eugene A. (1949) *Morphology: The Descriptive Analysis of Words*. University of Michigan Press, Ann Arbor.
- Noske, Manuela (1996) [ATR] harmony in Turkana, *Studies in African Linguistics* 25(1):61–99.
- Orgun, Cemil Orhan (1996) Sign-based Morphology and Phonology with Special Attention to Optimality Theory. PhD thesis, UC Berkeley.
- Orgun, Cemil Orhan & Sharon Inkelas (2002) Reconsidering bracket erasure, *Yearbook of morphology* 2001:115–46.
- Paschen, Ludger (2018) The Interaction of Reduplication and Segmental Mutation: A Phonological Account. PhD thesis, Leipzig University.
- Paschen, Ludger (2021) Trigger poverty and reduplicative identity in Lakota, *NLLT*.
- Passino, Diana (2013) A unified account of consonant gemination in external sandhi in Italian, *The Linguistic Review* 2013 30(2):313–346.
- Paster, Mary (2000) Issues in the Tonology of Gã. Columbus: Ohio State University, Undergraduate thesis.
- Paster, Mary (2003) Floating tones in Gã, *Studies in African Linguistics* 32(1):17–39.
- Paster, Mary & Yuni Kim (2011) Downstep in Tiriki, *Linguistic Discovery* 9(1):71–104.
- Pike, Kenneth L. (1944) Analysis of a Mixteco Text, *International Journal of American Linguistics* 10:113–138.
- Pike, Kenneth L. (1948) *Tone languages*. University of Michigan Press, Ann Arbor, MI.
- Prince, Alan (1983) Relating to the Grid, *Linguistic Inquiry* 4:19–100.
- Prince, Alan & Paul Smolensky (1993) 'Optimality Theory: Constraint Interaction in Generative Grammars'. Technical reports of the Rutgers University Center of Cognitive Science.
- Pulleyblank, Douglas (1986) *Tone in Lexical Phonology*. Reidel, Dordrecht.
- Pyatt, Elizabeth J. (2004) Relativized Mutation Domains in the Celtic Languages, *Proceedings from the Penn Linguistics Colloquium*. 26.
- Reilly, Ehren (2002) A Survey of Texistepec Popoluca Verbal Morphology. Master's thesis, Carleton College, Northfield, Minnesota.
- Remijsen, Bert (2021) Associative plural in Shilluk: in between a clitic and an inflection. Presentation at the 28th Manchester Phonology Meeting, Manchester, May 28.
- Remijsen, Bert & Otto Gwado Ayoker (2019) *Inflectional morphology and number marking in Shilluk nouns*. Language Documentation & Conservation, Honolulu, Hawai'i.

- Remijsen, Bert & Otto Gwado Ayoker (2020) Floating quantity in Shilluk, *Language* 96(3):135–156.
- Revithiadou, Anthi (1999) Headmost Accent Wins: Head Dominance and Ideal Prosodic Form in Lexical Accent Systems.. PhD thesis, Leiden University.
- Rolle, Nicholas (2018) Grammatical tone: Typology and theory. PhD thesis, UC Berkeley.
- Rose, Sharon (1997) Theoretical Issues in Comparative Ethio-Semitic Phonology and Morphology. PhD thesis, McGill University.
- Rose, Sharon (2000) Epenthesis positioning and syllable contact in Chaha, *Phonology* 17:397–425.
- Rose, Sharon (2004) Long distance Vowel-Consonant Agreement in Harari, *Journal of African Languages and Linguistics* 25(41-87).
- Rose, Sharon (2007) Chaha (Gurage) Morphology. In: A. S. Kaye, ed., *Morphologies of Asia and Africa*. Eisenbrauns, 399–424.
- Rose, Sharon & Rachel Walker (2011) Harmony Systems. In: J. Goldsmith, J. Riggle & A. C. L. Yu, eds., *The Handbook of Phonological Theory*. Blackwell, Cambridge MA, 240–290.
- Sande, Hannah (2018) Cross-word morphologically conditioned scalar tone shift in Guébie, *Morphology* 28:253–295.
- Sande, Hannah, Peter Jenks & Sharon Inkelas (2020) Cophonologies by Ph(r)ase, *NLLT* 38(4):1211–1261.
- Schiffman, Harold F. (1999) *A Reference Grammar of Spoken Tamil*. Cambridge University Press, Cambridge.
- Schrock, Terrill B. (2014) A grammar of Ik (Icé-tód): Northeast Uganda’s last thriving Kuliak language. PhD thesis, Universiteit Leiden.
- Schuh, Russell G. (1989) Long vowels and diphthongs in Miya and Hausa. In: P. Newman & R. Botne, eds., *Current Approaches to African Linguistics*. Foris, Dordrecht, 35–43.
- Selkirk, Elisabeth (1986) On derived domains in sentence phonology, *Phonology Yearbook* 3:371–405.
- Selkirk, Elisabeth (2011) The Syntax-phonology interface. In: J. Goldsmith, J. Riggle & A. C. L. Yu, eds., *The Handbook of Phonological Theory*. Wiley Blackwell, London, 435–484.
- Smolensky, Paul & Géraldine Legendre (2006) *The Harmonic Mind: From Neural Computation to Optimality-Theoretic Grammar*. MIT Press, Cambridge, MA.
- Solomiac, P (2014) *Phonologie et morphosyntaxe du Dúúngoo de Samogohiri*. Köppe, Köln.
- Stirtz, Timothy M. (2011) A grammar of Gaahmg. PhD thesis, Leiden University.
- Stump, G. T. (1988) Nonlocal spirantization in Breton, *Journal of Linguistics* 24(2):457–481.
- Stump, Gregory T. (2001) *Inflectional Morphology*. Cambridge University Press, Cambridge.
- Sundaresan, Sandhya & Thomas McFadden (2014) Tamil. Ms., Leipzig University.
- Tebay, Sören (2019) Outward mutation in Nimboran. Poster at mfm 27, May 25 2019.
- Tranel, Bernard (1995) On the Status of Universal Association Conventions: Evidence from Mixteco. In: *Proceedings of BLS 21*. 299–312.
- Trommer, Jochen (2011) Phonological Aspects of Western Nilotic Mutation Morphology. Habilitation Thesis, University of Leipzig.
- Trommer, Jochen (2022) The Stratal Structure of Kuria Morphological Tone, *Phonology* 679–723.
- Trommer, Jochen (2024) The concatenative structure of tonal overwriting, *Linguistic Inquiry* 55(1):95–151.
- Trommer, Jochen (to appear) Gestalt Contours, Polarity and Construction-specific Phonology in Gaahmg, *Phonological Data & Analysis*.
- Uchihara, Hiroto & Tiburcio Cano (2019) A phonological account of Tlapanec tonal alternation, *Journal of Linguistics* 1–57.

- van Oostendorp, Marc (2007) Derived Environment Effects and Consistency of Exponence. In: S. Blaho, P. Bye & M. Krämer, eds., *Freedom of Analysis?*. Mouton de Gruyter, Berlin, 123–148.
- van Oostendorp, Marc (2008) Incomplete Devoicing in Formal Phonology, *Lingua* 118(9):1362–1374.
- Wagner, Michael (2005) Prosody and recursion. Doctoral dissertation. PhD thesis, MIT.
- Wolf, Matthew (2005) An Autosegmental Theory of Quirky Mutations. In: J. Alderete, C.-H. Han & A. Kochetov, eds., *West Coast Conference on Formal Linguistics (WCCFL) 24*. Cascadilla Proceedings Project, Somerville, MA, 370–378.
- Wolf, Matthew (2007) For an Autosegmental Theory of Mutation. In: L. Bateman, M. O’Keefe, E. Reilly & A. Werle, eds., *University of Massachusetts Occasional Papers in Linguistics 32: Papers in Optimality Theory III*. GLSA, Amherst, 315–404.
- Yip, Moira (2002) *Tone*. Cambridge University Press, Cambridge.
- Yu, Kristine M. (2021) Tonal marking of absolutive case in Samoan, *Natural Language & Linguistic Theory* 39:291–365.
- Zimmermann, Eva (2017) *Morphological Length and Prosodically Defective Morphemes*. Oxford University Press, Oxford.
- Zimmermann, Eva & Jochen Trommer (2014) Generalized Mora Affixation and Quantity-Manipulating Morphology, *Phonology* 31:463–510.
- Zoll, Cheryl (1996) Parsing below the Segment in a Constraint-based Framework. PhD thesis, UC Berkeley.

Contents

A	The Language Sample	1
B	Basic Statistics of Language Sample	4
C	Language Data	6
C.1	Acoma – N America – Keresan [kjq] (6 patterns)	7
C.2	Afar – Africa – Afroasiatic (Cushitic) [aar] (1 pattern)	9
C.3	Aghem – Africa – Niger-Congo (Narrow Grassfields) [agq] (1 pattern)	10
C.4	Ainu – Eurasia – Ainu [ain] (3 patterns)	10
C.5	Aklán – Papuanesia – Austronesian [akl] (2 patterns)	10
C.6	Anywa – Africa – Eastern Sudanic (Western Nilotic) [anu] (2 patterns)	11
C.7	Awa – Papuanesia – Nuclear Trans New Guinea [imo] (2 patterns)	12
C.8	Bakweri – Africa – Niger-Congo (Bantu) [bri] (1 pattern)	12
C.9	Bangime – Africa – Isolate [dba] (2 Patterns)	12
C.10	Barasana – S America – Tucanoan [bsn] (1 Pattern)	12
C.11	Bari – Africa – Eastern Sudanic (Eastern Nilotic) [bfa] (3 Patterns)	13
C.12	Chaha – Africa – Afro-Asiatic (Semitic) [sgw] (1 pattern)	13
C.13	Cherokee (Oklahoma) – North America – Iroquoian [chr] (2 patterns)	14
C.14	Chichewa – Africa – Niger-Congo (Bantu) [chew1246] (4 patterns)	14
C.15	Chimila – South America – Chibchan [cbg] (1 pattern)	15
C.16	Chukchee – Eurasia – Chukotko-Kamchatkan [ckt] (1 pattern)	15
C.17	Ciyao – Niger-Congo (Bantu) [yao] (2 patterns)	16
C.18	Finnish – Eurasia – Uralic [fin] (3 patterns)	16
C.19	Fula Africa – Niger-Congo (Central Atlantic) [ful] (2 patterns)	17
C.20	Gã – Africa – Niger-Congo (Kwa) [gaa] (2 patterns)	17
C.21	Gaahmg – Africa – Eastern Sudanic [tbi] (1 pattern)	17
C.22	Gadsup – Papuanesia – Nuclear New Guinea [gaj] (1 pattern)	18
C.23	German – Eurasia – Indoeuropean (Germanic) [deu] (3 patterns)	18
C.24	Greek – Eurasia – Indoeuropean [ell] (2 patterns)	19
C.25	Guébie – Africa – Niger-Congo (Kru) [dic] (2 patterns)	19
C.26	Hungarian – Eurasia – Uralic – [hun] (1 pattern)	20
C.27	Igbo – Africa – Niger-Congo [ibo] (4 patterns)	20
C.28	Ik – Africa – Eastern Sudanic [ikx] (2 patterns)	21
C.29	Italian – Eurasia – Indoeuropean (Romance) [ita] (1 pattern)	21
C.30	Izon – Africa – Ijoid [ijc] (3 patterns)	22
C.31	Kashaya – N America – Hokan (Pomoan) [kju] (2 patterns)	22
C.32	Ket – Eurasia – Yeniseian – [ket] (1 pattern)	22
C.33	Kikuyu – Africa – Niger-Congo (Bantu) [kik] (1 pattern)	23
C.34	Kinande – Africa – Niger-Congo (Bantu) [nnb] (1 pattern)	23
C.35	Kipsigis – Africa – Eastern Sudanic (Southern Nilotic) [sgc] (1 pattern)	24
C.36	Kisi – Africa – Niger-Congo [kss] (1 pattern)	24
C.37	Kɔnnɪ – Africa – Niger-Congo (Gur) [kma](4 patterns)	24
C.38	Korean – Eurasia – Korean [kor] (4 patterns)	25
C.39	Koryak – Eurasia – Chukotko-Kamchatkan [kpy] (4 patterns)	26
C.40	Kpelle – Africa – Mande [kpe1252] (2 patterns)	27
C.41	Kunama – Africa – isolate [kun] (2 patterns)	27
C.42	Kuria – Africa – Niger-Congo (Bantu) [kuj] (3 patterns)	27
C.43	Lango – Africa – Eastern Sudanic (Western Nilotic) [lno] (1 pattern)	27
C.44	Makaa – Africa – Niger-Congo (Bantu) [mcp] (2 patterns)	28
C.45	Mao – Africa – Blue Nile Mao [myf] (4 patterns)	28
C.46	Margi – Africa – Afro-Asiatic (Chadic) [mrt] (1 pattern)	29
C.47	Mayo – N America – Uto-Aztecan [mfy] (2 patterns)	29
C.48	Mixtec – N America – Oto-Manguean [mig] (1 pattern)	29
C.49	Miya – Africa – Afro-Asiatic (West Chadic) [mkf] (1 pattern)	30
C.50	Moro – Africa – Kordofanian [mor] (2 patterns)	30
C.51	Neve’ei – Papuanesia – Austronesian (Oceanic) [vnm] (1 pattern)	31
C.52	Nimboran – Papuanesia – Nimboran [nir] (4 patterns)	31

C.53 Russian – Eurasia – Indoeuropean (Slavic) [rus] (2 patterns)	32
C.54 Samoan – Papuanesia – Austronesian (Oceanic) [smo] (1 pattern)	32
C.55 Seenku – Africa – Niger-Congo (Mande) [samo1309] (2 patterns)	33
C.56 Shilluk – Africa – Eastern Sudanic (Western Nilotic) [shk] (2 patterns)	33
C.57 Shoshoni – Uto-Aztecan [shh] (3 patterns)	34
C.58 Supyire – Africa – Niger-Congo (Senufo) [spp] (3 patterns)	35
C.59 Tagalog – Papuanesia – Austronesian (Malayo-Polynesian) [tgl] (2 patterns)	35
C.60 Tamil – Eurasia – Dravidian [tam] (1 pattern)	36
C.61 Tauya – Papuanesia – Nuclear Trans New Guinea [tya] (1 pattern)	36
C.62 Tenyidie – Eurasia – Sino-Tibetan [njm] (1 pattern)	36
C.63 Tetsót'iné – Na-Dene (Athabaskan) [chp] (1 pattern)	37
C.64 Tiriki – Africa – Niger-Congo (Bantu) [luy] (1 pattern)	37
C.65 Tlapanec – N America – Oto-Manguean [tcf] (1 pattern)	38
C.66 Turkana – Africa – Eastern Sudanic (Eastern Nilotic) [tuv] (1 pattern)	38
C.67 Urarina –South America – Isolate [urn] (4 patterns)	38
C.68 Uspanteko – N America – Mayan [usp] (1 pattern)	38
C.69 Welsh –Eurasia – Indoeuropean (Celtic) [cym] (2 patterns)	39
C.70 Yoruba – Africa – Niger-Congo [yor] (1 pattern)	39
D Abbreviations	40
E References (for the Appendix)	41

A The Language Sample

(96) *The language sample sorted alphabetically (genetic and geographic specifications according to WAL5)*

	Iso-639-3	Macroarea	Genus	Family
Acoma (6)	[kjq]	N America	Keresan	Keresan
Afar (1)	[aar]	Africa	Lowland East Cushitic	Afro-Asiatic
Aghem (1)	[agq]	Africa	Wide Grassfields	Niger-Congo
Ainu (3)	[ain]	Eurasia	Ainu	Ainu
Akalan (2)	[akl]	Papuanesia	Greater Central Philippine	Austronesian
Anywa (2)	[anu]	Africa	Western Nilotic	Eastern Sudanic
Awa (2)	[awb]	Papuanesia	Gauwa	Trans-New Guinea
Bakweri (1)	[bri]	Africa	Bantu	Niger-Congo
Bangime (2)	[dba]	Africa	Bangime	Bangime
Barasana (1)	[bsn]	S America	Tucanoan	Tucanoan
Bari (3)	[bfa]	Africa	Eastern Nilotic	Eastern Sudanic
Chaha (1)	[sgw]	Africa	Semitic	Afro-Asiatic
Cherokee (2)	[chr]	N America	Southern Iroquoian	Iroquoian
Chichewa (4)	[nya]	Africa	Bantu	Niger-Congo
Chimila (1)	[cbg]	S America	Chimila	Chibchan
Chukchee (1)	[ckt]	Eurasia	Northern Chukotko-Kamchatkan	Chukotko-Kamchatkan
Ciyao (2)	[yao]	Africa	Bantu	Niger-Congo
Finnish (3)	[fin]	Eurasia	Finnic	Uralic
Fula (2)	[ful]	Africa	Peul-Serer	Niger-Congo
Gā (2)	[gaa]	Africa	Ga-Ndangme	Niger-Congo
Gaahmg (1)	[tbi]	Africa	Eastern Jebel	Eastern Sudanic
Gadup (1)	[gaj]	Papuanesia	Gauwa	Trans-New Guinea
German (3)	[deu]	Eurasia	Germanic	Indo-European
Greek (2)	[ell]	Eurasia	Greek	Indo-European
Guébie (2)	[dic]	Africa	Kru	Niger-Congo
Hungarian (1)	[hun]	Eurasia	Ugric	Uralic
Igbo (4)	[ibo]	Africa	Igboid	Niger-Congo
Ik (2)	[ikx]	Africa	Kuliak	Eastern Sudanic
Italian (1)	[ita]	Eurasia	Romance	Indo-European
Kashaya (2)	[kju]	N America	Pomoan	Hokan
Ket (1)	[ket]	Eurasia	Yeniseian	Yeniseian
Kikuyu (1)	[kik]	Africa	Bantu	Niger-Congo
Kisi (1)	[kss]	Africa	Southern Mel	Niger-Congo
Kɔnnɪ (4)	[kma]	Africa	Oti-Volta	Niger-Congo
Korean (4)	[kor]	Eurasia	Korean	Korean
Koryak (4)	[kpy]	Eurasia	Northern Chukotko-Kamchatkan	Chukotko-Kamchatkan
Kpelle (2)	[xpe]	Africa	Western Mande	Mande
Kunama (2)	[kun]	Africa	Kunama	Kunama
Kuria (3)	[kuj]	Africa	Bantu	Niger-Congo
Lango (1)	[laj]	Africa	Western Nilotic	Eastern Sudanic
Makaa (2)	[mcp]	Africa	Bantu	Niger-Congo
Mao ¹ (4)	[myf]	Africa	Blue-Nile Mao	Blue-Nile Mao
Margi (1)	[mrt]	Africa	Biu-Mandara	Afro-Asiatic
Mayo (2)	[mfy]	N America	Cahita	Uto-Aztecan
Mixtec (1)	[mig]	N America	Mixtec	Otomanguean
Miya (1)	[mkf]	Africa	West Chadid	Afro-Asiatic
Moro (2)	[mor]	Africa	Heiban	Kordofanian
Neve'ei (1)	[vnm]	Papuanesia	Oceanic	Austronesian
Nimboran (4)	[nir]	Papuanesia	Nimboran	Nimboran
Russian (2)	[rus]	Eurasia	Slavic	Indo-European
Samoan (1)	[smo]	Papuanesia	Oceanic	Austronesian
Seenku (2)	[sos]	Africa	Mande	Niger-Congo

¹ ≈ Bambassi

Shilluk (2)	[shk]	Africa	Western Nilotic	Eastern Sudanic
Shoshoni (3)	[shh]	N America	Northern Uto-Aztecan	Uto-Aztecan
Supyire (3)	[spp]	Africa	Senufo	Niger-Congo
Tagalog (2)	[tgl]	Papuanesia	Greater Central Philippine	Austronesian
Tamil (1)	[tam]	Eurasia	Dravidian	Dravidian
Tauya (1)	[tya]	Papuanesia	Rai Coast	Trans-New Guinea
Tenyidie (1)	[njm]	Eurasia	Angami-Pochuri	Sino-Tibetan
Tetsót'iné (1)	[chp]	N America	Athapaskan	Na-Dene
Tiriki (1)	[luy]	Africa	Bantu	Niger-Congo
Tlapanec (1)	[tcf]	N America	Subtiaba-Tlapanec	Otomanguean
Turkana (1)	[tuv]	Africa	Eastern Nilotic	Eastern Sudanic
Urarina (4)	[urn]	S America	Urarina	Urarina
Uspanteko (1)	[usp]	N America	Mayan	Mayan
Welsh (2)	[cym]	Eurasia	Celtic	Indo-European
Yoruba (1)	[yor]	Africa	Defoid	Niger-Congo

(97) *The language sample sorted by macro-area and family (52 genera and 28 families)*

	Iso-639-3	Macroarea	Genus	Family
Margi (1)	[mrt]	Africa	Biu-Mandara	Afro-Asiatic
Miya (1)	[mkf]	Africa	West Chad	Afro-Asiatic
Afar (1)	[aar]	Africa	Lowland East Cushitic	Afro-Asiatic
Mao (3)	[myf]	Africa	Blue-Nile Ma	Afro-Asiatic
Chaha (1)	[sgw]	Africa	Semitic	Afro-Asiatic
Moro (2)	[mor]	Africa	Heiban	Kordofanian
Kpelle (2)	[xpe]	Africa	Western Mande	Mande
Seenku (2)	[sos]	Africa	Western Mande	Mande
Aghem (1)	[agq]	Africa	Wide Grassfields	Niger-Congo
Bakweri (1)	[bri]	Africa	Bantu	Niger-Congo
Chichewa (4)	[nya]	Africa	Bantu	Niger-Congo
Ciyao (2)	[yao]	Africa	Bantu	Niger-Congo
Kikuyu (1)	[kik]	Africa	Bantu	Niger-Congo
Kisi (1)	[kss]	Africa	Bantu	Niger-Congo
Kuria (3)	[kuj]	Africa	Bantu	Niger-Congo
Makaa (2)	[mcp]	Africa	Bantu	Niger-Congo
Tiriki (1)	[luy]	Africa	Bantu	Niger-Congo
Fula (2)	[ful]	Africa	Peul-Serer	Niger-Congo
Kɔnnɪ (4)	[kma]	Africa	Oti-Volta	Niger-Congo
Supyire (3)	[spp]	Africa	Senufo	Niger-Congo
Igbo (4)	[ibo]	Africa	Igbo	Niger-Congo
Guébie (2)	[dic]	Africa	Kru	Niger-Congo
Gã (2)	[gaa]	Africa	Ga-Ndangm	Niger-Congo
Yoruba (1)	[yor]	Africa	Yoruboid	Niger-Congo
Gaahmg (1)	[tbi]	Africa	Eastern Jebel	Eastern Sudanic
Ik (2)	[ikx]	Africa	Kuliak	Eastern Sudanic
Bari (3)	[bfa]	Africa	Eastern Nilotic	Eastern Sudanic
Turkana (1)	[tuv]	Africa	Eastern Nilotic	Eastern Sudanic
Anywa (2)	[anu]	Africa	Western Nilotic	Eastern Sudanic
Lango (1)	[laj]	Africa	Western Nilotic	Eastern Sudanic
Shilluk (2)	[shk]	Africa	Western Nilotic	Eastern Sudanic
Bangime (2)	[dba]	Africa	Bangime	Bangime
Kunama (2)	[kun]	Africa	Kunama	Kunama
Kashaya (2)	[kju]	N America	Pomoan	Hokan
Cherokee (2)	[chr]	N America	Southern Iroquoian	Iroquoian
Acoma (6)	[kjq]	N America	Keresan	Keresan
Uspanteko (1)	[usp]	N America	Mayan	Mayan
Tetsót'iné (1)	[chp]	N America	Athabaskan	Na-Dene
Mixtec (1)	[mig]	N America	Mixtec	Otomanguean
Tlapanec (1)	[tcf]	N America	Subtiaba-Tlapanec	Otomanguean
Shoshoni (3)	[shh]	N America	Northern Uto-Aztecan	Uto-Aztecan
Mayo (2)	[mfy]	N America	Cahita	Uto-Aztecan
Chimila (1)	[cbg]	S America	Chimila	Chibchan
Barasana (1)	[bsn]	S America	Tucanoan	Tucanoan
Urarina (4)	[urn]	S America	Urarina	Urarina
Chukchee (1)	[ckt]	Eurasia	Northern Chukotko-Kamchatkan	Chukotko-Kamchatkan
Koryak (4)	[kpy]	Eurasia	Northern Chukotko-Kamchatkan	Chukotko-Kamchatkan
Tamil (1)	[am]	Eurasia	Dravidian	Dravidian
Russian (2)	[rus]	Eurasia	Slavic	Indo-European
Welsh (2)	[cym]	Eurasia	Celtic	Indo-European
German (3)	[deu]	Eurasia	Germanic	Indo-European
Greek (2)	[ell]	Eurasia	Greek	Indo-European
Italian (1)	[ita]	Eurasia	Romance	Indo-European
Tenyidie (1)	[njm]	Eurasia	Angami-Pochuri	Sino-Tibetan
Finnish (3)	[fin]	Eurasia	Finnic	Uralic

Hungarian (1)	[hun]	Eurasia	Ugric	Uralic
Ket (1)	[ket]	Eurasia	Yeniseian	Yeniseian
Ainu (3)	[ain]	Eurasia	Ainu	Ainu
Korean (4)	[kor]	Eurasia	Korean	Korean
Akalan (2)	[akl]	Papuanesia	Greater Central Philippine	Austronesian
Tagalog (2)	[tgl]	Papuanesia	Greater Central Philippine	Austronesian
Samoan (1)	[smo]	Papuanesia	Oceanic	Austronesian
Neve'ei (1)	[vnm]	Papuanesia	Oceanic	Austronesian
Nimboran (4)	[nir]	Papuanesia	Nimboran	Nimboran
Gadup (1)	[gaj]	Papuanesia	Gauwa	Trans-New Guinea
Awa (2)	[awb]	Papuanesia	Gauwa	Trans-New Guinea
Tauya (1)	[tya]	Papuanesia	Rai Coast	Trans-New Guinea

B Basic Statistics of Language Sample

The following tables contain basic numerical statistics on our language sample which comprises 138 distinct anticyclic mutation patterns in 70 languages.² Note that we have included all patterns here which classify as anticyclic under the lexicalist approach presupposed in the main part of the paper (see section 1). Thus some phrase-level patterns of mutation listed here would not count as anticyclic under the alternative approaches to cyclicity discussed in section 5.1. The single patterns are identified by circled numbers in the language data summaries in appendix C. The linguistic macroareas in table (98) are the ones used in WALS (Dryer & Haspelmath 2013, see Miestamo et al. 2016 for possible alternatives). Modality (table (99)) and type (table (100)) of mutation follow the classifications used throughout the paper.

The sample is opportunistic in that it provides a collection of the maximal number of distinct anticyclic mutation patterns across the languages, accessible to us in published literature (with a number of limitations discussed below). Thus the numbers in the following tables are not intended as evidence for the typological distribution of (specific) anticyclic mutation (patterns), but as an indication of the sample's coverage of typological and grammatical diversity.

Two evident (and related) biases in the data are the predominance of tone and of African languages which both constitute roughly half of the sample. These reflect the fact that tonal morphophonology is considerably more variegated than other single phonological modalities (cf. Hyman's dictum that "tone can do everything that segmental and metrical phonology can do, but ... the reverse is not true" Hyman 2011:198), the density of tonal languages in Africa, and the much better documentation of tone in African languages compared to other parts of the world, especially the Americas and Papuanesia. The lack of morphological tone is also probably one of the reasons for the absence of data from Australian languages in the sample.³

(98) *Anticyclic mutation data from different linguistic macroareas*

	Africa	Eurasia	Papuanesia	North America	South America	Total
Number	70	29	14	19	6	138
Percentage	50.7%	21.0%	10.1%	13.8%	4.3%	

(99) *Phonological modality*

	Tone	Vocalic Features	Consonantal Features	Length	Stress	Total
Number	70	24	20	13	11	138
Percentage	50.7%	17.4%	14.5%	9.4%	8.0%	

On a global scale, our sample is also likely to underrepresent Root-to-Word mutation and tautomorphemic mutation. Since the structural interpretation of phrasal morphophonology is much more contentious than mor-

²We have identified mutation patterns as distinct if they differ either in the features they manipulate (e.g., High tone vs. Low tone), their directionality (inducing mutation to the left or right of the mutation trigger, or on the trigger itself). Thus alternating mutations are counted as different patterns here.

³Note that the typological distribution of *cyclic* mutation patterns discussed in the current literature is similar. Thus the two most comprehensive recent surveys, Akinlabi (2011) and Wolf (2007) both draw roughly 70% of their data from Africa and Eurasia, lack data from Australia, and have roughly 10% of data from South America and Papuanesia.

phophonology within the lexical phonology (see, e.g., the discussions in sections 2.5 and 5.1.2), our coverage of the documented cases is less comprehensive than for word-internal cases (for example, we have not included the Dogon data addressed shortly in section 5.1.2). Since they instantiate a kind of ‘mild’ form of anticyclicity, we also have not searched systematically for tautomorphemic mutation phenomena. The tautomorphemic patterns included here are thus mostly restricted to variants of (otherwise strictly anticyclic) alternating patterns.

(100) *Structural types of anticyclic mutation*

	Root to Affix	Affix to Affix	Affix to Word	Root to Word	Tauto- morphemic Root	Tauto- morphemic Affix	Total
Number	52	24	13	35	11	3	138
Percentage	37.7%	17.4%	9.4%	25.4%	8.0%	2.2%	

C Language Data

This appendix lists all languages of our language sample for anticyclic mutation and provides data for the patterns not explicitly discussed in the main text. Each language entry specifies the language family according to WALS (<https://wals.info/>) and provides the Isocode (ISO-639-3) for the language and a web link to its Glottolog page (<https://glottolog.org/>), which contains more detailed information on its genetic affiliation and geographic location.⁴

The single mutation patterns in this appendix are listed under schematic tables as in (101), where the first part of each entry (before the double bars “||”) specifies the general morphosyntactic type and directionality at different levels of concreteness, and the second part (after “||”) its phonological content, generically in the first column, and in the second one with the specific phonological features involved.

(101) ①

Root-to-Affix	Root-to-Suffix	→	Vocalic Features	+ATR
---------------	----------------	---	------------------	------

Alternating mutation is coded as in (102) (for Aklan below) where the tilde (“~”) separates the different alternating patterns.

(102) ①

Root-to-Affix	Root-to-Suffix	→	Stress
~	~	~	
② Tautomorphemic Root	Tautomorphemic Root	↻	

Circled numbers indicate the anticyclic mutation patterns we have identified as distinct either because they differ in the involved phonological feature, directionality (affecting left-adjacent or right-adjacent targets, marked by arrows), structural type (e.g., Root-to-Affix vs. Root-to-Word), or in having a segmental component or not. Alternations between different anticyclic mutation types as in (102) are thus counted as separate basic patterns.

If an alternation may affect different directions simultaneously, this is marked by “~ =” instead (see, e.g., Anywa in C.6 below). The entry for Acoma immediately below in C.1 illustrates a further aspect of the notation: Affixes or clitics which do not contain a segmental component are given in shaded rounded boxes (reminiscent of the classical autosegmental notation for floating features). If an anticyclic mutation pattern has a cyclic counterpart this is additionally specified in parentheses (see Acoma for an example). Recall also from section 2.2 how we indicate mutation triggers and targets in language data: target morphemes (or words) are marked by boxes and triggers by underlining (hence the pure underscore “_” in cases where the trigger doesn’t have segmental content). More information on the distribution of the sample is provided in appendix B.

⁴For the languages in the sample for which WALS doesn’t provide an entry (Afar, Gã, Mao, Seenku, Urarina and Uspanteko), we have substituted the family (and genus in tables (96) and (97)) of closely related languages where possible, or otherwise the affiliation given by Glottolog. Since Glottolog doesn’t specify genera, we have used the most low-level genetic grouping provided by Glottolog instead.

C.1 Acoma – N America – Keresan [kjq] (6 patterns)

Affix-to-Affix	Prefix-to-Prefix	←	Consonantal Features	① aspiration ② glottalization ③ palatalization
----------------	------------------	---	----------------------	--

([Affix-to-Root | Suffix-to-Root | ← | Consonantal Features | aspiration, glottalization, palatalization])

The Keresan language Acoma, spoken in Western New Mexico, exhibits Affix-to-Affix mutation and Root-to-Affix mutation for several consonantal features, as shown by Gleim (2015) based on the grammar of Miller (1965). Featural affixation expresses valency-changing categories such as Benefactive and Passive and TAM categories like Habitual and Continuous. Phonologically it involves aspiration glottalization and palatalization (partially in combination). (103) illustrates these with suffixes which – cyclically – modify root consonants. Note that the term ‘palatalization’ is an oversimplification. The process changes retroflex obstruents into palatals (e.g., /t/ → [c], into postalveolars (e.g., underlyings → [ʃ]), or into dentals (e.g., /tʂ/ → [t]), depending on their manner of articulation. It mutates velar stops into dental affricates (e.g., /k/ → [ts]), and palatal stops into dental stops (e.g., /c/ → [t]). Gleim analyses this by a floating complex of a Coronal feature dominating [+laminal]. In the following, we will abbreviate this simply as [+laminal]:

(103) Acoma cyclic Affix-to-Root mutation (Miller 1965, Gleim 2015)

a. Aspiration ([+s(pread) g(lottis)])

s'-úum'inaat-^[+sg]íʔit → s'-íum'inaat^h-i 'I spoke to him for her' p.34 ((/t/ → [t^h]))
1-speak-BEN

b. Glottalization ([+c(onstricted) g(lottis)])

k-áitseeša-^{[+lam][+cg]}áát'an → k-áitseeʃ-áát'a 'he is dreaming' p.74 ((/s/ → [ʃ]))
3-dream-HAB

c. Palatalization ([+lam(inal)])

- i. k-áitseeša-^{[+lam][+cg]}áát'an → k-áitseeʃ-áát'a 'he is dreaming' p.74 (/s/ → [ʃ])
3-dream-HAB
- ii. s-^{[+lam][+cg]}uts'aacan-^[+lam]aaajan → s'-its'aat-aaaja 'I am pinning it' p.74 (/c/ → [t])
1-pin-CONT

Prefixes trigger anticyclic mutation for the same features changing more outwards prefixes (pronominal prefixes show a complex allomorphy depending on tense, aspect, mood and root class, which we don't represent in the glosses):

(104) Acoma anticyclic Affix-to-Affix mutation (Miller 1965, Gleim 2015)

a. Aspiration ([+s(pread) g(lottis)])

k-^[+sg]-átʂik^han → k^h-átʂik^ha 'he smoked tobacco' p.74 (/k/ → [k^h])
3-ANTIP-smoke.tobacco

b. Glottalization ([+c(onstricted) g(lottis)])

- i. s-^[+cg]-uuk^hatʃ^han-^[+cg]aat → s'-uuk^hatʃ^han'-a 'We saw him' p.27 (/s/ → [s'])
1-TR-see-PL
- ii. ša-^[+cg]-úst'-^[+sg]íʔit → š'a-úst'-í 'I gave you water' p.71 (/s/ → [š'])
1>2-BEN-give.water-BEN

c. Palatalization [+lam(inal)]

- i. k-^{[+lam][+cg]}-ázəkaN-iš → ts'-á'təkán'i 'it has been smoked' p.121 (/k/ → [ts'])
3-PASS-smoke-COMPL
- ii. š-^[+lam]-úst'-íʔit → ʃ-íst'-íʔi 'you gave him water' p.71 (/s/ → [ʃ])
2-BEN-give-BEN

There are several locality effects indicating that the non-segmental diathesis affixes in (104) are in fact inside (to

the right) of the segmental affixes they modify. As shown by (105-c,d,e), the palatal ([+laminal]) feature cannot modify a consonant across an underlying vowel even if this is deleted as in (105-d). Whereas glottalization can skip an intervening vowel (e.g., (105-a)), it systematically modifies the last not the first (or both) consonants of a prefix (105-d). Note that Gleim assumes that in (105-e) association of [+cg] is blocked by an intervening zero affix marking dual (/sku-/ is a generic 3>1 person prefix).⁵

(105) *Acoma anticyclic Affix-to-Affix mutation* (Miller 1965, Gleim 2015)

- a. $\text{\textcircled{s}}\text{a-}^{[+cg]}\text{-}\text{\textcircled{u}}\text{st'}-^{[+sg]}\text{\textcircled{i}}\text{t}$ → $\text{\textcircled{s}}\text{'a-}-\text{st'}-\text{\textcircled{i}}$ 'I gave you water' p.71 (/s/ → [s'])
1>2-BEN-give.water-BEN
- b. $\text{s-}^{[+cg]}\text{-}\text{\textcircled{u}}\text{uk}^h\text{at}^h\text{\textcircled{a}}\text{n-}^{[+cg]}\text{aat}$ → $\text{\textcircled{s}}\text{'-}-\text{\textcircled{i}}\text{uk}^h\text{at}^h\text{\textcircled{a}}\text{n'}-\text{a}$ 'We saw him' p.27 (/s/ → [s'])
1-TR-see-PL
- c. $\text{cu-}^{[+lam]}\text{[+cg]}\text{-}\text{\textcircled{i}}\text{t'}\text{aa}$ → $\text{\textcircled{c}}\text{'u}\text{-}\text{\textcircled{i}}\text{t'}\text{a}$ 'You stepped on me' p.74 (/c/ → [c'])
2>1-TR-step
* $\text{\textcircled{t}}\text{'u}\text{-}\text{\textcircled{i}}\text{t'}\text{a}$
- d. $\text{\textcircled{s}}\text{ku-}^{[+lam]}\text{[+cg]}\text{\textcircled{a}}\text{a-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}\text{m}$ → $\text{\textcircled{s}}\text{k'}\text{-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}$ 'I was cut' p.57
3>1-PASS-cut
* $\text{\textcircled{c}}\text{'-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}$
* $\text{\textcircled{s}}\text{'k-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}$
* $\text{\textcircled{s}}\text{'k'}\text{-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}$
- e. $\text{\textcircled{s}}\text{ku-}\text{\textcircled{0}}\text{-}^{[+lam]}\text{[+cg]}\text{\textcircled{a}}\text{a-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}\text{m}$ → $\text{\textcircled{s}}\text{k-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}$ 'We two were cut' p.57
3>1-DU-PASS-cut
* $\text{\textcircled{s}}\text{k'}\text{-}\text{\textcircled{a}}\text{a}\text{faats}\text{\textcircled{i}}$

Finally, palatalizing affixes also have an effect on the following root: they front back vowels (106), again indicating that they are structurally the innermost affixes.

(106) *Acoma anticyclic Affix-to-Affix mutation* (Miller 1965, Gleim 2015)

- a. $\text{k-}^{[+lam]}\text{-}\text{\textcircled{u}}\text{ukat}^h\text{\textcircled{a}}\text{n}$ → $\text{\textcircled{ts}}\text{-}-\text{\textcircled{i}}\text{ukat}^h\text{\textcircled{a}}$ 'he saw him' p.75 (/k/ → [ts])
2-OBV-see
- b. $\text{\textcircled{s}}\text{-}^{[+lam]}\text{-}\text{\textcircled{u}}\text{st'}-\text{\textcircled{i}}\text{t}$ → $\text{\textcircled{j}}\text{-}-\text{\textcircled{i}}\text{st'}-\text{\textcircled{i}}\text{t}$ 'you gave him water' p.71 (/s/ → [j])
2-BEN-give-BEN

Root-to-Affix	Root-to-Prefix	←	Consonantal Features	④ aspiration ⑤ glottalization ⑥ palatalization
---------------	----------------	---	----------------------	--

In addition to Affix-to-Affix mutation, Acoma also has roots which trigger aspiration, palatalization and/or glottalization on prefixes, hence Root-to-Affix mutation. Note especially the minimal triple in (107-a-i,ii,b-i), where the 3sg prefix /k-/ appears in three different mutated shapes (cf. the minimally different /k-aipet'a/ 3-anus → [k-aipet'a] 'his anus' (p.144), where the prefix emerges in its underlying form).

⁵Miller notates the sibilant in 2>1 sku ((105)-d,e) as <s>. But since /s/ predictably becomes [s] in clusters where it precedes /k/, the morpheme is always realized as sku-, which we assume here as the underlying form

(107) *Acoma Root-to-Affix mutation (Miller 1965, Gleim 2015)*

a. *Aspiration* ([+spread] g(lottis))

i. k^{+[sg]}a:ma: → k^h-a:ma: 'his thigh' p.144 (/k/ → [k^h])
3-thigh

ii. k^{+[sg][+lam]}a:m'u:c'a → ts^h-a:m'u:c'a 'his toe' p.144 (/k/ → [ts^h])
3-toe

b. *Glottalization* ([+c(onstricted) g(lottis)])

i. k^{+[cg]}-u:pi → k'-u:pi 'his forehead' p.145 (/k/ → [k'])
3-forehead

ii. s^{+[lam][+cg]}-uts'aacan^{+[lam]}aajan → s'-its'aat-aaja 'I am pinning it' p.74 (/s/ → [s'])
1-pin-CONT

c. *Palatalization* (anticyclic, [+laminal])

i. s^{+[lam]}aʔáts'in → ʃ-aʔáts'i 'you arrived' p.74 (/s/ → [ʃ])
2-arrive

ii. ts^h^{+[lam]}aʔáts'in → tʃ^h-aʔáts'i 'did you arrive?' p.74 (/ts/ → [tʃ^h])
2-arrive

C.2 Afar – Africa – Afroasiatic (Cushitic) [aar] (1 pattern)

① [Affix]-to-Word [Suffix]-to-Word || → Length consonant gemination

The Cushitic language Afar has a Genitive suffix for feminine nouns which is realized as gemination on the following noun if the latter starts with a consonant (108-a) but as the segment [h] if it is vowel-initial (108-b).

(108) *Afar Genitive (Bliese 1981:165-167)*

a.	ruga:gé	+	galbó	→	ruga:gé__	g:albó	'calf's hide'
	calf		hide				
	ba:rá	+	bilá	→	ba:rá__	b:ilá	'woman's necklace'
	woman		necklace				
	bo:dá	+	cajsó	→	bo:dá__	c:ajsó	'meadow's grass'
	meadow		grass				
b.	sagá	+	íba	→	sagá__	híba	'a cow's foot'
	cow		foot				
	di:da:lé	+	amó	→	di:da:lé__	hámo	'a bee's head'
	bee		head				
	ba:dó	+	ul:ul:ú	→	ba:dó__	hul:ul:ú	'country's slope'
	country		slope				

That this is not just full assimilation of a final /h/ can be seen by the contrast to other homophonous formatives such as the case/postposition suffix /-h/ 'for' and the modal suffix /-h/ ('ought') which faithfully surface before consonants. This is shown for both markers in the sentence in (109).

(109) *Afar h-Infinitive (Bliese 1981:44)*

daj'lo-h lak'co ga:boj-'se-h me'ce-m dalejna
children-for money gather-INF-MOD good-NOM parents
'It is parents who ought to gather money for children'

C.3 Aghem – Africa – Niger-Congo (Narrow Grassfields) [agq] (1 pattern)

①	Root-to-Word	→	Tone	↓	↔ Hyman (1979, 2001)
---	--------------	---	------	---	----------------------

In the Grassfields Bantu language Aghem, specific high-toned nouns such as /wó/ ‘hand’ trigger downstep on following (high-toned) demonstratives (110-b’), under Hyman’s analysis triggered by a floating Low tone which is part of the root. That this downstep is an idiosyncratic effect of the noun is evident from the comparison with other high-toned nouns such as /fú/ ‘rat’ which don’t trigger downstep (110-b). Independent evidence for the floating Low comes from the fact that the same nouns also fail to trigger an otherwise general process of H-spreading ((110-c’) vs. (110-c), note also that noun class prefixes such as /kɛ-/ are obligatorily dropped if a noun is modified by a demonstrative as in (110-b)).

(110) *Root-to-Word downstepping in Aghem (Hyman 2001:3 Hyman 1979)*

- | | | | | | | | |
|----|---------|---------|------------|-----|-----------|---|-------------|
| a. | kɛ́-fú | | ‘rat’ | a’. | kɛ́-wó | | ‘hand’ |
| | CLS-rat | | | | CLS-hand | | |
| b. | fú | kɛ́n | ‘this rat’ | b’. | <u>wó</u> | ↓kɛ́n | ‘this hand’ |
| | rat | DEM | | | hand | DEM | |
| c. | kɛ́-fú | kɛ́-mɔ̃ | ‘one rat’ | c’. | kɛ́-wó | kɛ́-mɔ̃ | ‘one hand’ |
| | CLS-rat | CLS-one | | | CLS-hand | CLS-one | |

C.4 Ainu – Eurasia – Ainu [ain] (3 patterns)

Root-to-Affix	Root-to-Suffix	→	Vocalic Features	① [+high] ② [+back] ③ [-back]	↔ Ito (1984)
---------------	----------------	---	------------------	-------------------------------------	--------------

In Ainu, vocalic Root-to-Affix mutation involves multiple phonological features for different roots, [+high] and [±back] has been identified by Ito (1984). The Ainu transitivity suffix is a single vowel, which is identical to the root vowel for many roots (111-a). For specific non-low roots, however, it is a high vowel which systematically differs from the root vowel in backness (111-b).

(111) *Ainu Transitivity affixation: High mutation (Ito 1984:506)*

- a. *Copying*
- (i) tus-u 'to shake'
- (ii) pop-o 'to boil'
- (iii) ker-e 'to touch'
- (iv) mak-a 'to open'
- b. *'Polarity'*
- (i) mus-i 'to choke'
- (ii) hop-i 'to leave'

Under Ito's classical analysis of these data, this pattern is the result of a suffix which lacks vocalic features. These are in the default case (i.e., (111-a)) supplied by spreading from the root vowel. The roots in (111-b), on the other hand, have a floating [+high] feature feeding vowel dissimilation for [+/-back] (restricted to non-spreading contexts since dissimilation targets identical adjacent features not multiply associated single features). A second idiosyncratic group of roots with low vowels seems instead to specify [-back] (112-a,b) or [+back] (112-c,d) under height dissimilation as in (112).

(112) *Ainu Transitivity affixation: back mutation (Dettmer 1989:479f.)*

- a. an-i 'to have' b. car-i 'to rotate'
- c. ram-u 'to think' d. jak-u 'to destroy'

C.5 Aklan – Papuanesia – Austronesian [akl] (2 patterns)

①	Root-to-Affix ~	Root-to-Suffix ~	→ ~	Stress
②	Tautomorphemic Root	Tautomorphemic Root	↪	

The Aklan system of postaccentuation is similar to the one in Tagalog (appendix C.59), but has an interesting twist. As in Tagalog, roots determine whether stress is on the final or prefinal syllable extending this pattern to affixation (113). An example where penultimate stress falls on the affix is [haq-ambaq-án-un], ‘maxim’, Hayes 1980:183).

Following Hagberg’s 2006 analysis for Tagalog, we will assume that penultimate stress is the phonological default pattern while final stress is the result of anticyclic mutation.

(113) *Root-to-Affix stress in Aklan (Hayes 1980:181, Chai 1971)*

<i>Penultimate stress roots</i>		<i>Final stress roots</i>	
a. híkut	‘cook’	e. <u>bután</u>	‘place’
b. hikút-an	‘cook-REF.FOC.FUT’	f. <u>butaŋ</u> -án	‘place-REF.FUT’
c. sípaʔ	‘kick’	g. <u>bisá</u>	‘kiss’
d. sipáʔ-a	‘kick-GOAL.FOC.IMP’	h. <u>bisa</u> -hí	‘kiss-REF.IMP’

However, this generalization is overwritten by the requirement that penultimate closed syllables have primary stress. Thus the positional stress contrast is suppressed for un-affixed forms as in (114-e+g) but emerges under suffixation, additionally highlighting the abstract nature of the (pen-)ultimate contrast.

(114) *Root-to-Affix stress in Aklan (Hayes 1980:181, Chai 1971)*

<i>Penultimate stress roots</i>		<i>Final stress roots</i>	
a. bítbit	‘carry’	e. gásta	‘spend’
b. bitbít-a	‘carry-GOAL.FOC.IMP’	f. gasta- <u>hún</u>	‘spend-GOAL.FUT’
c. hámbag	‘speak’	g. ʔasírtar	‘lucky’
d. h-ag-am áb ag-un	‘that which should be said’	h. ʔasirta- <u>hí</u>	‘lucky-REF.IMP’

C.6 Anywa – Africa – Eastern Sudanic (Western Nilotic) [anu] (2 patterns)

①	Root-to-Affix ~ =	Root-to-Suffix ~ =	→ ~ =	Tone	High
②	Tautomorphemic Root	Tautomorphemic Root	↪		

(Affix-to-Root Prefix-to-Root → || Tone High)

Anywa is a case where mutation affects a different formative and the mutation trigger itself *simultaneously*: Many low-toned morphemes impose H on a following low-toned morpheme. This is shown for prefixes triggering raising of roots in (115). The example (115-b) also illustrates a further tonal regularity of the language: If a root or a suffix carry a H, this spreads to the other morpheme (H-spreading is plausibly restricted to this domain under the assumption that roots and suffixes as in many languages form a PWord to the exclusion of prefixes).⁶

(115) *Anywa Affix-to-Root mutation (Reh 1993:68)*

a. à ^H -	djàŋ	→	à- <u>djáŋ</u>
NMLZ-	durra:bird		‘durra bird’
b. à ^H -	càŋ	-jì	→ à- <u>cáŋ-jí</u>
PST-	eat	-it	‘you ate it’

Accordingly, if roots trigger H-mutation on following suffixes, the H not only surfaces on the affix, but also tautomorphemically on the root itself (116). Note that these nouns occur with L tones in other contexts, thus the root in (115-b) is [gàt̪] in isolation (in the singular).

⁶We have slightly adjusted Reh’s notation of examples: We have added predictable voicing where Reh gives underspecified archiphonemes. The syllables we write here as Low tones with floating H’s are notated as Mid tones by Reh, while she explicitly states that these are pronounced with low pitch (Reh 1993:45).

(116) Anywa Root-to-Affix mutation (Reh 1993:69)

- a. gát^H -è → gát-é
 ‘trade’ PL ‘types of trade’
- b. àcù:l^H -è → àcù:l-é
 island PL ‘islands’

C.7 Awa – Papuanesia – Nuclear Trans New Guinea [imo] (2 patterns)

①	Root-to-Affix ~	Root-to-Suffix ~	→	Tone	Low	↔ McPherson (2016)
②	Root-to-Word	Root-to-Word				

The Papuan language Awa exhibits a similar alternation as described for Kpelle in section 3.1. McPherson (2016, 42-47) based on data from Loving (1966) shows that roots with a final floating L impose this either on a following suffix (e.g., /tahnú/ ‘flea’ + /táté/ ‘du’ → [tahnú-táté] ‘fleas (du)’, p. 47) or – in absence of a suffix – on a following syntactic word. In this case, however, the morphophonological effect is slightly different: In the language, tones of non-initial words are independently raised to H after many syntactic categories (e.g., /kàpàntéh/ ‘sick’ + /kàpàtà/ ‘bird’ → [kàpàntéh kápátá] ‘sick bird’) For lexemes with a floating Low, this is realized on top of this neutralization process, e.g., /ànòqtâh/ ‘big’ + /kàpàtà/ ‘bird’ → [ànòqtâh kápátá] ‘big bird’ and /kàwèq/ ‘good’ + /nàh/ ‘house’ → [kàwèq nāh] ‘good house’. Compare this to a low-toned suffix after a floating-L base which simply remains low: /tahnú/ ‘flea’ + /-è/ ‘aug’ → [tahnú-è] ‘big flea’.

C.8 Bakweri – Africa – Niger-Congo (Bantu) [bri] (1 pattern)

①	Affix-to-Affix	Prefix-to-Prefix	←	Tone	↓ High
---	----------------	------------------	---	------	--------

An interesting variant of the prefix-to-prefix tone mutation in Chichewa (pattern ③ in appendix C.14) is found in another Bantu language, Bakweri, where the Past 2 prefix /-ma/ imposes a H followed by a downstep on preceding subject prefixes, no matter whether these have underlyingly a H or a L (117). The downstep must be an inherent part of the mutation process since it doesn’t depend on a L on the prefix and adjacent H’s in the language do not generally trigger downstep (cf. /ó-/ ‘Subj’ + /vá-/ ‘them’ + /fáf/ ‘hit’ + /-é/ (final vowel) → [’ó-vá-fáf-é] ‘hit them (Subj)’, p. 23).

(117) Bakweri (Marlo & Odden 2007:20+21)

	Future		Past 2	
a.	nà-zòz-à S1SG-wash-FV	‘I will wash’	ná⁺ -má-zòz-zà S1SG-PST2-wash-FV	‘I have washed’
b.	và-zòz-à S3PL-wash-FV	‘they will wash’	vá⁺ -má-zòz-zà S3PL-PST2-wash-FV	‘they have washed’

C.9 Bangime – Africa – Isolate [dba] (2 Patterns)

①	Root-to-Affix ~	Root-to-Suffix ~	→	Tone	High	↔ Hantgan (2009)
②	Tautomorphemic Root	Tautomorphemic Root	↻			

(see section 3.3 for data and discussion)

C.10 Barasana – S America – Tucanoan [bsn] (1 Pattern)

①	Root-to-Affix	Root-to-Suffix	→	Tone	High	↔ Gomez-Imbert & Kenstowicz (2000)
---	---------------	----------------	---	------	------	------------------------------------

Barasana is a restricted tone language, where under the analysis of Gomez-Imbert & Kenstowicz (2000) morphemes may either carry a High tone or a High-Low melody. With bimoraic roots, the High of both underlying tone types might either show up on the first mora (e.g., High [gáwá] ‘white people’ vs. [-wádi] ‘fish’, p.423)

or on the second mora, as shown in (118), where it leads to a mutation-like effect. In isolation (118-a), roots of both types have a Low-High profile on the surface (with an epenthetic initial Low under Gombert & Kenstowicz’ analysis). However, with tonally underspecified suffixes (118-b), final High spreads, and the Low of High-Low roots docks to the affix (see Gomez-Imbert & Kenstowicz for details on affixes which have underlying tone).

(118) *Barasana* (Gomez-Imbert & Kenstowicz 2000:427)

<i>Underlying High</i>		<i>Underlying High-Low</i>	
a.	gohé ‘hole’	wekó ‘parrot’	
b.	gohé-ré ‘to the hole’	wekór-è ‘to the parrot’	

C.11 Bari – Africa – Eastern Sudanic (Eastern Nilotic) [bfa] (3 Patterns)

①

Root-to-Affix	Root-to-Suffix	→	Vocalic Features	high, round
---------------	----------------	---	------------------	-------------

Several verb roots in the Eastern Nilotic language Bari trigger [+high +round] mutation on suffixes (Yokwe 1986). For example, the roots /fùt/ ‘to plant’ and /gír/ ‘to tattoo’ raise low back vowels of following suffixes to their high round counterpart [u], as for the Benefactive suffix /-akin/: /gír-ákin/ → [gír-úkin] ‘to tattoo for (BEN)’, (p.28) /fùt-ákin/ → [fùt-úkin] ‘to plant for (BEN)’ (p.52), the Antipassive suffix /-(j)a/: /gír-á/ → [gír-jú] ‘to tattoo (ANTIP)’ (p.33), /fùt-a/ → [fùd-ú] ‘to plant (ANTIP)’ (p.32), and the Ablative suffix /-àrà/: /fùt-àrà/ → [fùt-úrú] ‘to plant away (ABL)’ (p.52). Interestingly, in the latter case, the effect extends to both low vowels of the affix (*[fùt-úrà]). Note also that while all verbal roots which trigger this mutation are [+high], the majority of verbs of this shape doesn’t trigger raising, cf. the Benefactives [dìp-ákin]/*[dìp-úkin] ‘to support for (BEN)’ (p.27), [tún-ákin]/*[tún-úkin] ‘to gather for (BEN)’ (p.27).

②	Root-to-Affix ~	Root-to-Suffix ~	→ ~	Tone	(Low-)High(-Low)
③	Tautomorphemic Root	Tautomorphemic Root	↻		

Monosyllabic verb roots in Bari also evidence Root-to-Affix tone mutation alternating with tautomorphemic mutation. Several verbal affixes in the language are apparently tonally underspecified with their tone supplied by spreading from the verb root. Thus the Benefactive suffix which is realized as /-akin/ after consonants and as /-kin/ after vowels is typically H throughout after verb roots ending in a H and L after verb roots ending in a L tone (e.g., /dújám/ ‘cause to collapse’ + /-akin/ → [dújám-ákin] ‘cause to collapse for’ vs. /dílílì/ ‘winnow grain’ + /-akin/ → [dílílì-kin] ‘winnow grain for’, p.24+25). Although the language in principle allows falling contours (e.g., in the noun [têŋ] ‘herd’, p. 226), all monosyllabic verb roots have a single H in isolation, which for many verbs spreads to tone-less suffixes like the Benefactive (e.g., /dér/ ‘cook’ + /-akin/ → [dér-ákin] ‘cook for’). However, there is a second class of verbs which exhibit the melody LHL in these forms (e.g., /sút/ ‘bet’ + /-akin/ → [sút-ákin] ‘bet for’, p.25). Following Yokwe (1986) in the assumption that these roots underlyingly carry a floating Low-High-Low melody, the High component of this melody alternatingly shows up on suffixes and on the root itself.

C.12 Chaha – Africa – Afro-Asiatic (Semitic) [sgw] (1 pattern)

①

Affix-to-Affix	Suffix-to-Suffix	→	Consonantal Features	voicing and continuancy
----------------	------------------	---	----------------------	-------------------------

(119) shows a minimal pair where the object marker /-β/ shows up in its underlying form after the verb root with a 3sg subject (and no subject agreement marker) in (119-a) but in its hardened and devoiced form after the mutation-triggering 3pl subject agreement suffix /-o/ (119-b).

(119) *Object marking in Chaha* (Rose 2007, 40)

a.	jì-rəxɨβ-β-a	b.	jì-rəxɨβ-o- p -a
	S3-‘find’-MAL-O.3.SG.F-TNS.S		S3-‘find’-MAL-S3.PL-O3SG.F-TNS
	‘he finds (sth) to her detriment’		‘they find (sth) to her detriment’

(see section 2.3 for more data and discussion)

C.13 Cherokee (Oklahoma) – North America – Iroquoian [chr] (2 patterns)

①	Root-to-Affix	Root-to-Prefix	←	Tone	High
---	---------------	----------------	---	------	------

In Cherokee, specific verb roots such as /dlo:/ ‘strap’ trigger appearance of a High tone on a preceding prefix (120-a). (120-b) shows a verb not triggering H-tone as a minimal contrast:

(120) *Cherokee root-triggered High tone (Uchihara 2013:198 citing Feeling 1975:92+97)*

- a. ɔ́i-dlo:-híh-a b. ɔ́i-dʒagal-íh-a
 1SG-strap-PRS-IND 1SG-rip-PRS-IND
 ‘I am stapping it’ ‘I am ripping it’

②	Root-to-Affix	Root-to-Prefix	←	Length	Vowel Length
---	---------------	----------------	---	--------	--------------

Other Cherokee verb roots condition lengthening of a preceding pronominal prefix (121-a). (121-b) is an example for a non-lengthening verb root:

(121) *Cherokee root-triggered vowel lengthening (Scancarelli 1987:56+57)*

- a. kà:-ɔ́jò:híh-a b. kà-lò:nàhě:-ʔa
 1SG-prick-PRS 3SG-cheat-PRS
 ‘he is pricking it’ ‘he is cheating’

C.14 Chichewa – Africa – Niger-Congo (Bantu) [chew1246] (4 patterns)

①	Root-to-Affix ~ =	Root-to-Suffix ~ =	→	Tone	High
②	Tautomorphemic Root	Tautomorphemic Root	↪		

The Bantu language Chichewa exhibits an interesting length- and position-dependent variant of parallel tauto- and heteromorphemic mutation similar to the Anywa case in appendix C.6. The language has two tonal verb classes dubbed ‘H-toned’ and ‘toneless’ by Downing & Mtenje (2017). Abstracting away from the possible interference of tonally specified affixes, toneless verbs are low-toned throughout, whereas the high-toned class imposes a H on the ‘final vowel’ (122) – an obligatory final suffix in verb forms.⁷ While this describes the phrase-internal forms of verbs, two general phonological processes affect the position of this H for verb tones in isolation (which are hence at the end of a minimal prosodic phrase): lengthening of phrase penultimate vowels (indicated here by vowel doubling) and ‘retraction’ – spreading of a H on the phrase-final mora to the preceding one, i.e., here the second mora of the lengthened vowel. In effect, isolated verb forms of longer H-roots show pure anticyclic Root-to-Affix mutation (interestingly even across tone-less derivational suffixes such as Applicative /-il/ and Causative /-its/), whereas for shorter roots without suffixes such as /péz/, the H surfaces both on the root itself and the final vowel /-a/, as can be seen in (122).

(122) *Chichewa Imperatives (Downing & Mtenje 2017:139)*

Ø-Root		Phrase internally		H-Root		Phrase internally	
a. mèèj-à	‘hit!’	mèj-à	√-FV	<u>pèéz-á</u>	‘find!’	pèz- <u>á</u>	√-FV
b. jàṅààn-à	‘look!’	jàṅ-à	√-FV	<u>nàmííz-á</u>	‘deceive!’	nàmìz- <u>á</u>	√-FV
c. fòtòkòòz-à	‘explain!’	fòtòkòz-à	√-FV	<u>xùlùlùk-ìíí-á</u>	‘pardon!’	<u>xùlùlùk-il-á</u>	√-APPL-FV
d. tèmbènùùz-à	‘turn around!’	tèmbènùz-à	√-FV	<u>tàmàng-ìííts-á</u>	‘chase!’	<u>tàmàng-ìts-á</u>	√-CAUS-FV

③	Affix-to-Affix	Suffix-to-Suffix	→	Tone	High
---	----------------	------------------	---	------	------

The same type of alternation is not only found for roots but also for affixes. For example, the intensive suffix /-íts-/ has basically the same effect on the final vowel as a high-toned verb root. Thus the Intensive form of [jàṅààn-

⁷See Hyman & Mtenje (1999) for an explicit analysis where the H introduced by verb roots is underlyingly floating. This straightforwardly distinguishes this tone from preassociated tones which don’t shift to the final vowel, as in nouns (e.g., [címààngà]/*[címààngá] ‘maize’ and [kàtùndù]/*[kàtùndú] ‘luggage’, Downing & Mtenje 2017:117), and verbal prefixes such as the Simple Past marker /-ná/ in [mù-ná-mèèj-a] S2PL-TNS-hit-FV ‘you hit’ (Downing & Mtenje 2017:145). See Trommer (2024) for an explanation of the fact that some floating H tones also target the penultimate instead of the final vowel.

à] look-INT-FV in (122-b) is [jàŋààn-ìts-á] phrase-internally and [jàŋààn-ìts-á] phrase-finally (Downing & Mtenje 2017:141). See South Kyungsang Korean C.38 for a similar system with more pitch-accent-like properties.

④

Affix-to-Affix	Prefix-to-Prefix	←	Tone	High
----------------	------------------	---	------	------

The Remote Past prefix which itself has a consistent L enforces a H on preceding subject agreement affixes. In (123), this is contrasted with the Simple Past where the agreement prefixes have L, and the Recent Past which has a low-toned tense prefix but (cyclically) introduces a H on the following stem vowel.

(123) *Chichewa prefix-to-prefix mutation (Remote Past) (Downing & Mtenje 2017:159)*

Remote Past	Simple Past	Recent Past	
<u>ndí</u> - <u>naa</u> -jaŋán-a [ndínáajajána]	ndi- <u>ná</u> -jaŋan-a [ndinájajana]	ndi- <u>na</u> - <u>já</u> ŋan-a [ndinajájaana]	S1SG-TNS-look.at-FV 'I looked at'
<u>á</u> - <u>naa</u> -fotokóz-a [ánáfotokóza]	a- <u>ná</u> -fotokoz-a [anáfótokooza]	a- <u>na</u> - <u>fót</u> okoz-a [anafótókooza]	S3PL-TNS-explain-FV 'they explained'
<u>tí</u> - <u>naa</u> -tembenúz-a [tínáatembenúza]	ti- <u>ná</u> -tembenuz-a [tinátémbenuza]	— —	S1PL-TNS-explain-FV 'we explained'

C.15 Chimila – South America – Chibchan [cbg] (1 pattern)

①

Root-to-Affix	Root-to-Suffix	→	Consonantal Features	nasal	↵ Malone (2010)
---------------	----------------	---	----------------------	-------	-----------------

Specific roots trigger nasalization of initial stop consonants in adjacent suffixes such as the suffix /-ta/ 'towards/up', as shown in (124).

(124) *Chimila Root-to-Affix nasalization (Malone 2010:11-12)*

Plain Roots	Nasalizing Roots
a. ūú- <u>ta</u> 'to sit up'	<u>ú</u> - <u>n</u> -à 'drink toward'
b. há- <u>ta</u> 'to come'	<u>ʔa</u> - <u>n</u> -à 'to return'
c. moo- <u>ta</u> 'to carry upslope'	<u>kúū</u> - <u>n</u> -à 'milk toward'
d. ká- <u>ta</u> 'to open toward'	<u>tó</u> - <u>n</u> -à 'to shuck corn'

Nasalizing roots also seem to systematically affect other suffixes which are susceptible to nasalization. For example, the plural suffix /-ke/, which shows up faithfully after plain roots (cf. [ūú-ta] 'to sit up' (124-a) and [ū-ke-wáŋa?] sit-PL-MPPL-up), roots nasalizing /-ta/ also nasalize /-ke/ (e.g., [hí-nā-kri] go-up-elevated 'to go up a slope' and [hí-ŋē-tá-kri] go-PL-up-elevated 'let's go up!').

At the same time, nasalization is codependent on the identity of the suffix, i.e., there are suffixes which consistently resist nasalization by nasalizing roots such as the stem formative /-ko/-ku/ (e.g., [ú-n-à] 'drink toward' (124-a) vs. [u-k-u] 'drink' and [ⁿdi-na] 'remove hair from goat carcass' vs. [ⁿdi-ko-ta] 'take the hair off!', p.15. The last example also shows that nasalization of /-ta/ is blocked if other suffixes intervene).

Malone captures the double conditioning of nasalization by the assumption that affixes may be specified or unspecified for nasality and nasalization is triggered by floating phonological material. Malone also provides detailed arguments that this pattern is not allomorphy (based, e.g., on the fact that nasalizing roots trigger the effect on all affixes which are in principle eligible for nasalization).

C.16 Chukchee – Eurasia – Chukotko-Kamchatkan [ckt] (1 pattern)

①

Root-to-Affix	Root-to-Suffix	→	Vocalic Features	Vowel height	↵ Wolf (2005)
---------------	----------------	---	------------------	--------------	---------------

(see section 2.2 for data and discussion)

C.17 Ciyao – Niger-Congo (Bantu) [yao] (2 patterns)

①

Affix	-to-Word	Suffix	-to-Word	→	Tone	High	↔ Hyman & Ngunga (1994)
-------	----------	--------	----------	---	------	------	-------------------------

(

Affix	-to-Root	Prefix	-to-Root	→	Tone	High	↔ Hyman & Ngunga (1994)
-------	----------	--------	----------	---	------	------	-------------------------

)

Hyman & Ngunga (1994) is a classical study on cases of purely tonal Affix-to-Word morphology without concomitant segmental affixes in the Bantu language Ciyao. Ciyao verbs have cyclic as well as anticyclic TAM inflection whose docking seems to be governed mainly by directionality. Tonal prefixes associate to the left edge of the verb stem (e.g., the LH in the Negative Remote Past (P2) tense /nga-ni-ju-_L _H-telek-a NEG-TNS-S3SG-TNS-forget-FV → [nga-ni-ju-_L tèlék-a] ‘he didn’t forget’, p.29), whereas the (suffixal) H of the Negative Imperative consistently shows up on a following word as shown in (125) for the noun /cíló/ ‘night’.⁸

(125) *Ciyao Negative Imperative (Hyman & Ngunga 1994:43)*

Bare Verb	Verb + noun		
a. n-ká-mìl-à	n-ká-mìl-à__	<u>cíló</u>	‘don’t swallow (at night)!’
b. n-ká-kàmùl-à	n-ká-kàmùl-à__	<u>cíló</u>	‘don’t hold (at night)!’
c. n-ká-dìvədìl-à	n-ká-dìvədìl-à__	<u>cíló</u>	‘don’t forget (at night)!’
IMP-NEG-V-FV	IMP-NEG-V-FV	night	

②

(Affix)	-to-Root	(Suffix)	-to-Root	↔	Tone	High	↔ Hyman & Ngunga (1994)
~		~		~			
Affix	-to-Word	Suffix	-to-Word	→			

C.18 Finnish – Eurasia – Uralic [fin] (3 patterns)

①

Root-to-Affix	Root-to-Suffix	→	Length	consonant gemination	↔ Kiparsky (2003)
~	~	~			
Root-to-Word	Root-to-Word	→			

(data are given in Finnish orthography, apart from segmental length which we mark by the IPA symbol ‘:’)

In Finnish, specific nouns such as *li:ke* ‘movement, shop’ have a final ‘ghost consonant’, i.e., they trigger gemination of following consonant-initial affixes, clitics (e.g., /li:ke/ + /-tä/ ‘part:sg’ → *li:ke-tä* ‘of a shop’ and /li:ke/ + /-ko/kö/ ‘(question enclitic)’ → *li:kek:ö* ‘the shop?’, Kiparsky 2003:119), and independent words (e.g., /li:ke/ + /menestyi/ ‘flourished’ → *li:ke m:enestyi* ‘the shop flourished’, Kiparsky 2003:119).

③

Affix-to-Word	Suffix-to-Word	→	Length	consonant gemination	↔ Kiparsky (2003)
---------------	----------------	---	--------	----------------------	-------------------

The Finnish ghost consonant triggering gemination on following consonants is also found on suffixes. An example is the Infinitive *-ta* which results in initial gemination on following words (126-a+b). As can be seen in the examples (126-c+d), the suffix has no effect if it does not precede a consonant.

⁸The affirmative P2 in Ciyao has a pattern where verb forms have a H consistently associated to the final vowel, e.g., [jw-à: cí-tèm-àg-á] TNS-S3SG-TNS-break-IPFV-FV ‘he was breaking it’, p.35 which seems to contradict this generalization. Since the final vowel itself is an exponent of TAM, we assume that this H is preassociated to the final vowel, and hence not subject to the constraints governing association of floating tones. See Hyman & Ngunga (1994) for controversial discussion on the morphotonological structure of this paradigm.

(126) *Finnish Affix-to-Word gemination (Kiparsky 2003:119)*

- a. ei se tuo-tta^C voi-ta → ei se tuo-ta ^[v:]oita
 Aux.Neg.3sg it produce-Inf butter-PRT ‘it does not produce butter’
- b. ei se voi-tta^C tuo-ta → ei se voi-ta ^[t:]uota
 Aux.Neg.3sg it beat-Inf that-PRT ‘it won’t beat that one’
- c. ei se voi-ta tuo-tta^C → ei se voita tuota
 Aux.Neg.3sg it butter-PRT produce-INF ‘it does not produce *butter*’
- d. ei se tuo-ta voi-tta^C → ei se tuota voita
 Aux.Neg.3sg it that-PRT beat-INF ‘it won’t beat *that one*’

C.19 Fula Africa – Niger-Congo (Central Atlantic) [ful] (2 patterns)

Root-to-Affix	Root-to-Suffix	→	Consonantal Features	① continuancy ② nasalization	↔ Paradis (1992)
---------------	----------------	---	----------------------	---------------------------------	------------------

Fula is well-known for its consonant gradation or mutation pattern that is prominent in the class marking for nouns (Paradis 1992). As can be seen in (127), the initial consonants of the nouns alternate depending on the noun class and are realized as either stops or nasals. Crucially, the initial consonant of the following noun class suffix alternates as well and is realized as stop, nasal, continuant, or zero. This choice of consonant grade for the suffix depends idiosyncratically on the stems preceding them and cannot be predicted from any additional semantic or morphological factor. We can hence conclude that the nouns in (127) trigger an anticyclic consonantal mutation on the class suffix directly following them, in parallel to the Korean data discussed in section 2.2.

(127) *Alternating noun class suffixes in Fula (Churma 1988, 40)*

wor-	wa:-	hufine-	da:g-
‘man’	‘monkey’	‘cap’	‘sleeping mat’
<i>stop</i>	<i>nasal</i>	<i>continuant</i>	<i>zero</i>
gor- ^{cls.3}	ba:- ^{cls.3}	kufine- ^{cls.3}	da:g-el ^{cls.3}
^[g] gel	^[ŋ] ngel	^[j] jel	
gor- ^{cls.5}	ba:- ^{cls.5}	kufine- ^{cls.5}	da:g-um ^{cls.5}
^[g] gum	^[ŋ] ngum	^[j] jum	
ngor- ^{cls.7}	mba:- ^{cls.7}	kufine- ^{cls.7}	nda:g-a ^{cls.7}
^[g] ga	^[ŋ] nga	^[w] wa	
gor- ^{cls.1}	wa:- ^{cls.11}	hufine- ^{cls.9}	da:g-o ^{cls.14}
^[d] dó	^[n] ndu	^[r] re	

C.20 Gã – Africa – Niger-Congo (Kwa) [gaa] (2 patterns)

Affix-to-Affix	Prefix-to-Prefix	←	Tone	① High ② Low	↔ Paster (2003)
----------------	------------------	---	------	-----------------	-----------------

((Affix)-to-Root (Suffix)-to-Root ← Tone High ↔ Paster (2003))

(see section 2.3 for data and discussion)

C.21 Gaahmg – Africa – Eastern Sudanic [tbi] (1 pattern)

① (Affix)-to-Affix (Suffix)-to-Suffix → Tone Mid

The Eastern Sudanic language Gaahmg (Stirtz 2011) has tonal Affix-to-Affix mutation similar to the Gã case discussed in section 2.3, but more restricted in scope. The Continuous suffix /-án/ has a consistent H (e.g., /fír/ ‘smell’ + /án/ + ^(M) ‘1sg’ → [fírón] ‘I smell’, and /fír/ ‘smell’ + /án/ + ^(L) ‘3pl’ → [fír-^(H)ón] ‘they smell’, p.199). The Past form of the Continuous carries an additional Mid tone *preceding* the H of Continuous /-án/. (e.g., /fír/ ‘smell’ + ^(M) ‘Past’ + /án/ + ^(M) ‘1sg’ → [fírón] ‘I smelled’, and /fír/ ‘smell’ + ^(M) ‘Past’ + /án/ + ^(L) ‘3pl’ → [fír-^(M)ón] ‘they smelled’, p.196). If this M is not a tonal infix (a phenomenon, to our knowledge, not attested otherwise), this is again anticyclic inside-out mutation.

C.22 Gadsup – Papuanesia – Nuclear New Guinea [gaj] (1 pattern)

①

Root-to-Word	→	Tone	High-Low
--------------	---	------	----------

In Gadsup, specific modifiers trigger a change in the tone pattern of following nouns (Cahill 2011:18, see also Frantz & Frantz 1973). Cahill describes this as an initial High and concomitant simplification of underlying contour tones, but the data he provides exhibit more specifically a H⁺ L pattern, as can be seen in (128).

(128) *Gadsup Root-to-Word mutation (Cahill 2011:18)*

Word in Isolation	After mutation trigger
nòmí ‘water’	éná nómì ‘another water’
ùnámì ‘bag’	àmú?nà únámì ‘many bags’
jòkànì ‘work’	nùrànkéná jókànì ‘yesterday’s work’
kùkùnì ‘fence’	kàmàn kúkùnì ‘sweet potato garden (lit. fence)’

Other prenominal modifiers don’t trigger tonal changes on nouns (e.g., [tǐǎ nkàn kùkùnì] ‘ten fences’).

C.23 German – Eurasia – Indo-European (Germanic) [deu] (3 patterns)

①

(Affix)-to-Root ~	(Suffix)-to-Root ~	← ~ ↺	Vocalic Features) ~	Coronal	↔ Trommer (2021)
Tautomorphemic Affix	Tautomorphemic Suffix		Consonantal Features		

As shown in detail by Trommer (2021), Standard German has a complex pattern, where noun plurals are expressed by affixing an underspecified segment and a coronal feature. The segment may surface either as a mid central vowel ([ma:t] ~ [ma:tə] ‘steward(s)'), a central low vowel ([bɪlt] ~ [bɪldə] ‘picture(s)’) or as a coronal nasal consonant ([p^ha:tə] ~ [p^ha:tən] ‘godfather(s)'). The coronal feature, on the other hand, triggers umlaut (vowel fronting) when possible ([ʀat] ~ [ʀədə] ‘wheel(s)'). Now, whereas both the segmental shape and umlaut exhibit a complex distribution due to morphological and phonological factors, a central restriction of the system is that the [n]-realization of the segmental suffix never cooccurs with umlaut (thus monosyllabic feminine nouns might show [n]-affixation ([za:t] ~ [za:tɳ] ‘seed(s)’) or umlaut ([na:t] ~ [nɛ:tə] ‘seam(s)’) but not both. There are hence no alternations such as *[la:t] ~ [lɛ:tɳ]. This fact follows naturally if the [n]-allomorph can only be formed with the tautomorphemic CORONAL feature, unavailable whenever this docks to the base vowel in umlaut.

②

Root-to-Affix	Root-to-Suffix	→	Vocalic Features	Pharyngeal	↔ Trommer (2021)
---------------	----------------	---	------------------	------------	------------------

③

Root-to-Affix	Root-to-Suffix	→	Consonantal Features	Nasal	↔ Trommer (2021)
---------------	----------------	---	----------------------	-------	------------------

In certain cases, the specific segmental realization of plural also result from floating root material docking anti-cyclically to outer affixes even in the absence of alternation with mutation. This is found in a number of noun roots which idiosyncratically select the ‘wrong’ allomorph. Thus the expected plural realization for a monosyllabic masculine noun is suffixed [ə] ([ma:t] ~ [ma:tə] ‘steward(s)'). However there are exceptional roots which take [ɐ] and [n] instead (e.g., [gɔt] ~ [gœtɐ] ‘god(s)’ and [gra:f] ~ [gra:fɳ] ‘count(s)'). Trommer (2021) shows that this can be naturally derived by positing specific floating features (nasal for [n], and pharyngeal for [ɐ]) which are part of the lexical specification of these nouns and associate to the featurally underspecified segment of the plural affix, as shown in (129) (abstracting away from the [Coronal] feature of the affix triggering cyclic mutation on the root):

(129) a. /gra:f^{nas} + ●/ → [gra:f-n]
b. /gɔt^{phar} + ●/ → [gœt-ɐ]

C.24 Greek – Eurasia – Indoeuropean [ell] (2 patterns)

①	Root-to-Affix	Root-to-Suffix	→	Stress	↔ Revithiadou (1999)
---	---------------	----------------	---	--------	----------------------

(see section 2.2 for data and discussion)

②	Affix-to-Affix	Suffix-to-Suffix	→	Stress	↔ Revithiadou (1999)
---	----------------	------------------	---	--------	----------------------

(see section 2.3 for data and discussion)

C.25 Guébie – Africa – Niger-Congo (Kru) [dic] (2 patterns)

①	Root-to-Word	Proclitic-to-Word	→	Tone	Lowering	↔ Trommer (to appear)
②	Root-to-Word	Enclitic-to-Word	←	Tone	Raising	

(see Sande 2018 for an alternative analysis in terms of construction-specific phonology)

A case of alternating different word-target mutation outside of the DP domain is identified in work by Sande (2018) in Guébie, a language with an underlying 4-tone system (in Sande's notation, 4 is the highest and 1 the lowest tone). Most tenses of the language are expressed by second-position auxiliaries, with the verb in clause-final position, as can be seen in (130).

(130) *Guébie Future and Irrealis auxiliaries* (Sande 2018:259)

a.	e ⁴	ji ³	ja ³¹	li ³	b.	e ⁴	ka ³	ja ³¹	li ²
	1SG.NOM	FUT	COCONUTS	eat		1SG.NOM	IRR	COCONUT	eat
	'I will eat coconuts'					'I would eat coconuts'			

In perfective and imperfective forms, there is no segmental auxiliary and the verb appears in the V2 position. While the Perfective is thus effectively zero-marked (131-a), the imperfective is realized as a complex tonal modification (131-b). For most inputs the initial tone of the verb undergoes a chain shift where every tone is lowered by one degree (4 → 3, 3 → 2, 2 → 1).

(131) *Guébie: Imperfective verb tone lowering* (Sande 2018:260)

a.	e ⁴	li ³	ja ³ -6ə ¹	b.	e ⁴	—	li ²	ja ³ -6ə ¹
	1SG.NOM	PFV	eat		1SG.NOM	IPFV	eat	coconuts-SG
	'I ate a coconut'				'I am eating a coconut'			

The only exception are verbs with the lowest tone, 1, where – apparently to avoid a superlow tone, there is instead raising of the final tone in the preceding subject (132).

(132) *Guébie: Imperfective subject tone raising* (Sande 2018:263)

a.	ja ^{23.1}	pa ¹	b.	ja ^{23.2}	—	pa ¹
	Djatchi	PFV		Djatchi	IPFV	run
	‘Djatchi ran’					‘Djatchi runs’

Just as verb lowering, subject raising is chain-shifting. Every tone is replaced by the next-higher tone in the inventory, summarized in (133). A natural interpretation of this pattern worked out in Trommer (to appear) is that the imperfective auxiliary consists only of tonal register features. These are realized on the following or preceding tonal root node, dependent on phonological well-formedness, in the latter case a clear instance of anticyclic mutation.

(133) *Guébie tonal chain shifts* (Sande 2018:260+261)

a. Verb tone lowering		b. Subject tone raising	
Underlying tone	Imperfective tone	Underlying tone	Imperfective tone
4	3	4	5
3	2	3	4
2	1	2	3
1	1	1	2

C.26 Hungarian – Eurasia – Uralic – [hun] (1 pattern)

①	Root-to-Affix	Root-to-Suffix	→	Vocalic Features	low	↔	Abrusan (2005), Siptár & Törkenczy (2000)
---	---------------	----------------	---	------------------	-----	---	---

Hungarian has vocalic [+low] mutation. Noun stems in the language can be classified into different types that trigger idiosyncratic changes on the suffixes that may follow them (Vago 1980, Wunderlich & Stiebels 1999). The two classes of interest are the so-called shortening and lowering stems (Abrusan 2005) that both cause an otherwise mid suffix vowel to be low⁹, shown in (134-a+b). (134-c) shows stems that are phonotactically quite similar to those in (134-a+b) but do not trigger any change: this vowel lowering is hence an idiosyncratic feature of certain stems and not predictable from general phonological restrictions.

(134) *Stems triggering lowering in Hungarian (Abrusan 2005, 1+2)*

Plural			
a. Shortening stems			
nyar	'summer'	nyar-	ak
madar	'bird'	madar-	ak
b. Lowering stems			
ház	'house'	ház-	ak
fog	'tooth'	fog-	ak
c. Non-shortening/non-lowering stems			
par	'pain'	par-	ok
taná	'teacher'	taná-	ok
jog	'law'	jog-	ok

C.27 Igbo – Africa – Niger-Congo [ibo] (4 patterns)

①	Root-to-Word	Proclitic-to-Word	→	~ =	Tone	High	(Associative Marking)
②	Root-to-Word	Enclitic-to-Word	←	~ =			

(see section 2.5 for data and discussion)

③	Root-to-Word	Proclitic-to-Word	→	~ =	Tone	Low	(Question Lowering)
④	Root-to-Word	Enclitic-to-Word	←	~ =			

The second case of alternating tone is similar to the Igbo patterns ①+②, but applies at the clausal level: Question formation lowers the initial H of a verb (135-a), the immediately preceding subject (135-b), or both (135-c,d,e). Again this pattern includes not only pronouns (135-b), but also full noun phrases.

⁹The shortening stems additionally undergo shortening of their final long vowel before certain suffixes.

(135) *Igbo tonal question marking (Clark 1990:48-49,175)*

- a. ó-mèchìrì ánjá → — ò-mèchìrì ánjá
3sg-closed eyes ‘Did (s)he close his/her eyes?’
- b. há mèchìrì ánjá → hà — mèchìrì ánjá
they closed eyes ‘Did they close their eyes?’
- c. ófé ò-dì ù-tó → òfé — ò-dì ù-tó
soup 3sg-is good.tasting ‘Is the soup good?’
- d. ùwé óchá ò-dì né-bé à → ùwé òchò — ò-dì né-bé à
shirt clean 3sg-is in.place this ‘Is there a clean shirt here?’
- e. ómù tàkírì há ázàálá á⁺má → ómù tàkírì há — àzàálá á⁺má
children they have.swept yard.ASS ‘Have the children swept the yard?’

The crucial datum is (135-e). It shows that questions trigger not simply lowering of complete noun phrases, but of the final H-span in a noun phrase (here: *kírì*). The initial *ó* is immune to lowering because it is separated from this High-span by the sequence *mù tà*, which is underlyingly (and on the surface) Low, and thus interrupts this span.

C.28 Ik – Africa – Eastern Sudanic [ikx] (2 patterns)

- ① Affix-to-Affix Suffix-to-Suffix → Vocalic Features [+ATR] ↔ Schrock (2014)

(see section 2.3 for data and discussion)

- ② Root-to-Affix Root-to-Suffix → Vocalic Features [+ATR] ↔ Schrock (2014)

In addition to pattern ①, suffixes triggering anticyclic [+ATR] mutation, Ik also exhibits this phenomenon for roots. Thus whereas many roots with [-ATR] [a] trigger vowel harmony on following suffixes such as Intransitive Infinitive [-ɔn/-on] and Transitive Infinitive [-és/-és] (136-a), a subset triggers changing the affix vowel to [+ATR] (136-b). We give here only examples for verbs, but Schrock provides similar data for nouns.

(136) *Ik Root-to-affix [+ATR] mutation (Schrock 2014:85)*

a. Harmonic a-verbs: [-ATR] suffixes

- áts' → áts'-és 'to gnaw'
kán → kán-és 'to lick'
tam → tam-és 'to think'
bar → bar-ɔn 'to be rich'
ɲwaf → ɲwaf-ɔn 'to be lame'

b. Mutating a-verbs: [+ATR] suffixes

- tsáj → tsáj-és 'to smear'
táb → táb-és 'to touch'
raf → raf-es 'to return'
ák'áf → ák'áf-on 'to yawn'
ats → ats-on 'to come'

C.29 Italian – Eurasia – Indo-European (Romance) [ita] (1 pattern)

- ① Root-to-Word → Length consonant gemination ↔ Chierchia (1982), Passino (2013)

(see section 2.5 for data and discussion)

C.30 Izon – Africa – Ijoid [ijc] (3 patterns)

Root-to-Word	→	Tone	① High ② Low High ③ Low
--------------	---	------	-------------------------------

(see section 2.5 for data and discussion)

C.31 Kashaya – N America – Hokan (Pomoan) [kju] (2 patterns)

① Root-to-Word → Stress

Postaccentuation across word boundaries is diagnosed for the Pomoan language Kashaya by Buckley (2018). Kashaya has regular iambic stress with initial syllable extrametricality spanning phonological phrases as in (137-a+b) (only the head of the first foot is phonetically marked by pitch accent). Specific roots such as /k'abat/ 'madrone' trigger postaccentuation, i.e., shifting of the pitch accent to the following foot (137-c+d).

(137) *Kashaya postaccentuation* (Buckley 2018:180)

- a. bihfə 'deer' + boʔoʔ-k^he 'will hunt' → <bih>(fəbó)(ʔoʔ)k^he 'will hunt deer'
b. bihfə 'deer' + hc^hojic'-ʔ 'died' → <bih>(fəh)(c^hojiʔ) 'the deer died'
c. k'abat 'madrone' + fihp^ha 'leaf' → <k'a>(baʔ)(fih)p^ha 'madrone leaf'
d. k'abat 'madrone' + q^hale 'tree' → <k'a>(baʔ)(q^halé) 'madrone tree'

②	Affix-to-Root ~ = Affix-to-Affix	Suffix-to-Root ~ = Suffix-to-Suffix	← ~ = →	Consonantal	palatalization
---	--	---	---------------	-------------	----------------

Kashaya also has a segmental anticyclic mutation pattern. The language expresses plural agents by palatalization which applies to the segment [d] in suffixes (138-a) and final root consonants (138-b). Mutation also can affect multiple suffix consonants (138-c):¹⁰

(138) *Plural agent palatalization in Kashaya* (Buckley 1994:140-141 based on Oswalt 1961:153,186)

- a. p^hil-___[ač]-mé=ʔ
p^hila-Ø-ad-me=ʔ
come-PL-here-IMP=PL
'come here!'
b. du-hlu-___[č-ač]-
du-^hlud-Ø-ad-
INSTR-pick-PL-DUR
'keep picking'
c. du-hlu-___[č-ač-č-č]-
du-^hlud-Ø-ad-ad-
INSTR-pick-PL-DIR-DUR
'keep picking while moving'

C.32 Ket – Eurasia – Yeniseian – [ket] (1 pattern)

① Affix-to-Affix Prefix-to-Prefix ← Vocalic Features round

In Ket (Vajda 2001), past tense prefixes trigger labialization (rounding) of preceding low-vowel prefixes to [o] (139-b,d). (139-a,c) show the corresponding present forms, where the underlying /a/ surfaces. The affected prefix-

¹⁰Glosses are provided by us based on the detailed prose description by Buckley. We have also added a line with underlying forms. Following Buckley (2023), we posit /d/, not /n/ as the underlying phoneme undergoing palatalization.

/a/ is an agreement affix in (139-a,b), and a prefixal stem formative in (139-c,d) (glossed as ‘L’ following Vajda). Epenthetic segments ([j],[i], [ɣ]) are glossed ‘Ø’ (see Vajda’s paper on the intricate morphophonological conditioning of these elements).

(139) *Ket Labialization* (Vajda 2001:380)

- | | |
|---|---|
| a. d-a-j-s-i-n
1-3-Ø-dress-Ø-PL.AN
‘we dress him’ | b. d- o -l-s-i-n
1-3-PST-dress-Ø-PL.AN
‘we dressed him’ |
| c. di-ɣ-a-daɣ
1-Ø-L-live
‘I live’ | d. d- o -l-daɣ
1-L-PST-live
‘I lived’ |

C.33 Kikuyu – Africa – Niger-Congo (Bantu) [kik] (1 pattern)

①

Root-to-Word	→	Tone	High
--------------	---	------	------

 ↵ Clements & Ford (1981), Clements (1984)

Besides the basic patterns of Root-to-Word tone described in section 2.5, Kikuyu also provides a striking demonstration that anticyclic mutation might have rather differentiated and global effects. If the last syllable of a floating-L noun has a H and is followed by a low-toned syllable, the language doesn’t simply downstep the latter (plausibly to avoid a maximal phonetic pitch fall from a H to a downstepped L). Instead the L and all immediately following L’s are raised to H, and the downstep is realized on the first following high-toned syllable even if this involves multiple words (140-c,d).

(140) *Kikuyu raising + downstep* (Clements & Ford 1981:321+322)

- | | | |
|--|--|--------------------------------|
| a. kàŋɛrí L òníré
Kangeri saw | → kàŋɛrí ɔ́níré | ‘Kangeri saw’ |
| b. mwàɣáhìjá L àrètáhá
weakling will.draw | → mwàɣáhìjá áré⁺táhá | ‘a weakling will draw (water)’ |
| c. mòcé:ɣèní L nà kèŋàŋí
‘blackjack and crocodile’ | → mòcé:ɣèní ná kéŋá⁺ŋí | ‘blackjack and crocodile’ |
| d. mwàɣáhìjá L nà mòɣèràŋjá
‘weakling and examiner’ | → mwàɣáhìjá ná móɣérán⁺já | ‘weakling and examiner’ |

If all following syllables are Low, these are raised without an overt downstep (141). However, Clements & Ford argue that the displaced L is still at the right edge of the H-span since words with non-downstepped final H in Kikuyu which are not followed by a floating tone are regularly lowered (e.g., ‘examiner’ with an underlying final H surfacing in [mòɣèràŋjá ná mò:ŋgàì] ‘examiner and Mungai’ which is lowered if pronounced in isolation: [mòɣèràŋjá], Clements & Ford 1981:321+322).

(141) *Kikuyu raising without downstep* (Clements & Ford 1981:315+333)

- | | | |
|--|--|---------------------------------------|
| a. áhèiré mwàɣáhìjá L ßiríßirí
he.gave weakling chillies | → áhèiré mwàɣáhìjá ßiríßirí (L) | ‘he gave the weakling chillies’ |
| b. áhèiré mwàɣáhìjá L ìrìyò ìrìtò
he.gave weakling banana heavy | → áhèiré mwàɣáhìjá íríyó írító (L) | ‘he gave the weakling a heavy banana’ |

C.34 Kinande – Africa – Niger-Congo (Bantu) [nnb] (1 pattern)

Root-to-Affix	Root-to-Prefix	←	Tone	High
---------------	----------------	---	------	------

 ↵ Black (1995)

(see section 2.2 for data and discussion)

C.35 Kipsigis – Africa – Eastern Sudanic (Southern Nilotic) [sgc] (1 pattern)

①	Root-to-Affix	Root-to-Prefix	←	Length	vowel lengthening	↔ Kouneli (2022)
---	---------------	----------------	---	--------	-------------------	------------------

(see section 2.2 for data and discussion)

②	Affix-to-Affix	Prefix-to-Prefix	←	Length	vowel lengthening	↔ Kouneli (2022)
---	----------------	------------------	---	--------	-------------------	------------------

(see section 2.3 for data and discussion)

Note that Kouneli interprets both patterns ① and ② as reflexes of a mora realizing the little v-head assumed as a general verbal category marker in Distributed Morphology. For Kouneli, this is structurally closer to the root in the cases in ① without transparent semantics, and merged outside of a second little v with predictable causative meaning in the cases in ②. We believe that our interpretation of the data, where the intransparent combination v+root is understood as a single root morpheme, is the closest possible translation of Kouneli's DM-analysis into the less abstract terms of a lexicalist approach.

C.36 Kisi – Africa – Niger-Congo [kss] (1 pattern)

①	Affix-to-Word ~	Suffix-to-Word ~	→	Tone	High
	Affix-to-Root	Suffix-to-Root	←		

(see section 3.2 for data and discussion)

C.37 Kɔnnɪ – Africa – Niger-Congo (Gur) [kma](4 patterns)

①	Root-to-Affix	Root-to-Suffix	→	Tone	↓High	(142-c)
---	---------------	----------------	---	------	-------	---------

Root-to-Affix	Root-to-Suffix	→	Tone	② ↓High ~ ③ Low	(142-d)
---------------	----------------	---	------	-----------------------	---------

The Gur language Kɔnnɪ provides a case where register and full-tone mutation form a partial contrast with two classes of lowering nouns, shown here with the high tone suffix /-ká/ and the tonally variable (underspecified) suffix /-sɪ/. With non-mutating noun roots (142-a,b), /-ká/ remains unchanged, and /-sɪ/ adopts the H or L root tone. The first type of lowering roots (142-c) leads for both suffixes to a downstepped H, the second class downsteps a H-toned suffix but imposes a simple L on a tonally variable one.

(142) 2 Types of root-final floating tones in Kɔnnɪ (Cahill 1999:340-41)

	Singular Definite	Plural	
a. Low	jùlì-ká	jùlì-sì	'whydah' (bird)
b. High	kpá-ká	kpá-sí	'chicken'
c. High+ ①	cí- +ká	cí- +sí	'squirrel'
d. High+ ②	líá- +ká	líá- sì	'axe'

A natural interpretation of this difference in terms of tonal features is that the root in (142-c) carries a low-register feature which modifies the underlying H of suffixes like /-ká/ and triggers epenthesis of a H for underspecified suffixes like /-sɪ/ since only H's can carry downstep in Kɔnnɪ. If a root as in (142-d) carries a full L instead, this explains immediately the L on /-sɪ/. Under the standard feature-geometric assumption that the l-register feature is part of the representation of a full L (Yip 1989, 2002, Snider 2018), the downstep on /-ká/ can then be understood as the partial residue of this tone. Note finally that besides the lowering nouns there are also raising nouns in a different inflectional class argued by Cahill to confer a floating H. Thus the plural suffix /-a/ is low-toned after most

high-toned nouns (e.g., /múg-ìŋ/ ‘river’ → [múg-à] ‘rivers’), but raising nouns exhibit a Fall (/ǵág-íŋ/ ‘shade’ → [ǵág-á] ‘shades’ (Cahill 2004:7).

④

Root-to-Word	←	Tone	High	↓
--------------	---	------	------	---

Kɔnnɪ also exhibits a minor pattern of tonal root-to word mutation. 1sg, 2sg, and 2pl possessive pronouns in the language usually surface with L’s (143-a). A small set of kinship nouns, however, imposes a H with a following downstep on the preceding pronoun (143-b).¹¹

(143) *Kɔnnɪ Possessor High + downstep* (Cahill 1999:375)

1sg	2sg	1pl	
a. ò dàáŋ	fì dàáŋ	tì dàáŋ	‘stick’
b. <u>ní</u> ná:	<u>fí</u> ná:	<u>tí</u> ná:	‘mother’

C.38 Korean – Eurasia – Korean [kor] (4 patterns)

Root-to-Affix	Root-to-Suffix	→	Consonantal Features	① aspiration ② glottalization
---------------	----------------	---	----------------------	----------------------------------

(see section 2.2 for data and discussion)

③	Root-to-Affix	Root-to-Suffix	→		
④	Tautomorphemic Root	Tautomorphemic Root	~ U	Tone	Low

South Kyungsang Korean exhibits root-to-affix tone mutation alternating with tautomorphemic tone docking on the root in a system with pitch-accent-like properties. In this dialect, both monosyllabic high-toned roots (144) and bisyllabic LH-roots (145) trigger minimally different tonal patterns on specific suffixes such as Nominative *-i* and Conditional *-imjèn*. Where the H1 and LH1 roots extend their final H to the first syllable of these suffixes, H2 and LH2 roots trigger a L. A natural assumption in a concatenativist account is that the initial TBUs of these affixes are underlyingly toneless and receive a tone either by spreading of the root-H or by association to a final floating L which is part of the lexical specification of H2 and LH2 roots. Crucially, with monosyllabic roots and a suffix such as *-mánkhùm* which doesn’t alternate (plausibly because its initial syllable is associated to a H), the L appears on the root itself.

(144) *Monosyllabic H-roots in South Kyungsang Korean* (Lee & Zhang 2014:73)

H1		H2 ①	
a. nwún	‘eye’	mún	‘door’
b. nwún-í	‘eye’ (nom.)	<u>mún</u> - <u>ì</u>	‘door (nom.)’
c. nwún-ímjèn	‘if eye’	<u>mún</u> - <u>ímjèn</u>	‘if door’
d. nwún-mánkhùm	‘as much as eye’	<u>mùn</u> -mánkhùm	‘as much as door’

(145) *Bisyllabic LH-roots in South Kyungsang Korean* (Lee & Zhang 2014:74)

LH1		LH2 ①	
a. sàlám	‘person’	pàlám	‘wind’
b. sàlám-í	‘person’ (nom.)	<u>pàlám</u> - <u>ì</u>	‘wind (nom.)’
c. sàlám-ímjèn	‘if person’	<u>pàlám</u> - <u>ímjèn</u>	‘if wind’
d. sàlám-mánkhùm	‘as much as person’	<u>pàlùm</u> -mánkhùm	‘as much as wind’

¹¹ Interestingly with other possessive pronouns, Kɔnnɪ uses suppletive forms for these kinship nouns, which don’t have a mutating effect (e.g., [ù nùŋ]/*[ú⁺ nùŋ] ‘his mother’), which shows that anticyclic mutation (floating features) are characteristic of allomorphs, not necessarily of morphemes. This is similar to cyclic mutation as in the German plural, where the plural suffix allomorph [e] triggers almost exceptionless umlaut (e.g., [ra:t] ~ [rɛ:dɐ] ‘wheel(s)’) whereas the allomorph [n] never does (e.g., [na:d] ~ [na:d|n] ‘needle(s)’).

C.39 Koryak – Eurasia – Chukotko-Kamchatkan [kpy] (4 patterns)

①	Root-to-Affix ~ =	Root-to-Prefix ~ =	→	Vocalic Features	Height	↔ Abramovitz (2021)
②	Root-to-Affix	Root-to-Suffix ←	←			

Koryak is a language with an especially pervasive vocalic mutation system both from affixes to roots and from roots to affixes, as revealed by recent fieldwork of Abramovitz (2021). Abramovitz shows stringently that Koryak has two classes of mutating affixes, where the first class lowers non-reduced mid vowels to low (146-b), while the second class adds to this effect the lowering of high to mid vowels (146-c). (146-a) shows the same root emerging faithfully with a non-trigger morpheme.

(146) *Affix-to root mutation in Koryak (Abramovitz 2021:56+57+60)*

a. No Mutation	b. Mutation 1	c. Mutation 2
[ujetikik]	[<u>u</u> jatik-piλ]	[<u>o</u> jateke-ŋ]
ujetiki-k	ujetiki-piλ	ujetiki-ŋ
sled-LOC	sled.DIM.ABS.SG	sled-DAT
‘in a sled’	‘little sled’	‘to a sled’

Abramovitz captures this difference by assigning to the first class a floating [+low] feature, and to the second class the features [-high +low] (see Abramovitz 2021 for his detailed OT analysis which also derives the chain-shift effect in pattern 2 where /e/ shifts to [a] but /i/ shifts to [e] and not further to [a]). Crucially the same classes of mutation-triggering morphemes are also found among roots, thus instantiating anticyclic Root-to-Affix mutation on prefixes and suffixes (147).

(147) *Root-to-Affix mutation in Koryak (Abramovitz 2021:60+64, 2019:5)*

a. No Mutation	b. Mutation 1	c. Mutation 2
[yečečkejɯŋlinet]	[<u>na</u> kun-kaɭičit- <u>a</u> ŋŋəɲaw]	[<u>na</u> kon-vet- <u>a</u> ŋŋəɲaw]
ye-čečkejɯŋ-line-t	ne-ku-n-kaɭičit-et-ŋ-ne-w	ne-ku-n-vet-ew-ŋ-ne-w
UW.PST-think-3.UW.PST-3DU	INV-PRS-CAUS-study-VBLZ-PRS-O3-3PL	INV-PRS-CAUS-work-3DU-VBLZ-PRS-O3-3PL
‘they two thought’	‘they are making them study’	‘they are making them work’

③	Affix-to-Affix ~ =	Suffix-to-Suffix ~ =	→	Vocalic Features	Height	↔ Abramovitz (2021)
	Affix-to-Root	Suffix-to-Root	←			

④	Affix-to-Affix ~ =	Prefix-to-Prefix ~ =	←	Vocalic Features	Height	↔ Abramovitz (2021)
	Affix-to-Root	Prefix-to-Root	→			

Both classes of lowering affixes may also trigger the same types of mutation in more peripheral affixes than they do in roots. Thus the diminutive prefix /qaj-/ in (148-b) (carrying [+low] in Abramovitz’ analysis) besides lowering the two mid vowels of the root /-čečkejɯŋ-/ also triggers lowering in the more peripheral ‘Unwitnessed’ circumfix /ye- -line/. The same outwards effects is shown for a suffix in the second lowering pattern (carrying [-high +low] under Abramovitz’ analysis).

(148) *Affix-to-Affix (+concomitant Affix-to-Root) mutation (Abramovitz 2021:60 +61)*

a. No Mutation	b. Mutation 1	c. Mutation 2
[yečečkejɯŋlinet]	[y <u>a</u> -qaj- <u>ča</u> čkajɯŋlinat]	[y <u>a</u> čačkajɯŋ-ŋəvo- <u>len</u> at]
ye-čečkejɯŋ-line-t	ye-qaj-čečkejɯŋ-line-t	ye-čečkejɯŋ-ŋvo-line-t
UW.PST-think-3.UW.P-3DU	UW.PST-DIM-think-3.UW.PST-3DU	UW.PST-think-INC-3.UW.PST-3DU
‘they two thought’	‘they two thought a bit’	‘they two began to think’

C.40 Kpelle – Africa – Mande [kpel1252] (2 patterns)

①	Root-to-Affix ~	Root-to-Suffix ~	→		
②	Root-to-Word	Root-to-Word	→	Tone	High

(see section 3.1 for data and discussion)

C.41 Kunama – Africa – isolate [kun] (2 patterns)

Root-to-Affix	Root-to-Suffix	→	Tone	① High ② Mid	↔ Ashkaba et al. (2000)
---------------	----------------	---	------	-----------------	-------------------------

Kunama in the description of Ashkaba et al. (2000) exhibits Root-to-Affix mutation for tone. Nouns in the language are directly followed either by an obligatory theme vowel (/ə/ in the singular and /-ē/ in the plural) or by a determiner suffix such as /-àm/ ‘this’ (149-a,b). Nouns like /tôm/ ‘fire’ exhibit a high tone which appears on the suffix vowel immediately following the root (149-c). As Ashkaba et al. point out, the most parsimonious account for these nouns is a final floating H-tone which is part of their lexical specification. Strikingly, there are also nouns of this class which consist only of consonants (149-d), highlighting that this H-component is likely to be underlyingly floating. Realization of this floating H is fully transparent for the singular (⊕ + M → HM) whereas tonal sandhi processes apply in the two other forms. On the plural theme vowel, the Mid component is deleted between H and L (⊕ + ML → HML → HL) since the language disallows 3-tone contours. On the demonstrative affix, the resulting HL is flattened to HM triggered by a following M (⊕ + L + M → HL + M → HM + M), another regular sandhi process (Ashkaba et al. 2000:21).

Finally, there is an additional class of vowel-less roots which appear to have a contrastive floating Mid-tone which emerges on the demonstrative suffix (149-e) but also on possessive suffixes (cf. e.g., [el-à:ŋ-ā] ‘my tree’ and [m-ā:ŋ-ā] ‘my tooth’, p.26+27).

(149) *Root-to-suffix tone mutation in Kunama (Ashkaba et al. 2000:20+21+23)*

	a. ‘land’	b. ‘water pot’	c. ‘fire’	d. ‘love’	e. ‘tooth’
<i>Sg</i>	làg-ā	āgūd-ā	tôm-ā̃	m-ā̃	m-ā̃
<i>Pl</i>	làg-ē	āgūd-ē	tôm-ē	m-ē	m-ē
<i>‘this’</i>	làg-àm-ā	āgūd-àm-ā	tôm-ā̃m-ā	m-ā̃m-ā	m-ā̃m-ā

C.42 Kuria – Africa – Niger-Congo (Bantu) [kuj] (3 patterns)

Affix-to-Root ~ =	Prefix-to-Root ~ =	→		① Low High ② Low Low High ③ Low Low Low High	↔ Trommer (2024)
Affix-to-Word	Prefix-to-Word	→	Tone		

(see section 3.2 for data and discussion, and Trommer 2024 for a detailed OT-analysis.)

C.43 Lango – Africa – Eastern Sudanic (Western Nilotic) [lno] (1 pattern)

①	Root-to-Word	→	Tone	↓
---	--------------	---	------	---

Monosyllabic H-initial nouns in Lango (Noonan 1992) show a 3-way contrast in tonal behavior. Besides H (150-a) and HL (falling) tone nouns (150-c), there is a third group which Noonan calls the H(L) class (150-b). Phrase-finally (and hence also in isolation), these nouns exhibit a simple H but triggers downstep on a following high-toned noun. Furthermore, they block H-spreading to a low-toned noun just as for HL-nouns do (150-c).

(150) *Lango monosyllabic H-initial nouns (Noonan 1992:63+64)*

	Phrase-final	+ H-noun	+ L-noun	
a. H	dóg	dóg gór	dóg ɲòŋ	‘mouth’
b. H(L)	wúm	wúm [↓] gór	wúm ɲèŋ	‘nose’
c. HL	gwók	gwók [↓] gór	gwók ɲèŋ	‘dog’
		‘ground squirrel’	‘crocodile’	

C.44 Makaa – Africa – Niger-Congo (Bantu) [mcp] (2 patterns)

① Root-to-Word	Proclitic-to-Word	→	Tone	High	↔ Cahill (2000)
② Root-to-Word	Enclitic-to-Word	←			

In Makaa, the H exponent of the possessive relation appears on the second noun if this has a prefix (151-a) (hence, presumably a prosodically weak position), and otherwise on the first noun when this ends in an open syllable (151-b).

(151) *Possessor-possessum alternation in Makaa (Cahill 2000:59, Heath 1991)*

- a. bùdè + mèn-kwì:ndjè → bùdè — mé-kwì:ndjè
 potato posts ‘potato of the posts’
- b. bùdè + jùgà → bùdè — jùgà
 potato plug ‘potato of the plug’

See Cahill (2000) for more data and a detailed analysis in terms of a floating H clitic.

C.45 Mao – Africa – Blue Nile Mao [myf] (4 patterns)

Root-to-Affix	Root-to-Suffix	→	Tone	① Low ② High
③ Root-to-Word	Root-to-Word	→		Downstep

Nouns in Mao, as described by Ahland (2012), provide a further simple tonal example for Root-to-Affix mutation. Most nominal affixes in Mao don’t carry distinctive tone but are ‘assigned’ tone by the root to which they attach. Besides nominals such as (152-a-c) where the final (or only) tone of the root-final syllable spreads to these affixes, there are other nouns for which the affix tone is not identical to the tone appearing on the root itself (152-d-g). However, it is consistent across tone-variable suffixes including the ‘thematic affix’ /-e/ which is attached to nominal roots in most contexts where they don’t carry a more specific suffix. Note that all these affixes are even segmentally maximally uniform.

(152) *Root-to-Affix mutation in Mao (Ahland 2012:133)*

	Citation Form	Subject Form	Object Form	
a. L	kàl-è	kàl-íʃ	kàl-là	‘porridge’
b. M	p’íʃ-ē	p’íʃ-íʃ	p’íʃ-nā	‘child’
c. H	íʃ-é	íʃ-íʃ	íʃ-ná	‘that’
d. H ⊕	íʃ-è	íʃ-íʃ	íʃ-nà	‘(s)he’
e. H ⊕	múnts’-è	múnts’-íʃ	múnts’-nà	‘woman’
f. M ⊕	kān-é	kān-áʃ	kān-ná	‘dog’
g. M ⊕	ēs-è	ēs-íʃ	ēs-nà	‘person’

A straightforward interpretation under a GNA approach is to posit that nouns such as /múnts’/ and /kān/ carry a floating tone at their right edge which associates to the affix syllable to provide it with a tone. Since Mao allows contour tones only under highly specific contexts (in compounding), the floating tone is otherwise deleted (or stays floating). Independent evidence for the floating tones is found in possessive constructions where the possessor nominals appear without any thematic suffix. Here, high-toned nominals with floating L trigger downstep on following high and mid tone nouns as is shown in (153-b) with the 3sg pronoun /íʃ⊕/ (152-d). It forms a minimal pair with the distal demonstrative /íʃ/ (153-c) that doesn’t carry a floating tone.¹²

¹²As shown in detail by Ahland, downstep in Mao, as in many other languages, appears only after overt H’s and is triggered by floating L elements.

(153) *Downstep triggered by floating Low (Ahland 2012:110+111)*

- a. íʃ p'íʃ-íʃ hā-jé:ts'-[↓]á b. íʃ [↓]p'íʃ-íʃ hā-jé:ts'-[↓]á
that child-SBJ AFF-RUN-DECL 3sg child-SBJ AFF-RUN-DECL
'the child ran' 'her/his child ran'

④ [Affix-to-Word | Suffix-to-Word | → || Tone | Downstep]

The Mao (Omoti) Genitive suffix /-iŋ/ triggers downstep on following words, as can be seen in (154).

(154) *Mao Genitive (Ahland 2012:328)*

- tí-ŋ [↓]māgèw ... 'my friend ...'
1SG-GEN 'friend'

C.46 Margi – Africa – Afro-Asiatic (Chadic) [mrt] (1 pattern)

① [Affix-to-Affix | Suffix-to-Suffix | → || Tone | Low]

(see section 2.3 for data and discussion, and section 4.2 for an OT-analysis)

C.47 Mayo – N America – Uto-Aztecan [mfy] (2 patterns)

①	Root-to-Affix	Root-to-Prefix	←	Stress	↔ Hagberg (2006)
②	Tautomorphemic Root	Tautomorphemic Root	↔		

While postaccentuation is a relatively familiar phenomenon from Indo-European languages, the Uto-Aztecan language Mayo (Hagberg 2006) instantiates a second slightly different type of Root-to-Affix stress mutation. In Mayo, roots are idiosyncratically specified for stress on the initial or the second syllable (155). However, this contrast is not a property of underlying accent prespecified for particular syllables but positions. This can be seen under prefixation where stress is enforced onto the resulting first and second syllable respectively, even where both of them are affixal. Following Hagberg (2006), we will assume that peninitial accent in the language is the default pattern and initial accent is the effect of mutation – under Hagberg's analysis association of floating accent gridmarks which are part of the lexical representations of specific roots (the forms in (155-f) exhibit habitual reduplication).

(155) *Mayo accent (Hagberg 2006:73)*

Accent on 1st σ (accented)			Accent on 2nd σ (unaccented)		
a.	chúpna [↓] ke	'will harvest' TRNS	ponná [↓] ke	'will play' TRNS	
b.	hí [↓] -chupna [↓] ke	'will harvest' INTR	hi-pón [↓] na [↓] ke	'will play' INTR	
c.	hí [↓] -hi-chupna [↓] ke	'will always harvest' INTR HAB	hi-hí [↓] -ponna [↓] ke	'will always play' INTR HAB	
d.	chí [↓] kna [↓] ke	'will sweep' TRNS	wisé [↓] ka	'sawing' TRNS	
e.	hí [↓] -chíkna [↓] ke	'will sweep' INTR	hi-wí [↓] seka	'sawing' INTR	
f.	hí [↓] -hi-chíkna [↓] ke	'will always sweep' INTR HAB	hi-hí [↓] -wiseka	'always sawing' INTR HAB	

C.48 Mixtec – N America – Oto-Manguean [mig] (1 pattern)

① [Root-to-Word | → | Tone | High | ↔ McKendry (2013)]

(see section 2.5 for data and discussion)

([Affix]-to-Root | [Prefix]-to-Root | → || Tone | High | ↔ McKendry (2013))

(see section 1 for data and discussion)

C.49 Miya – Africa – Afro-Asiatic (West Chadic) [mkf] (1 pattern)

①	Affix-to-Word	Suffix-to-Word	→	Tone	Low
---	---------------	----------------	---	------	-----

Similarly to the Gã case discussed in the main text (section 2.3), the Afroasiatic language Miya (Schuh 1998) has a functional morpheme triggering Affix-to-Word tone mutation without a segmental affix component. All fully ‘verbal’ verb forms (i.e., those not employing a participle in their formation) enforce a L-tone on a following object noun (156-b) as can be seen from the contrast to the nominal/participial subjunctive forms (e.g., the Subjunctive form [dà màrá mǎdǎ] ‘that he get a goat’). An interesting complication of the Miya pattern is that it is restricted to verbs with a lexical H (156-c).¹³

(156) *Miya Suffix-to-Word Low (Schuh 1998:56)*

	<i>a. Underlying Noun</i>	<i>b. Perfective H-Verb + Noun</i>	<i>c. Perfective L-Verb + Noun</i>	
H	mǎdǎ	à már— mǎdǎ	à zàr mǎdǎ	‘castrated goat’
H-H	dérwétlí	à már— dèrwétlí	à zàr dér wétlí	‘leopard’
L	mbǎrgù	à már— mbǎrgù	à zàr mbǎrgù	‘ram’
Ø	zà:kə	à már— zà:kə	à zàr zà:kə	‘donkey’
		‘he got a ...’	‘he called a ...’	

A natural interpretation of this dependency in a floating-feature account would be to assume that in forms with a L-toned verb, the affix-L can fuse phonetically vacuously with this root-L whereas this option is blocked for high-toned roots where the floating L alternatively docks on the following noun (if there is one).

C.50 Moro – Africa – Kordofanian [mor] (2 patterns)

①	Root-to-Affix ~ =	Root-to-Prefix ~ =	←	Vowel Features	Height
②	Root-to-Affix	Root-to-Suffix	→		

The Kordofanian language Moro (Ritchart & Rose 2017) instantiates vowel height harmony as can be seen in (157). Considerably simplifying, morphemes of the higher set {ɜ, i, u} trigger raising of morphemes of the lower set of vowels {a, e, o}. See Ritchart & Rose 2017 on detailed discussion on whether the involved feature is [ATR] or [high] (the prefix /a-/ RTCL marks finite root clauses, and CLS.g /g-/ concord with nouns of the g-class).

(157) *Moro verb roots (Ritchart & Rose 2017:170)*

a.	é-g-a-vaǎ-ó	‘I shaved’	b.	í-g-3-vǜg-ú	‘I miscarried’
	é-g-a-veǎ-ó	‘I knocked’		í-g-3-kiǎ-ú	‘I opened’
	1SG-RTCL-CLS.g-√-PFV			1SG-RTCL-CLS.g-√-PFV	

Unexpectedly, the same root contrast is also found with verbal and adjectival roots consisting of single consonants such as /-k-/ ‘hear’ or /-p-/ ‘beat’ both triggering higher vowels (158-a). The roots in (158-c,d) form a minimal pair with /t/ ‘drink’ triggering raising and /t/ ‘get lost’ failing to do so.

(158) *Moro vowelless roots (Ritchart & Rose 2017:170)*

a.	é-g-a-ni-ó	‘I heard’	b.	í-g-3-p-ú	‘I beat’
c.	é-g-a-t-ó	‘I got lost’	d.	í-g-3-t-ú	‘I drank’
	1SG-RTCL-CLS.g-√-PFV			1SG-RTCL-CLS.g-√-PFV	

Since this process extends to all affixes also undergoing regular harmony, this is unlikely to be suppletive allomorphy and provides thus compelling evidence for Root-to-Affix mutation.

¹³The underlying forms assumed here follow the analysis of Schuh, implying default insertion of Low on toneless syllables, and insertion of a H on low-toned words (“Low-Raising”, p.54).

C.51 Neve’ei – Papuanesia – Austronesian (Oceanic) [vnm] (1 pattern)

①	Root-to-Affix	Root-to-Prefix	←	Vocalic Features	[+low]
---	---------------	----------------	---	------------------	--------

In Neve’ei (Musgrave 2007), specific prefixes show complete harmony with either the first vowel of the base (159-a) or an initial glide (159-b) (making the standard assumption that [i] ≈ [j], and [u] ≈ [w]). However, for an idiosyncratic group of roots with an initial high-vowel syllable, the prefix is low, not high (159-c).

(159) *Neve’ei 1sg Realis prefix nV-* (Musgrave 2007:23-25)

- a. ni-vi ‘I make’
ne-veh ‘I carry’
na-gah ‘I chase’
b. ni-johjoh ‘I vomit’
nu-wahan ‘I look for’
c. na-sido ‘I remember’
na-tur ‘I stand’

C.52 Nimboran – Papuanesia – Nimboran [nir] (4 patterns)

①	Affix-to-Affix ~ =	Suffix-to-Suffix ~ =	→			
②	Tautomorphemic Affix ~ =	Tautomorphemic Suffix ~ =	↺	Vocalic Features	[-back]	↔ Inkelas (1993)
③	Affix-to-Affix	Suffix-to-Suffix	←			

In section 2.3 we discuss data for the purely consonantal particle suffix /-N/ (see section 4.2 for an OT-analysis). Here we give additional data from syllabic and non-segmental affixes triggering vowel mutation, also providing evidence that the patterns are iterative and bidirectional.

Across-the-Board Mutation: (160) shows an example where another affix triggering the same mutations, Durative /-ta/ has a vowel causing fronting of vowels in multiple affixes (and of its own vowel) and additional raising of the final vowel.

(160) *Affix-to-Affix mutation in Nimboran* (Anceaux 1965:186+234)

- a. ŋgedúo-man-t-ám → ŋgedúo-man-t-ám
draw-SINCL.DU-PRS-INCL ‘you (sg.) and I draw here’
b. ŋgedúo-ta-man-t-ám → ŋgedúo-te-men-t-ím
draw-DUR-SINCL.DU-PRS-INCL ‘you (sg.) and I are drawing here’

Bidirectional Mutation: Our next example shows that Affix-to-Affix mutation in Nimboran is bidirectional, causing fronting also in preceding suffixes. This is shown for the mutating Dual agreement suffix /-se/ in (161) that also causes fronting of the preceding Inclusive /-man/ (cf. the faithful appearance of /-man/ in (160-a)). This fronting, however, does not extend to verbal roots (cf. (20)-b) in the main text) which doesn’t surface as *[seimberi]. As pointed out by Tebay (2019), this is plausibly due to the fact that the root forms an independent initial foot which is protected by positional faithfulness.

(161) *Bidirectional mutation in Nimboran* (Anceaux 1965:188, Inkelas 1993:565)

- ŋgedúo-man-sa-t-ám → ŋgedúo-men-se-t-ím
draw.sg-SINCL.DU-7.LOC-INCL ‘you and I draw from here to below’

④	Affix -to-Affix	Suffix -to-Suffix	→	Vocalic Features	[-back]	↔ Inkelas (1993)
---	---	---	---	------------------	---------	------------------

Inkelas (1993) shows that there is also a derivational particle suffix without segmental content, consisting only of the mutating features. This is shown in (162-b+d) that contrast with non-mutating contexts for the same final suffixes (162-a+c). Since particles appear relatively close to the verb stem, there are no examples showing whether this mutation also (expectedly) affects preceding suffixes.

(162) *Mutation-trigger without segmental content in Nimboran (Anceaux 1965:87+124+125, Inkelas 1993:564+565+576)*

- a. dí-k-d-u → díkedú
roast-SDU-FUT-1 ‘we two will roast (here)’
- b. u-k-[-back]-d-u → uk-__-edí
say-SDU-PART-FUT-1 ‘we two will say (here)’
- c. krí-<i>-dár-k-e → krídiáke
close-<SPL>-PART-PST-2 ‘you (many) closed (here)’
- d. krí-k-[-back]-t-e → krík-__-etí
flee-SDU-PART-PRES-2 ‘you two flee (here)’

C.53 Russian – Eurasia – Indoeuropean (Slavic) [rus] (2 patterns)

①	Root-to-Affix ~	Root-to-Suffix ~	→	Stress	↔ Revithiadou (1999)
②	Tautomorphemic Root	Tautomorphemic Root	↺		

It is well known that Russian has roots where stress is fixed on one of the root syllables (163-a,b), a second class where it alternates between the initial root and a suffix syllable (163-c), and a third ‘postaccenting’ class where stress is on the post-root suffix if there is one and otherwise on the root-final syllable (163-d). Under a classical interpretation (Melvold 1989), both roots and affixes may have underlyingly specified accent, where root (or equivalently: leftmost) specifications systematically win out over later/suffixal ones. Thus, all stresses in (163-a,b) correspond to underlying root stress, the stressed affixes in (163-c) to underlying affix stress (e.g., Genitive Plural -óv) that surfaces in combination with roots like *ostrov* are underlyingly accent-less. The postaccenting class in (163-d) is the most contentious in the literature but a straightforward interpretation is that it triggers Root-to-Affix mutation for stress (Revithiadou 1999).

(163) *Russian stress classes (Molczanow et al. 2019:64)*

	a. Fixed σ1 stress	b. Fixed σ2 stress	c. Alternation Affix/σ1	d. Postaccenting
NOM SG	ávtor	moróz	óstrov	<u>kulák</u>
GEN SG	ávtor-a	moróz-a	óstrov-a	<u>kulak</u> -á
DAT SG	ávtor-u	moróz-u	óstrov-u	<u>kulak</u> -ú
ACC SG	ávtor-a	moróz	óstrov	<u>kulák</u>
INSTR SG	ávtor-om	moróz-om	óstrov-om	<u>kulak</u> -óm
LOC SG	ávtor-e	moróz-e	óstrov-e	<u>kulak</u> -é
NOM PL	ávtor-i	moróz-i	ostrov-á	<u>kulak</u> -í
GEN PL	ávtor-ov	moróz-ov	ostrov-óv	<u>kulak</u> -óv
DAT PL	ávtor-am	moróz-am	ostrov-ám	<u>kulak</u> -ám
ACC PL	ávtor-ov	moróz-i	ostrov-á	<u>kulak</u> -í
INSTR PL	ávtor-ami	moróz-ami	ostrov-ámi	<u>kulak</u> -ámi
LOC PL	ávtor-ax	moróz-ax	ostrov-áx	<u>kulak</u> -áx
	‘author’	‘frost’	‘island’	‘fist’

C.54 Samoan – Papuanesia – Austronesian (Oceanic) [smo] (1 pattern)

①	(Root)-to-Word	(Enclitic)-to-Word	←	Tone	High	↔ Yu (2021)
---	----------------	--------------------	---	------	------	-------------

(see section 2.5 for data and discussion)

C.55 Seenku – Africa – Niger-Congo (Mande) [samo1309] (2 patterns)

①	Root-to-Word ~	→	Consonantal Features	nasal	← McPherson (2017)
②	Tautomorphemic Root	↪			

Seenku has nouns for which McPherson (2017) argues that they have a final floating [nasal] feature which nasalizes following sonorant consonants such as /sâ/ ‘rabbit’ in (164-b) which nasalizes the initial /l/ of the subordinator /lě/ (which surfaces faithfully in (164-a)).

(164) *Seenku nasalization* (McPherson 2017:245)

- a. sǝ-bé-sǎ lě bǝǝŋ lé ně
 beautiful-NOM-thing SUBORD bag REL in
 ‘the bag in which there is an important thing’
- b. sâ ně njǝǝ né jǝ
 rabbit SUBORD honey REL SEC. REAL.PFV
 ‘the honey that the rabbit found’

Stops are prenasalized if preceded by the floating [nasal] (165-b). If there is no following word, the nasal feature is lost (166-a). The examples in (165-a’,b’,c’) show the behaviour of a minimally distinct noun which lacks floating [nasal]. In plural forms of nasalizing nouns (165-c), there is no nasalization of a following word, and the nasal feature is realized instead on the vowel of the noun itself:

(165) *Seenku nasalization* (McPherson 2017:244)

- | <i>Nasalizing</i> | <i>Non-nasalizing</i> |
|--|--------------------------------|
| a. kâ
‘hut’ | a’. kâ
‘griot’ |
| b. kâ mbələ
‘big hut’ | b’. kâ bələ
‘big griot’ |
| c. kě bələ
‘big huts’ | c’. kě bu-bələ
‘big griots’ |

See McPherson (2017) for an analysis which derives the plural pattern from a generalized ban on tautomorphemic docking. The nasal feature is not allowed to associate to the purely lexical vowel of a singular noun, but may do so in the plural where the stem vowel of the noun has already associated to floating material of the plural affix resulting in raising.

C.56 Shilluk – Africa – Eastern Sudanic (Western Nilotic) [shk] (2 patterns)

①	(Affix)-to-Word	(Suffix)-to-Word	→	Length	vowel length	← Remijsen & Ayoker (2020)
---	-----------------	------------------	---	--------	--------------	----------------------------

See section 2.4 for data and discussion of nominal plural cases. (166) shows analogous data for nominal demonstrative forms.

(166) *Shilluk demonstrative Affix-to-Word lengthening* (Remijsen & Ayoker 2020:145)

- | | | | | | |
|----------------------------|---------------|---|----------------|----------------------------------|-----------|
| a. kùl | á-lî:d-à | b. kùl | ú-lî:d-ò | c. kùl | ù-lî:d-ò |
| warthog | PAST-look-1SG | warthog | NEVP-look-NEVP | warthog | IPFV-look |
| ‘I looked at the warthog’ | | ‘Sb. apparently looked at the warthog’ | | ‘Sb. is looking at the warthog’ | |
| d. kùl-__ | ǎ:lî:d-à | e. kùl-__ | ũ:lî:d-ò | f. kùl-__ | û:lî:d-ò |
| warthog-DEM | PST-look-1SG | warthog-DEM | NEVP-look-NEVP | warthog-DEM | IPFV-look |
| ‘I looked at this warthog’ | | ‘Sb. apparently looked at this warthog’ | | ‘Sb. is looking at this warthog’ | |

②

Root-to-Word	→	Length	vowel length
--------------	---	--------	--------------

 ↪ Remijsen & Ayoker (2020)

Shilluk not only has Affix-to-Word mutation but also mutation triggered by noun roots on other words. This is shown by the examples in (167-a,b,c) which trigger lengthening on the initial vowel of the following numeral modifier. The nouns in (167-d,e,f) show that this effect is not a general property of this construction but lexeme-specific.

(167) *Root-to-Word length mutation: Shilluk* (Remijsen & Ayoker 2020:147)

- | | | | | | |
|-------------------|-----------|----------------|-----------|---------------|-----------|
| a. pūk | á:kjèl | b. kīt | á:kjèl | c. kǎ:k | á:kjèl |
| storage.pot | CRD-first | mountain | CRD-first | harpoon | CRD-first |
| ‘one storage pot’ | | ‘one mountain’ | | ‘one harpoon’ | |
| d. bàt | á:kjèl | e. wāṽ | á:kjèl | f. úgĩ:k | á:kjèl |
| arm | CRD-first | eye | CRD-first | buffalo | CRD-first |
| ‘one arm’ | | ‘one eye’ | | ‘one buffalo’ | |

C.57 Shoshoni – Uto-Aztecan [shh] (3 patterns)

①

Root-to-Word	→	Consonantal Features	[-voice]
--------------	---	----------------------	----------

Shoshoni has lexical roots which are vowel-final in isolation but trigger devoicing (in addition to regular phonological spirantization) on obstruents in following words, as can be seen in (168).

(168) *Shoshoni devoicing (and spirantization)* (Crum & Dayley 1993:250)

- | | | | | | | |
|-----------|---|-------|---|-------|-------|---------------------------|
| a. kap:i | + | paʔa | → | kap:i | ɸaʔa | ‘on top of the pine cone’ |
| b. kap:i | + | tuk:a | → | kap:i | θuk:a | ‘under the pine cone’ |
| c. kap:i | + | kup:a | → | kap:i | xup:a | ‘in the pine cone’ |
| pine.cone | | POSTP | | | | |

That this behavior is root-specific can be seen from minimal pairs of devoicing noun roots (169-a) with other vowel-final noun roots where spirantization applies to the same incorporating verb /paʔi/ ‘have’, but the intervocalic consonant is voiced (169-b).

(169) *Shoshoni spirantization with and without devoicing* (Crum & Dayley 1993:251)

- | | | | | | | |
|----------|---|------|---|--------|------|----------------------|
| a. hai | + | paʔi | → | hai- | ɸaʔi | ‘have an uncle/crow’ |
| si:pɛ | + | paʔi | → | si:pɛ- | ɸaʔi | ‘have urine’ |
| b. kahni | + | paʔi | → | kahni- | βaʔi | ‘have a house’ |
| tso: | + | paʔi | → | co:- | βaʔi | ‘have a grandparent’ |
| N | | have | | | | |

(170) shows that the [-voice] feature of roots which trigger this alternation surfaces as the full segment [h] before sonorants.¹⁴

¹⁴The [-voice] feature remains inert before geminates and consonant clusters, which seem to be generally voiceless (Crum & Dayley 1993:251)

(170) *Shoshoni segmental realization of mutation feature (Crum & Dayley 1993:250)*

a.	kap:i	+	ma	→	kap:i	hma	‘with the pine cone’
b.	kap:i	+	ni	→	kap:i	hni	‘like a pine cone’
c.	kap:i	+	waʔih	→	kap:i	hwaʔih	‘like a pine cone’
	pine.cone		POSTP				

③	Root-to-Affix ~	Root-to-Suffix ~	→	Length	Gemination
④	Root-to-Word	Root-to-Word	→		

Sapir (1930:111,142) shows that specific nouns and verbs in Shoshoni also trigger unpredictable gemination on following affixes or independent words. Thus the root /ja:/ ‘carry, take’ induces gemination on different following suffixes (171).

(171) *Root-to-Affix gemination in Shoshoni (Pycha 2008:150, Crum & Dayley 1993:253)*

ja: ‘carry/take-sg’

ja:	+	-taipeh (COMPL)	→	ja:-	taipeh	‘taken completely’
ja:	+	-kan (STAT)	→	ja:-	kan	‘hold’
ja:	+	-kin (here)	→	ja:-	kin	‘bring here’
ja:	+	-kwantoʔin (going.to)	→	ja:-	kwantoʔin	‘going to take away’

The noun /tua/ ‘son triggers gemination on following postpositions:

(172) *Root-to-Word gemination in Shoshoni (Crum & Dayley 1993:248+249)*

a.	kap:i	+	ma	→	tua	maʔai	‘with the son’
b.	tua	+	ni	→	tua	ni	‘like a son’
c.	tua	+	kup:an	→	tua	kup:an	‘in side the son’
	son		POSTP				

C.58 Supyire – Africa – Niger-Congo (Senufo) [spp] (3 patterns)

①	Affix-to-Word	Suffix-to-Word	→	Tone	Low	↔ Carlson (1994)
---	---------------	----------------	---	------	-----	------------------

(see section 2.4 for data and discussion)

②	Root-to-Word ~	→	Tone	Low	↔ Carlson (1994)
③	Tautomorphemic Root	↪			

(see section 3.3 for data and discussion)

C.59 Tagalog – Papuanesia – Austronesian (Malayo-Polynesian) [tgl] (2 patterns)

①	Root-to-Affix ~	Root-to-Suffix ~	→	Stress	↔ Hagberg (2006)
②	Tautomorphemic Root	Tautomorphemic Root	↪		

Tagalog shows the mirror image of Mayo (appendix C.47). Roots determine whether stress is final or penultimate (173), a pattern which is preserved under suffixation, as shown in (174). Following Hagberg (2006), we assume that penultimate accent is the default pattern, and final accent is the effect of Root-to-Affix mutation alternating with tautomorphemic root mutation triggered by floating accent grid marks.

(173) *Lexical root stress in Tagalog (Hagberg 2006:176, French 1988)*

Penultimate Stress Roots

- a. pú:noʔ ‘trunk of a tree’
 b. bá:soh ‘drinking glass’
 c. bú:kas ‘tomorrow’

Final Stress Roots

- d. punóʔ ‘full’
 e. basóh ‘target practice’
 f. bukás ‘open’

(174) *Root-to-Affix stress in Tagalog (Hagberg 2006:176, French 1988)*

Penultimate Stress Roots

- a. ʔá:ral ‘study’ ʔa rá:l-an ‘place for studying’
 b. bá:sa ‘read’ basá:h-in ‘to read’
 c. pá:sok ‘enter’ pa sú:k-an ‘(someone) places (something) inside of X’

Final Stress Roots

- d. dugóʔ ‘blood’ dugu- án ‘bloody’
 e. ʔupóʔ ‘sit’ ʔupoʔ- án ‘(someone) sits on X’
 f. bilih ‘to buy’ bilih- ín ‘to buy X’

C.60 Tamil – Eurasia – Dravidian [tam] (1 pattern)

①	<u>Affix</u> -to-Affix ~ <u>Affix</u> -to-Root	<u>Suffix</u> -to-Suffix ~ <u>Suffix</u> -to-Root	→ ~ ←	Length	consonant gemination
---	--	---	-------------	--------	----------------------

(see section 2.3 for data and discussion)

C.61 Tauya – Papuanesia – Nuclear Trans New Guinea [tya] (1 pattern)

①	Root-to-Affix	Root-to-Suffix	→	Consonantal Features	[-continuant]
---	---------------	----------------	---	----------------------	---------------

Tauya shows Root-to-Affix hardening. The initial /r/ of the topic suffix /-ra/ is plosivized to [t] after certain pronominal bases, as is shown in (175).

(175) *Tauya personal pronoun topic forms (MacDonald 2011:62)*

	1	2	3	‘this (one)’	‘that (one)’
sg.	ja-ra	<u>na-</u> <u>ta</u>	<u>ne-</u> <u>ta</u>	<u>me-</u> <u>ta</u>	ʔe-ra
pl.	<u>se-</u> <u>ta</u>	<u>te-</u> <u>ta</u>	<u>ne-</u> <u>ta</u>	—	

C.62 Tenyidie – Eurasia – Sino-Tibetan [njm] (1 pattern)

①	Root-to-Affix	Root-to-Suffix	→	Tone	high register	↔ Meyase (2020)
---	---------------	----------------	---	------	---------------	-----------------

(see section 2.2 for data and discussion)

C.63 Tetsóť'iné – Na-Dene (Athabaskan) [chp] (1 pattern)

①	Affix-to-Affix	Prefix-to-Prefix	←	Tone	High	↔ Jaker & Kiparsky (2020)
---	----------------	------------------	---	------	------	---------------------------

Jaker & Kiparsky (2020) describe and analyze in detail the tonal behavior of two conjugational markers in Tetsóť'iné, /θe-/ and /ɲe-/, which trigger a H-tone on preceding prefixes with rhythmically and stratically conditioned alternations. (176) shows data for /θe-/ but the pattern for /ɲe-/ is mostly parallel. If the affix is preceded by two syllables, its High surfaces on the immediately preceding syllable (176-a), if preceded by three syllables, two syllables to the left (176-b). If only one syllable precedes, the H is deleted (176-c).

(176) *Tetsóť'iné Affix-to-Affix H-tone (Jaker & Kiparsky 2020:630)*

<i>Input</i>	<i>Intermediate output</i>	<i>Surface</i>	
<i>a. Preceded by 2 syllables: left-adjacent association</i>			
/ʔede- ^H (θ-'ká:r)/ REFL-CONJ-slap	(ʔe.'déh)(θ-'ká:r)	(ʔe.'déh)(-'ká:r)	'he slapped himself'
/ʔede- ^H (θ-'ts'ór)/ REFL-CONJ-scratch	(ʔe.'déh)(θ-'ts'ór)	(ʔe.'déh)(-'ts'ór)	'he scratched himself'
/ʔede- ^H (θ-tθí:/) REFL-CONJ-pinch	(ʔe.'déh)(θ-tθí:/)	(ʔe.'déh)(-tθí:/)	'he pinched himself'
<i>b. Preceded by 3 syllables: left-association and shifting</i>			
/ʔede-he- ^H (θ-'ká:r)/ REFL-S3PL-CONJ-slap	(ʔe.'déh)(heθ.'ká:r)	(ʔe.'déh)(heh.'ká:r)	'they slapped themselves'
/ʔede-he- ^H (θ-'ts'ór)/ REFL-S3PL-CONJ-scratch	(ʔe.'déh)(heθ.'ts'ór)	(ʔe.'déh)(heh.'ts'ór)	'they scratched themselves'
/ʔede-he- ^H (θ-tθí:/) REFL-S3PL-CONJ-pinch	(ʔe.'déh)(heθ.'tθí:/)	(ʔe.'déh)(heh.'tθí:/)	'they pinched themselves'
<i>c. Preceded by 1 syllable: H-deletion</i>			
/he- ^H (θe-'ká:r)/ S3PL-CONJ-slap	(he.θe)(-'ká:r)	(-hee)(-'ká:r)	'they slapped'
/he- ^H (θe-'ts'ór)/ S3PL-CONJ-scratch	(he.θe)(-'ts'ór)	(-hee)(-'ts'ór)	'they scratched'
/he- ^H (θe-'gor)/ S3PL-CONJ-stab	(he.θe)(-'gor)	(-hee)(-'gor)	'they stabbed'

Note that these data don't violate phonological locality of anticyclic mutation as formulated in section 5.3 since, as shown by Jaker & Kiparsky, the dislocation of the H-tone to a non-adjacent syllable follows from a general phonological process in the language, the attraction of H-tones to stressed syllables. See their paper for an analysis in terms of floating H's and further effects of stratal organization.

C.64 Tiriki – Africa – Niger-Congo (Bantu) [luy] (1 pattern)

①	Affix-to-Word	Prefix-to-Word	←	Tone	High	↔ Paster & Kim (2011)
---	---------------	----------------	---	------	------	-----------------------

(see section 2.4 for data and discussion)

C.65 Tlapanec – N America – Oto-Manguean [tcf] (1 pattern)

①	Affix-to-Root ~	Prefix-to-Root ~	→ ~ ↺	Tone	Low	↔ Uchihara & Cano (2019)
	Tautomorphemic Affix	Tautomorphemic Prefix				

(see section 3.3 for data and discussion)

C.66 Turkana – Africa – Eastern Sudanic (Eastern Nilotic) [tuv] (1 pattern)

①	Root-to-Affix	Root-to-Suffix	→	Vocalic Features	[+ATR]	↔ Noske (1996)
---	---------------	----------------	---	------------------	--------	----------------

(see section 2.2 for data and discussion)

C.67 Urarina –South America – Isolate [urn] (4 patterns)

Root-to-Word	→	Tone	① High ② Low High Low ③ Low High ④ Low
--------------	---	------	---

In Urarina (Olawsky 2006), there are four classes of nouns which have different tonal mutation effects on following words such as verbs and adjectives (177). Under Olawsky’s classification, type A nouns trigger an initial High on the target word, type B nouns a High on the second or third syllable (depending on the length of the word), type C a final High, and type D an all-Low pattern. Note that this difference in H-positioning is similar to the one triggered by word-internal affixes in Kuria.

(177) *Urarina phrasal tone (Olawsky 2006:128)*

Type A	Initial High	<u>rà:nà</u>	<u>hérià</u>	‘he wants the white-lipped peccary’
Type B	Medial High	òbà <u>nà</u>	hè <u>rì</u> à	‘he wants the collared peccary’
Type C	Final High	rè:mà <u>è</u>	hèrì <u>á</u>	‘he wants the dog’
Type D	All Low	màkùsàjà <u>rí</u>	hèrìà	‘he wants the pepper’
		N	want.3sg	

C.68 Uspanteko – N America – Mayan [usp] (1 pattern)

①	Affix-to-Affix ~	Prefix-to-Prefix ~	← ~ →	Tone	High
	Affix-to-Root	Prefix-to-Root			

Bennett & Henderson (2013) show that the Mayan language Uspanteko has a general nominal possessive head realized exclusively by a High Tone while specific phi-features of the possessor are marked by segmental prefixes (178). As all H’s in the language, it shows up on the penultimate vocalic mora of the word (1sg here is [w-] before vowel-initial and [in-] before consonant-initial roots, Bennett & Henderson 2013:605).

(178) *Uspanteko Possessor High (Bennett & Henderson 2013:604)*

a. aqan	b. w- <u>á</u> qan	c. aw- <u>á</u> qan
‘leg’	‘my leg’	‘your leg’
d. pix	e. <u>ín</u> -pix	f. <u>qá</u> -pix
‘tomato’	‘my tomato’	‘our tomato’

That the H is independent from the pronominal prefixes is particularly clear from the fact that the same affixes are used in predicative constructions but *without* a concomitant High, as can be seen in (179).

(179) *Uspanteko: lack of High in predicative position (Bennett & Henderson 2013:604)*

- a. kar b. ín—kar c. in-kar
 ‘fish’ ‘my fish’ ‘I am a fish’

In Bennett & Henderson’s syntactic analysis, (but, as far as we can see, also in plausible lexicalist alternatives), the general Possessor morpheme is structurally inside the pronominal prefixes it licenses. Thus with monomoraic nouns where the Possessive High appears on the prefix, this instantiates anticyclic mutation (178-b-c). With longer nouns mutation cyclically affects the noun itself (178-e-f).

C.69 Welsh –Eurasia – Indoeuropean (Celtic) [cym] (2 patterns)

Root-to-Word	→	Consonantal Features	① lenition ② spirantisation	↔ Iosad (2012) and references cited there
--------------	---	----------------------	--------------------------------	---

(see section 2.5 for data and discussion)

C.70 Yoruba – Africa – Niger-Congo [yor] (1 pattern)

①	Root-to-Affix	Root-to-Prefix	←	Vocalic Features	[-ATR]	↔ Archangeli & Pulleyblank (1994)
---	---------------	----------------	---	------------------	--------	-----------------------------------

In Yoruba, prefixes harmonize in [ATR] with the first vowel of their base, as shown in (180) with prefixes nominalizing verbs.

(180) *[ATR]-harmony in Yoruba (Archangeli & Pulleyblank 1989:188,210)*

Verb	Nominalization
a. rò ‘think’	è-rò ‘thought’
b. rọ ‘fabricate’	è-rọ ‘machine’
c. tá ‘shoot’	ó-tá ‘person who is a good shot’
d. kú ‘die’	ò-kú ‘corpse of a person’
e. pín ‘come to an end’	ò-pín ‘end/termination’

However, there is a handful of roots with an initial high [+ATR] vowel which render their prefixes [-ATR], illustrated in (181). Archangeli & Pulleyblank (1994) capture this by assuming a floating [-ATR] feature.

(181) *[-ATR]-Mutation in Yoruba (Archangeli & Pulleyblank 1994:148+149)*

Verb	Nominalization
a. mú ‘drink’	ò -mú ‘drinker’
b. bí ‘give birth to’	è -bí ‘birth’
c. wù ‘please person’	é -wù ‘a pleasurable feeling’
d. rí ‘exhibit a certain appearance’	è -rí ‘evidence’

D Abbreviations

AFF	Affirmative
ASSOC	Associative marker
AUG	Augmentative
CONJ	Conjugation marker
CONT	Continuous
CLS	(Noun) class
COOR	Coordination
CRD	Cardinal
CURR.PST	Current past
DEC	Declension marker
DER	derivational affix
FV	Final vowel
HAB	Habitual
IPFV	Imperfective
INT	Intensive
IV	Initial vowel
INV	Inverse
MAL	Malefactive
MOD	Modal
NEVP	nonevidential past
O	Object
OBV	Obviative
PART	Particle
POSTP	Postposition
PF	Prefinal
RPST	Remote past
RTCL	Root clause
S	Subject
SPST	Simple past
SUBORD	subordinator
TNS	Tense
UW.PST	Unwitnessed past
VBLZ	Verbalizer
√	lexical root
Ø	(in glosses) epenthetic material

E References (for the Appendix)

- Abramovitz, Rafael (2019) Against the Richness of the Base: Two Koryak Case Studies. Talk at the IGRA Colloquium Series, February 6, 2019.
- Abramovitz, Rafael Meghani (2021) Topics in the Grammar of Koryak. PhD thesis, MIT.
- Abrusan, Marta (2005) Underspecified precedence relations and vowel-zero alternations in Hungarian. In: *Proceedings of BLS 31*. 1–12.
- Ahland, Michael Bryan (2012) A Grammar of Northern Mao. PhD thesis, University of Oregon.
- Akinlabi, Akinbiyi (2011) Featural Affixes. In: M. van Oostendorp, C. J. Ewen, E. Hume & K. Rice, eds., *The Blackwell companion to phonology*. Wiley-Blackwell, Hoboken, NJ, 1945–1971.
- Anceaux, J. C. (1965) *The Nimboran Language: Phonology and Morphology*. Martinus Nijhof, 's Gravenhage.
- Archangeli, Diana & Douglas Pulleyblank (1989) Yoruba Vowel Harmony, *Linguistic Inquiry* 20(2):173–217.
- Archangeli, Diana & Douglas Pulleyblank (1994) *Grounded Phonology*. MIT Press, Cambridge MA.
- Ashkaba, John Abraha, Bruce A. Connell & Richard J. Hayward (2000) Observations on Kunama tone, *Studies in African Linguistics* 29(1):1–42.
- Bennett, Ryan & Robert Henderson (2013) Accent in Uspanteko, *NLLT* 31:589–645.
- Black, Cheryl A. (1995) Boundary Tones on Word-Internal Domains in Kinande, *Phonology* 12(1):1–38.
- Bliese, Loren F. (1981) *A generative grammar of Afar*. Vol. 65 of *Summer Institute of Linguistics Publications in Linguistics*, Summer Institute of Linguistics and the University of Texas at Arlington, Dallas.
- Buckley, Eugene (1994) *Theoretical aspects of Kashaya phonology and morphology*. CSLI Publications, Stanford, California.
- Buckley, Eugene (2018) Kashaya foot extrametricality as post-accentuation. Talk at Annual Meeting on Phonology UC San Diego 7 October 2018.
- Buckley, Eugene (2023) M-words, P-words, and accent phrases in Kashaya. In: K. Bogomolets & H. van der Hulst, eds., *Word Prominence in Languages with Complex Morphologies*. Oxford University Press, Oxford, 248–273.
- Cahill, Michael (1999) *Aspects of the morphology and phonology of Konni*. Vol. 141, SIL International and the University of Texas at Arlington publications in linguistics.
- Cahill, Michael (2000) Tonal Associative Morphemes in Optimality Theory, *OSU Working Papers in Linguistics* 53:31–70.
- Cahill, Michael (2004) Tone Polarity in Konni nouns, *Studies in African Linguistics* 33(1):1–33.
- Cahill, Michael (2011) Tonal Diversity in Languages of Papua New Guinea. Technical Report Electronic Working Paper 2011-008, SIL.
- Carlson, Robert (1994) *A Grammar of Supyire*. Walter de Gruyter.
- Chai, N. M (1971) A grammar of Aklan. PhD thesis, UMI, Ann Arbor.
- Chierchia, Gennaro (1982) An Autosegmental Theory of Raddoppiamento. In: *Proceedings of NELS*. Vol. 12, p. 1.
- Churma, Donald G. (1988) Consonant gradation in Fula suffixes: the ugly truth, *Studies in African Linguistics* 19(1):35–74.
- Clark, Mary M. (1990) *The tonal system of Igbo*. Foris, Dordrecht.
- Clements, G. N. & K. C. Ford (1981) On the phonological status of downstep in Kikuyu. In: D. L. Goyvaerts, ed., *Phonology in the 1980's*. Story-Scientia, Ghent, 309–357.
- Clements, G.N. (1984) Principles of Tone Assignment in Kikuyu. In: G. Clements & J. Goldsmith, eds., *Autosegmental Studies in Bantu Tone*. Foris Publications, Dordrecht, 281–340.
- Crum, Beverly & Jon Dayley (1993) *Western Shoshoni grammar*. Boise State University, Boise.
- Dettmer, Hans Adalbert (1989) *Ainu-Grammatik. Volume 1: Texte und Hinweise*. Harrassowitz, Wiesbaden.
- Downing, Laura J. & Al Mtenje (2017) *The Phonology of Chichewa*. Oxford University Press, Oxford.
- Dryer, Matthew S. & Martin Haspelmath, eds. (2013) *The World Atlas of Language Structures Online*. Max Planck Institute for Evolutionary Anthropology. (Available online at <http://wals.info>), Leipzig.
- Feeling, Durbin (1975) *Cherokee-English dictionary*. William Pulte ed. Tahlequah, OK: Cherokee Nation of Oklahoma.
- Frantz, Chester & Marjorie Frantz (1973) Gadsup Phoneme and Toneme Units. In: H. McKaughan, ed., *The languages of the eastern family of the East New Guinea Highland stock*. The University of Washington Press, Seattle, 406–413.
- French, Koleen Matsuda (1988) *Insights into Tagalog reduplication, infixation, and stress from nonlinear phonology*. Vol. 84, Summer Institute of Linguistics and the University of Texas at Arlington Publications in Linguistics.
- Gleim, Daniel (2015) Palatalization in Acoma. Ms., Universität Leipzig.
- Gomez-Imbert, Elsa & Michael Kenstowicz (2000) Barasana tone and accent, *International Journal of American Linguistics* 66:419–63.

- Hagberg, Lawrence Raymond (2006) *An Autosegmental Theory of Stress*. SIL International.
- Hantgan, Abbie (2009) Does Tone Polarity Exist? Evidence From Plural Formation among Bangime Nouns, *Indiana University Working Papers in Linguistics* 8:69–83.
- Hayes, Bruce (1980) Aklan stress: disjunctive ordering or metrical feet?. In: *Cahiers linguistiques d'Ottawa 9, Papers from the Tenth Meeting of the North East Linguistic Society*. University of Ottawa, 179–181.
- Heath, Daniel (1991) Tone in the Makaa Associative Construction. In: S. C. Anderson, ed., *Tone in Five Languages of Cameroon*. Summer Institute of Linguistics, Dallas, 3–28.
- Hyman, L. M. & A. Mtenje (1999) Prosodic morphology and tone: the case of Chichewa. In: R. Kager, H. van der Hulst & W. Zonneveld, eds., *The prosody-morphology interface*. Cambridge University Press, Cambridge, 90–133.
- Hyman, Larry M. (2001) Privative tone in Bantu. In: K. Shigeki, ed., *Proceedings of the Symposium "Crosslinguistic Studies of Tonal Phenomena: Tonogenesis, Japanese Accentology, and Other Topics"*. Tokyo University of Foreign Studies, Tokyo, 237–258.
- Hyman, Larry M. (2011) Tone: Is it Different?. In: J. Goldsmith, J. Riggle & A. Yu, eds., *The Handbook of Phonological Theory*. 2nd edn, Blackwell, 197–239.
- Hyman, Larry M. & Armindo Ngunga (1994) On the non-universality of tonal association conventions: evidence from Ciyao, *Phonology* 11:25–68.
- Hyman, Larry. M, ed. (1979) *Aghem grammatical structure*. University of Southern California. SCOPIL 7, Los Angeles.
- Inkelas, Sharon (1993) Nimboran position class morphology, *NLLT* 11:559–624.
- Iosad, Pavel (2012) Representation and variation in substance-free phonology A case study in Celtic. PhD thesis, University of Tromsø.
- Ito, Junko (1984) Melodic Dissimilation in Ainu, *Linguistic Inquiry* 15(3):505–513.
- Jaker, Alessandro & Paul Kiparsky (2020) Level ordering and opacity in Tetsôt'iné: a Stratal OT account, *Phonology* 37:617–655.
- Kiparsky, Paul (2003) Finnish Noun Inflection. In: D. Nelson & S. Manninen, eds., *Generative Approaches to Finnic Linguistics*. CSLI, Stanford, 109–161.
- Kouneli, Maria (2022) Inflectional classes in Kipsigis, *Glossa: a journal of general linguistics* 7(1):1–33.
- Lee, Hyunjung & Jie Zhang (2014) The nominal pitch accent system of South Kyungsang Korean, *Journal of East Asian Linguist* 23:71–111.
- Loving, Richard (1966) Awa phonemes, tonemes, and tonally differentiated allomorphs, *Papers in New Guinea Linguistics* 5:23–32.
- MacDonald, Lorna (2011) *A Grammar of Tauya*. Mouton de Gruyter, Berlin.
- Malone, Terry (2010) Nasality and Nasal Prosody in Chimila, *IJAL* 76(1):1–41.
- Marlo, Michael R. & David Odden (2007) The exponence of TAM in Bakweri, *SOAS Working Papers in Linguistics* 15:19–31.
- McKendry, Inga (2013) Tonal Association, Prominence and Prosodic Structure in South-eastern Nochixtlán Mixtec. PhD thesis, University of Edinburgh.
- McPherson, Laura (2016) Cumulativity and ganging in the tonology of Awa suffixes, *Language – Phonological Analysis* 92.(1):e38–e66.
- McPherson, Laura (2017) Multiple feature affixation in Seenku plural formation, *Morphology* 27:217–252.
- Melvold, Janis (1989) Structure and stress in the phonology of Russian. PhD thesis, MIT.
- Meyase, Savio Megolhuto (2020) Tone Polarity, Tone Features, and Tonal Representation in the Four-level Tone System of Tenyidie. PhD thesis, Universität Leipzig.
- Miestamo, Matti, Dik Bakker & Antti Arppe (2016) Sampling for variety, *Linguistic Typology* 20(2):233–296.
- Miller, Wick R. (1965) *Acoma Grammar and Texts*. University of California Press, Berkeley.
- Molczanow, Janina, Ekaterina Iskra Olga Dragoy, , Richard Wiese & Ulrike Domahs (2019) Default stress assignment in Russian: evidence from acquired surface dyslexia, *Phonology* 36:61–90.
- Musgrave, Jill (2007) *A grammar of Neve'ei, Vanuatu*. Vol. 587 of *Pacific Linguistics*, Australian National University, Canberra.
- Noonan, Michael (1992) *A grammar of Lango*. Mouton, Berlin and New York.
- Noske, Manuela (1996) [ATR] harmony in Turkana, *Studies in African Linguistics* 25(1):61–99.
- Olawsky, Knut (2006) *A Grammar of Urarina*. : Mouton de Gruyter, Berlin, New York: Berlin.
- Oswalt, Robert L. (1961) *A Kashaya Grammar (Southern Pomo)*. PhD thesis, UC Berkeley.
- Paradis, Carol (1992) *Lexical Phonology and Morphology: The Nominal Classes in Fula*. New York: Garland.
- Passino, Diana (2013) A unified account of consonant gemination in external sandhi in Italian, *The Linguistic Review* 30(2):313–346.
- Paster, Mary (2003) Floating tones in Gã, *Studies in African Linguistics* 32(1):17–39.

- Paster, Mary & Yuni Kim (2011) Downstep in Tiriki, *Linguistic Discovery* 9(1):71–104.
- Pycha, Anne (2008) Morphological Sources of Phonological Length. PhD thesis, UC Berkeley.
- Reh, Mechthild (1993) *Anywa Language*. Rüdiger Köppe Verlag, Köln.
- Remijsen, Bert & Otto Gwado Ayoker (2020) Floating quantity in Shilluk, *Language* 96(3):135–156.
- Revithiadou, Anthi (1999) Headmost Accent Wins: Head Dominance and Ideal Prosodic Form in Lexical Accent Systems.. PhD thesis, Leiden University.
- Ritchart, Amanda & Sharon Rose (2017) Moro vowel harmony: implications for transparency and representations, *Phonology* 34:163–200.
- Rose, Sharon (2007) Chaha (Gurage) Morphology. In: A. S. Kaye, ed., *Morphologies of Asia and Africa*. Eisenbrauns, 399–424.
- Sande, Hannah (2018) Cross-word morphologically conditioned scalar tone shift in Guébie, *Morphology* 28:253–295.
- Sapir, Edward (1930) The Southern Paiute Language (Part I): Southern Paiute, a Shoshonean Language. In: *Proceedings of the American Academy of Arts and Sciences*. Vol. 65, 1–296.
- Scancarelli, Janine (1987) Grammatical relations and verb agreement in Cherokee. PhD thesis, UCLA.
- Schrock, Terrill B. (2014) A grammar of Ik (Icé-tód): Northeast Uganda's last thriving Kuliak language. PhD thesis, Universiteit Leiden.
- Schuh, Russell G. (1998) *A Grammar of Miya*. Vol. 130. of *University of California Publications in Linguistics*, UC Press, Berkeley.
- Siptár, P. & M. Törkenczy (2000) *The Phonology of Hungarian*. Oxford University Press, Oxford.
- Snider, Keith L. (2018) *Tone Analysis for Field Linguists*. SIL International Publications.
- Stirtz, Timothy M. (2011) A grammar of Gaahmg. PhD thesis, Leiden University.
- Tebay, Sören (2019) Outward mutation in Nimboran. Poster at mfm 27, May 25 2019.
- Trommer, Jochen (2021) The subsegmental structure of German plural allomorphy, *Natural Language and Linguistic Theory* 39:601–656.
- Trommer, Jochen (2024) The concatenative structure of tonal overwriting, *Linguistic Inquiry* 55(1):95–151.
- Trommer, Jochen (to appear) Gestalt Contours, Polarity and Construction-specific Phonology in Gaahmg, *Phonological Data & Analysis*.
- Uchihara, Hiroto (2013) Tone and Accent in Oklahoma Cherokee. PhD thesis, Buffalo State University of New York.
- Uchihara, Hiroto & Tiburcio Cano (2019) A phonological account of Tlapanec tonal alternation, *Journal of Linguistics* 1–57.
- Vago, Robert M. (1980) *The sound pattern of Hungarian*. Georgetown University Press, Washington.
- Vajda, Edward J. (2001) The role of position class in Ket verb morphophonology, *Word* 52(3):369–436.
- Wolf, Matthew (2005) An Autosegmental Theory of Quirky Mutations. In: J. Alderete, C.-H. Han & A. Kochetov, eds., *West Coast Conference on Formal Linguistics (WCCFL) 24*. Cascadia Proceedings Project, Somerville, MA, 370–378.
- Wolf, Matthew (2007) For an Autosegmental Theory of Mutation. In: L. Bateman, M. O'Keefe, E. Reilly & A. Werle, eds., *University of Massachusetts Occasional Papers in Linguistics 32: Papers in Optimality Theory III*. GLSA, Amherst, 315–404.
- Wunderlich, Dieter & Barbara Stiebels (1999) Second stems in Hungarian nouns, *The Linguistic Review* 16:253–294.
- Yip, Moira (1989) Contour tones, *Phonology* 6:149–174.
- Yip, Moira (2002) *Tone*. Cambridge University Press, Cambridge.
- Yokwe, Eluzai M. (1986) The tonal grammar of Bari. PhD thesis, University of Illinois at Urbana-Champaign.
- Yu, Kristine M. (2021) Tonal marking of absolutive case in Samoan, *Natural Language & Linguistic Theory* 39:291–365.